

ATHENA ULTIMATE V15.3.3

A Unified Topo-Geometric Framework for Fundamental Physics

Complete Revised Monograph: Volumes 0–IV

Mathematical Foundation Update

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Mathematical Foundation Update

Abstract and Scope

ATHENA Ultimate V15.3.3 is a controlled mathematical-physics research monograph and reproducibility release. Its frozen lower-order core consists of Einstein–Hilbert gravity, one real compact scalar $\Phi \sim \Phi + 2\pi f_a$ with a periodic potential, and matter coupled through a pure-disformal metric. The internal flat T^2 is configuration space rather than an additional spacetime dimension.

This edition separates proved algebra, conditional dynamics, calibrations, fits, historical targets, open problems, and no-go results. It records the rank-one disformal inverse and determinant, ADM decomposition, local degree count, matter exchange, homogeneous and dimensionless FLRW equations, source-free stability limits, exact scalar backgrounds, a Derrick obstruction, and a conditional mismatch between sustained kinetic lapse scaling and the phenomenological collapsed-fraction profile. Reproducibility code and tests verify algebraic identities and source-free limits without observational parameter fitting.

The V15.3.3 mathematical update additionally records the exact pure-disformal connection and curvature-difference identities, the associated geodesic relation, a global limit-set theorem for the closed expanding scalar-only FLRW system, self-dual derivative and asymptotic information for the exact Epstein continuation, and the explicit hypersurface-deformation algebra of the closed Einstein–scalar Hamiltonian constraints.

The release is not experimentally validated and does not claim a completed unified theory, a Standard Model derivation, emergent gravity, a complete matter sector, a particle-mass construction, or a solved cosmological likelihood. Unspecified matter, neutrino, topological, higher-operator, perturbation, lensing, growth, and waveform sectors remain explicitly open.

Keywords: mathematical physics; theoretical physics; compact scalar; pure-disformal metric; scalar–tensor theory; FLRW cosmology; effective field theory; reproducibility.

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V15.3.1 Controlled Consistency Patch Note

V15.3.1 is a minimal consistency patch of the frozen V15.3 baseline. It preserves the single real compact scalar $\Phi \sim \Phi + 2\pi f_a$, its winding classification, and all independent torus mathematics. The physical matter metric remains the pure-disformal ansatz

$$\tilde{g}_{\mu\nu} = g_{\mu\nu} - \frac{\alpha}{M_*^4} \partial_\mu \Phi \partial_\nu \Phi.$$

The former conformal factor $(\Phi/M_*)^2$ is not part of the V15.3.1 physical metric. The candidate organizational functional Ω_{cand} is retained only as nondynamical mathematical data; it is not a field and is not globally identified with Φ . A useful equivalent representation of the compact scalar is $U = \exp(i\Phi/f_a)$ with $|U| = 1$; no radial field or second scalar degree of freedom is introduced.

Only the verified compact-shift corrections to two phenomenological expressions and the global-status clarification of the optional $\Phi F \tilde{F}$ operator are introduced. In particular, this patch does not remove the Derrick obstruction, derive particle masses, derive emergent gravity, or derive the phenomenological Hubble ansatz from the covariant action.

V15.3.2 Stage 5 Observational Release Note

V15.3.2 reproduces the mathematical and physical content of the frozen V15.3.1 baseline and adds only mathematical closure audits, observational context, provenance, explicit claim classifications, falsification criteria, and release documentation. No field, interaction, free parameter, fitted parameter, cosmological ansatz, gravity law, particle mechanism, or physical assumption is added. No observational result is used to tune an existing parameter.

ATHENA V15.3.2 is not experimentally validated. It is not a Standard Model derivation, not an emergent-gravity derivation, not a complete cosmological model, not a complete particle-mass theory, and not a complete perturbation theory. Several mappings from the frozen action to observables remain open. The Stage 5 additions therefore classify evidence; they do not repair or complete the theory.

Part I

Volume 0: Mathematical Framework

Chapter 1

Topological Foundations

1.1 Introduction to Volume 0

ATHENA is a unified topo-geometric research framework containing mathematically defined structures, conditional derivations, phenomenological hypotheses, proposed empirical tests, and conditional predictions. This volume presents its proposed mathematical foundation. Physical interpretation is introduced only where it is required to state the model, and every result that depends on an additional assumption or calibration is marked explicitly.

The reader is assumed to be familiar with standard differential geometry, topology, and mathematical physics at the graduate level.

Notation 1.1 (Derivation-status convention). Every mathematical calculation in Volume 0 is assigned one of the following statuses. *Proved* means that the conclusion follows from the displayed definitions and standard mathematical results. *Derived* means that the algebra follows once an explicitly identified ATHENA postulate or ansatz is accepted; it does not prove that input. *Conditional* means that one or more additional assumptions, normalizations, or calibrations are required. *Conjectural* means that a proposed statement has not been proved. *Open Problem* identifies a missing construction or theorem. A postulate, axiom, or phenomenological ansatz is never promoted to a proved result merely because its algebraic consequences are calculated.

1.2 The Toroidal Manifold T^2

Definition 1.1 (Toroidal Manifold). The toroidal manifold used in the spectral calculations is the flat product $T^2 = S_R^1 \times S_r^1$ of two circles with cycle radii R and r . These are not the major and minor radii of a doughnut-shaped surface embedded in $\mathbb{R}^3$. The aspect ratio is defined as:

$$\beta \equiv \frac{r}{R}, \quad 0 < \beta < \infty.$$

Points on T^2 are parameterized by angles satisfying $0 \leq \theta < 2\pi$ and $0 \leq \phi < 2\pi$.

Remark 1.1 (Intrinsic versus Embedded Torus). An embedded ring torus has the non-flat metric $ds^2 = r^2 d\theta^2 + (R + r \cos \theta)^2 d\phi^2$. That geometry has a different Laplacian and spectrum. Unless explicitly stated otherwise, ATHENA uses the intrinsic flat product metric below; embedded-torus arguments must therefore be treated as separate model assumptions.

Remark 1.2. The flat torus is adopted because it is the simplest compact product geometry with two independent periodic coordinates and non-trivial one-cycles. Its use as a foundation for particle spin is an ATHENA hypothesis examined in Chapter 3, not a standard consequence of torus topology.

Notation 1.2. Throughout this volume, we use the following conventions:

- The metric on T^2 is $ds^2 = R^2 d\theta^2 + r^2 d\phi^2$.
- The area element is $dA = Rr d\theta d\phi$.
- Functions on T^2 are 2π -periodic in both arguments.

Proposition 1.1 (Metric, connection, curvature, and area of the product torus). Let $x = R\theta$ and $y = r\phi$ be arc-length coordinates on the two circle factors. Pulling back $dx^2 + dy^2$ gives

$$ds^2 = R^2 d\theta^2 + r^2 d\phi^2, \quad (g_{ab}) = \begin{pmatrix} R^2 & 0 \\ 0 & r^2 \end{pmatrix}, \quad (g^{ab}) = \begin{pmatrix} R^{-2} & 0 \\ 0 & r^{-2} \end{pmatrix}.$$

Consequently

$$\det g = R^2 r^2, \quad \sqrt{g} = Rr, \quad \text{Area}(T^2) = \int_0^{2\pi} \int_0^{2\pi} Rr d\theta d\phi = 4\pi^2 Rr.$$

Because R and r are constant, $\partial_c g_{ab} = 0$. Hence

$$\Gamma^a_{bc} = \frac{1}{2} g^{ad} (\partial_b g_{dc} + \partial_c g_{db} - \partial_d g_{bc}) = 0,$$

and substitution into the coordinate formula for the Riemann tensor gives $R^a_{bcd} = 0$. Thus this intrinsic product torus is flat.

Dimension and normalization check. $[R] = [r] = L$, $[g_{ab}] = L^2$ in angular coordinates, $[dA] = L^2$, and the integrated area has dimension L^2 .

Status: Proved. The result follows from the declared product metric. The corresponding quotient construction and coordinate audit are recorded in Appendix 14.4.

Definition 1.2 (Laplace–Beltrami Operator). For a scalar field $\Phi(\theta, \phi)$ on T^2 , the scalar Laplace–Beltrami operator is defined by

$$\Delta_{T^2} \Phi = \frac{1}{\sqrt{g}} \partial_a \left(\sqrt{g} g^{ab} \partial_b \Phi \right).$$

Proposition 1.2 (Flat-torus Laplacian and scalar spectrum). For the declared product metric,

$$\begin{aligned} \Delta_{T^2} \Phi &= \frac{1}{Rr} \left[\partial_\theta \left(\frac{r}{R} \partial_\theta \Phi \right) + \partial_\phi \left(\frac{R}{r} \partial_\phi \Phi \right) \right] \\ &= \frac{1}{R^2} \partial_\theta^2 \Phi + \frac{1}{r^2} \partial_\phi^2 \Phi. \end{aligned}$$

Single-valuedness under both 2π shifts requires Fourier labels $m, n \in \mathbb{Z}$. With the normalized modes

$$u_{mn}(\theta, \phi) = \frac{1}{2\pi\sqrt{Rr}} e^{i(m\theta + n\phi)},$$

one has

$$\int_{T^2} \overline{u_{mn}} u_{pq} dA = \delta_{mp} \delta_{nq}.$$

Indeed,

$$\partial_\theta^2 u_{mn} = -m^2 u_{mn}, \quad \partial_\phi^2 u_{mn} = -n^2 u_{mn},$$

so

$$-\Delta_{T^2} u_{mn} = \left(\frac{m^2}{R^2} + \frac{n^2}{r^2} \right) u_{mn}.$$

Therefore

$$\lambda_{mn} = \frac{m^2}{R^2} + \frac{n^2}{r^2} = \frac{1}{R^2} \left(m^2 + \frac{n^2}{\beta^2} \right), \quad m, n \in \mathbb{Z}.$$

The zero mode is $(m, n) = (0, 0)$, and every eigenvalue is non-negative because it is a sum of squares. The Fourier basis is complete in $L^2(T^2, dA)$ by the standard Fourier theorem on the quotient $\mathbb{R}^2/(2\pi\mathbb{Z})^2$.

Dimension and normalization check. $[u_{mn}] = L^{-1}$ under unit L^2 normalization and $[\lambda_{mn}] = L^{-2}$, as required for a second-order differential operator.

Status: Proved. No particle interpretation is used. Further details are in Appendix 14.4.

1.3 Scalar Field Definition

Definition 1.3 (Scalar Field Φ). The scalar field has canonical mass dimension one, $[\Phi] = 1$, and the compact identification $\Phi \sim \Phi + 2\pi f_a$. On a local branch, or in the topologically trivial sector, it admits a real-valued representative with Fourier expansion

$$\Phi(\theta, \phi) = \sum_{m,n \in \mathbb{Z}} c_{mn} e^{i(m\theta + n\phi)},$$

where $c_{-m,-n} = \bar{c}_{mn}$ ensures that the chosen representative is real. Non-trivial winding sectors require local representatives related by shifts of $2\pi f_a$ and are classified below.

Remark 1.3 (Equivalent unit-phase representation). The compact scalar may equivalently be represented by

$$U = \exp(i\Phi/f_a), \quad |U| = 1.$$

This notation introduces no radial field and no additional propagating degree of freedom. The construction $\Phi = \rho e^{i\theta}$ is not part of ATHENA V15.3.

Remark 1.4. The scalar has canonical mass dimension one, $[\Phi] = 1$. Consequently $M_*^{-4} \partial_\mu \Phi \partial_\nu \Phi$ is dimensionless, as required for the disformal metric.

Definition 1.4 (Two-Dimensional Configuration Energy). On a chosen effective-field-theory branch near a vacuum, a dimensionally normalized quadratic energy is

$$E_2[\Phi] = \frac{1}{2M_*} \int_{T^2} (|\nabla \Phi|^2 + m^2 \Phi^2) dA,$$

where ∇ is the gradient on T^2 , m is a mass parameter, and the explicit scale M_* converts the dimension-two internal functional to an energy in natural units. A dimensional reduction may replace $1/M_*$ by a coefficient calculated from the higher-dimensional measure.

1.4 Winding Number

Definition 1.5 (Compact Scalar and Winding Pair). For topologically non-trivial sectors, the scalar is treated as a compact field $\Phi : T^2 \rightarrow \mathbb{R}/(2\pi f_a \mathbb{Z}) \simeq S^1$. Let C_θ and C_ϕ denote the two fundamental cycles of T^2 . The winding pair is

$$(w_\theta, w_\phi) = \frac{1}{2\pi f_a} \left(\oint_{C_\theta} d\Phi, \oint_{C_\phi} d\Phi \right) \in \mathbb{Z}^2.$$

For a cycle $C = pC_\theta + qC_\phi$, its winding is $w_C = pw_\theta + qw_\phi$. When a single physical cycle is selected, the abbreviation $w \equiv w_C$ is used.

Remark 1.5. The pair (w_θ, w_ϕ) represents the homotopy class of the map $T^2 \rightarrow S^1$ and is naturally identified with an element of $H^1(T^2; \mathbb{Z}) \simeq \mathbb{Z}^2$.

Lemma 1.1 (Integrality of the Winding Pair). For any continuous compact scalar $\Phi : T^2 \rightarrow S^1$, both w_θ and w_ϕ are integers.

Proof. Restrict Φ to either fundamental cycle and choose a real-valued lift $\tilde{\Phi}(t)$ along that cycle. Although the compact field returns to the same point of S^1 , its lift may satisfy $\tilde{\Phi}(1) - \tilde{\Phi}(0) = 2\pi f_a n$ for some $n \in \mathbb{Z}$. Therefore $\frac{1}{2\pi f_a} \oint d\Phi = n$. Applying this argument independently to C_θ and C_ϕ gives $(w_\theta, w_\phi) \in \mathbb{Z}^2$. $\square$

Proposition 1.3 (Lift, winding decomposition, and harmonic representative). Fix a winding pair $(w_\theta, w_\phi) \in \mathbb{Z}^2$. A lift of the compact field to the universal cover may be written

$$\tilde{\Phi}(\theta, \phi) = f_a(w_\theta \theta + w_\phi \phi) + \varphi(\theta, \phi),$$

where φ is 2π -periodic in both variables. The first term gives

$$\frac{1}{2\pi f_a} \int_0^{2\pi} \partial_\theta \tilde{\Phi} d\theta = w_\theta, \quad \frac{1}{2\pi f_a} \int_0^{2\pi} \partial_\phi \tilde{\Phi} d\phi = w_\phi.$$

For the gradient part of the quadratic functional, periodic integration makes the cross term between the linear lift and φ vanish. Hence

$$\begin{aligned} E_{\text{grad}}[\tilde{\Phi}] &= \frac{1}{2M_*} \int_{T^2} |\nabla \tilde{\Phi}|^2 dA \\ &= \frac{2\pi^2 f_a^2}{M_*} \left(\beta w_\theta^2 + \frac{w_\phi^2}{\beta} \right) + \frac{1}{2M_*} \int_{T^2} |\nabla \varphi|^2 dA. \end{aligned}$$

The minimum within that sector is therefore attained by constant φ and equals

$$E_{\text{grad}}^{\min}(w_\theta, w_\phi) = \frac{2\pi^2 f_a^2}{M_*} \left(\beta w_\theta^2 + \frac{w_\phi^2}{\beta} \right).$$

Dimension and normalization check. In natural units $[f_a] = [M_*] = M$, so f_a^2/M_* has dimension M and the bracket is dimensionless. A non-periodic term $m^2\Phi^2$ is not globally defined for a compact scalar in a non-zero winding sector; it must be replaced by a periodic potential or restricted to a local trivial branch.

Status: Proved for the lift decomposition and gradient minimum under the displayed quadratic gradient functional. The choice of that functional as physical energy is **Conditional**. See Appendix 14.5.

The variation of this sector minimum with respect to the aspect ratio, including the mixed- and single-winding cases and its separation from the postulated β_0 , is proved in Proposition K.2.

Example 1.1 (Trivial Configuration). For a constant field $\Phi = \Phi_0$, we have $d\Phi = 0$ and therefore $(w_\theta, w_\phi) = (0, 0)$. This is the topologically trivial configuration.

Example 1.2 (Single Winding). Consider the compact field represented locally by $\tilde{\Phi}(\theta) = \Phi_0 + f_a \theta$. It returns to the same point of the target circle after one poloidal circuit and has winding pair $(1, 0)$.

1.5 Stability Classes

Postulate 1.1 (Selected-Cycle Stability Rule). For the physical cycle selected by the model, stable winding configurations are postulated to have:

$$w = 1, 2, 4, 8, \dots \quad (\text{powers of two}).$$

All other selected-cycle windings $w = 3, 5, 6, 7, \dots$ are assumed to decay or reorganize into stable configurations. This is a dynamical model postulate, not a consequence of the homotopy classification alone.

Remark 1.6 (What topology and the quadratic energy do establish). Continuous nonsingular evolution cannot move between distinct winding pairs, so every pair labels a topologically disconnected sector. This is topological protection, not a proof of energetic or dynamical stability. Moreover, the derived quadratic minimum is proportional to $\beta w_\theta^2 + \beta^{-1} w_\phi^2$ and contains no arithmetic mechanism that selects powers of two. TEMP, developed in Chapter 5, also does not currently supply such a mechanism.

Status of the power-of-two rule: Conditional postulate.

Open Problem 1.1 (Power-of-Two Stability). Specify a globally defined energy or transition-rate functional, its admissible configuration space, and topology-changing dynamics. Then compute the first and second variations and prove that precisely the selected-cycle sectors $w = 2^k$ are stable while the remaining sectors possess allowed decay channels. No such proof follows from $H^1(T^2; \mathbb{Z})$ or from the displayed quadratic energy.

1.6 Topological Invariants

Definition 1.6 (Topological Charge). The topological charge of a configuration is the winding pair $\mathbf{w} = (w_\theta, w_\phi)$. A scalar charge w refers to its projection onto a specified physical cycle.

Lemma 1.2 (Topological Charge Conservation). For any continuous evolution through compact scalar configurations, the winding pair is invariant:

$$\frac{d\mathbf{w}}{dt} = 0.$$

Proof. Let $\tilde{\Phi}(t)$ be a smooth local lift along either closed cycle C . Then

$$\frac{d}{dt} \oint_C d\Phi = \oint_C d(\partial_t \tilde{\Phi}) = 0$$

because the integral of an exact one-form around a closed cycle vanishes. Thus both integer periods are constant during every smooth evolution for which the compact field and the cycles remain well defined. A singularity, boundary, or topology-changing event lies outside these assumptions. $\square$

Status: Proved for continuous nonsingular evolution on a fixed torus.

Definition 1.7 (Linking Number). Let $T^2 \hookrightarrow \mathbb{R}^3$ be a specified embedding and let C_1 and C_2 be disjoint oriented closed curves in the embedded torus. Their Gauss linking number is:

$$L(C_1, C_2) = \frac{1}{4\pi} \oint_{C_1} \oint_{C_2} \frac{d\mathbf{x} \times d\mathbf{y} \cdot (\mathbf{x} - \mathbf{y})}{|\mathbf{x} - \mathbf{y}|^3}.$$

Lemma 1.3 (Linking Number Conservation). The linking number of two disjoint embedded soliton curves is conserved under ambient isotopies that keep the curves disjoint.

Proof. The Gauss linking number is invariant under ambient isotopy. It can change only if the curves intersect, a singularity forms, or the embedding conditions are otherwise violated. $\square$

Chapter 2

Topological Constants

2.1 Introduction

This chapter introduces the parameter relations used by the ATHENA framework. Some relations follow algebraically after the model assumptions are specified; they should not be interpreted as consequences of topology alone.

2.2 Casimir Energy on T^2

Definition 2.1 (Formal zero-point energy and spectral zeta). Assume a real massless scalar with periodic boundary conditions on the flat product torus and frequencies $\omega_{mn} = \sqrt{\lambda_{mn}}$ in units $\hbar = c = 1$. Omitting the zero mode, its formal zero-point energy is

$$E_{\text{vac}}^{\text{formal}} = \frac{1}{2} \sum_{(m,n) \neq (0,0)} \sqrt{\lambda_{mn}} = \frac{1}{2R} \sum_{(m,n) \neq (0,0)} \left(m^2 + \frac{n^2}{\beta^2} \right)^{1/2}.$$

The associated Epstein spectral zeta is

$$\mathcal{Z}_\beta(s) = \sum'_{(m,n) \in \mathbb{Z}^2} \left(m^2 + \frac{n^2}{\beta^2} \right)^{-s/2}, \quad \text{Re } s > 2,$$

where the prime deletes $(0,0)$. Zeta regularization defines

$$E_{\text{vac}}^\zeta = \frac{1}{2R} \text{AC}_{s=-1} \mathcal{Z}_\beta(s),$$

provided a renormalization prescription has been fixed.

Proposition 2.1 (Analytic continuation at the zero-point argument). Let $Q_\beta = \text{diag}(1, \beta^{-2})$. The two-dimensional Epstein functional equation gives

$$\pi^{-u} \Gamma(u) Z_{Q_\beta}(u) = (\det Q_\beta)^{-1/2} \pi^{-(1-u)} \Gamma(1-u) Z_{Q_\beta^{-1}}(1-u),$$

where $Z_Q(u) = \sum' [(m,n)Q(m,n)^T]^{-u}$ and $\mathcal{Z}_\beta(s) = Z_{Q_\beta}(s/2)$. Setting $u = -1/2$ and using

$$(\det Q_\beta)^{-1/2} = \beta, \quad \Gamma(-1/2) = -2\sqrt{\pi}, \quad \Gamma(3/2) = \frac{\sqrt{\pi}}{2},$$

yields

$$\boxed{\mathcal{Z}_\beta(-1) = -\frac{\beta}{4\pi^2} \sum'_{(m,n) \in \mathbb{Z}^2} (m^2 + \beta^2 n^2)^{-3/2}}.$$

Therefore the zeta-regularized spectral sum is

$$E_{\text{vac}}^{\zeta} = -\frac{\beta}{8\pi^2 R} \sum'_{(m,n) \in \mathbb{Z}^2} (m^2 + \beta^2 n^2)^{-3/2}.$$

The sum in the boxed expression converges absolutely. Restoring units multiplies the right-hand side by $\hbar c$, so it has the dimension of energy.

Status: Proved as the analytic continuation of the explicitly defined Epstein zeta. The physical subtraction prescription remains **Conditional**. The Mellin-transform derivation is given in Appendix 14.6.

Remark 2.1 (V15.3.3 Poisson-decomposition cross-reference). The explicit separation of zero and nonzero modes, including both equivalent K_1 expansions and the square-torus numerical check, is given in Proposition K.1. Exact self-dual derivatives, fixed-area normalization, and endpoint asymptotics are recorded in Proposition K.9.

Hypothesis 2.1 (Reduced Model Casimir Ansatz). *ATHENA separately assumes the reduced expression*

$$\mathcal{Z}_{\beta}^{\text{model}}(-1) = -\frac{\pi}{12} \left(\beta + \frac{1}{\beta} \right), \quad E_{\text{vac}}^{\text{model}} = -\frac{\pi}{24R} \left(\beta + \frac{1}{\beta} \right).$$

This is not the analytic continuation of $\mathcal{Z}_{\beta}$ defined above. For example, at $\beta = 1$ the exact Epstein continuation is approximately -0.228824 , whereas the model expression gives $-\pi/6 \approx -0.523599$. The reduced formula must therefore be treated as an independent spectral normalization or truncation ansatz, not as an Epstein-zeta theorem.

Status: Conditional ansatz.

Postulate 2.1 (Equilibrium Condition for β_0). The ATHENA equilibrium ansatz relates the dimensionless pressure parameter to the chosen Casimir normalization by

$$\beta_0(2 - \beta_0) = \frac{\pi}{12}.$$

Moving all terms to one side gives

$$\beta_0^2 - 2\beta_0 + \frac{\pi}{12} = 0.$$

The quadratic formula therefore gives the two roots

$$\beta_{\pm} = 1 \pm \sqrt{1 - \frac{\pi}{12}}.$$

Both are positive and satisfy $\beta_+ + \beta_- = 2$; the additional branch choice $0 < \beta_0 < 1$ selects

$$\beta_0 = 1 - \sqrt{1 - \frac{\pi}{12}} = 0.1408139828 \dots$$

Every term is dimensionless and direct substitution returns $\pi/12$.

Status: Derived algebraically from the equilibrium postulate; the equilibrium equation and branch selection are **Conditional model inputs**, not theorems of torus topology.

Proposition 2.2 (The displayed Casimir ansatz does not derive β_0). At fixed R , differentiation of the reduced model energy gives

$$\frac{dE_{\text{vac}}^{\text{model}}}{d\beta} = -\frac{\pi}{24R} \left(1 - \frac{1}{\beta^2} \right).$$

Its only positive stationary point is $\beta = 1$, and

$$\left. \frac{d^2 E_{\text{vac}}^{\text{model}}}{d\beta^2} \right|_{\beta=1} = -\frac{\pi}{12R} < 0.$$

Thus neither the equation $\beta_0(2 - \beta_0) = \pi/12$ nor the value $\beta_0 \simeq 0.140814$ follows by minimizing the displayed reduced Casimir energy. Varying at fixed area or with additional stresses is a different problem and requires a complete dimensionally normalized effective potential.

Status: Proved as a consistency check; derivation of the equilibrium condition is an **Open Problem**.

The precise inversion law of the exact Epstein continuation and its distinction from the reduced symmetric ansatz are recorded in Proposition K.3.

2.3 Conditional Parameter Relations

Proposition 2.3 (Conditional Parameter Relations). Given the equilibrium postulate for β_0 , introduce a phenomenological reference length L_* and a dimensionless scale-matching coefficient κ_f . A further dimensionless acceleration calibration κ_a is required for the historical MOND-scale target. The framework then defines:

$$\begin{aligned} Z_0 &= \frac{5}{3}\beta_0 \approx 0.2347, \\ \gamma_0 &= \frac{\beta_0^2}{2} \approx 0.009914, \\ R_c &= L_* \frac{10}{3\beta_0}, \\ f_a &= \kappa_f \frac{\hbar c}{R_c}, \\ a_0 &= \kappa_a \frac{c^2}{R_c} \frac{\beta_0}{2\pi\alpha_{\text{eff}}^{\text{gal}}}. \end{aligned}$$

The numerical values quoted in this monograph use $L_* = 1 \text{ kpc}$, $\kappa_f = 0.10$, $\alpha_{\text{eff}}^{\text{gal}} = 0.36$, and $\kappa_a \simeq 1.44 \times 10^{-5}$, giving $R_c \approx 23.67 \text{ kpc}$ and $f_a \approx 2.7 \times 10^{-29} \text{ eV}$ and matching the historical $a_0 \approx 1.1 \times 10^{-10} \text{ m/s}^2$ target.

Derivation 2.1 (Numerical substitutions and unit audit). Starting from the selected root,

$$\begin{aligned} Z_0 &= \frac{5}{3}(0.1408139828\dots) = 0.2346899713\dots, \\ \gamma_0 &= \frac{1}{2}(0.1408139828\dots)^2 = 0.009914288872\dots, \\ \frac{R_c}{L_*} &= \frac{10}{3\beta_0} = 23.67189158\dots \end{aligned}$$

Thus $L_* = 1 \text{ kpc}$ gives

$$R_c = 23.67189158\dots \text{ kpc} = 7.30438252 \times 10^{20} \text{ m}.$$

Using $\hbar c = 1.973269804 \times 10^{-7} \text{ eV m}$ and $\kappa_f = 0.10$,

$$f_a = 0.10 \frac{1.973269804 \times 10^{-7} \text{ eV m}}{7.30438252 \times 10^{20} \text{ m}} = 2.70149 \times 10^{-29} \text{ eV}.$$

For the galactic screened-fit input $\alpha_{\text{eff}}^{\text{gal}} = 0.36$, the uncalibrated acceleration factor is

$$\frac{c^2}{R_c} \frac{\beta_0}{2\pi\alpha_{\text{eff}}^{\text{gal}}} = 7.65987 \times 10^{-6} \text{ m s}^{-2}.$$

Multiplication by $\kappa_a = 1.44 \times 10^{-5}$ gives

$$a_0 = 1.10302 \times 10^{-10} \text{ m s}^{-2}.$$

Conversely, the exact coefficient required to target $1.1 \times 10^{-10} \text{ m s}^{-2}$ with the displayed inputs is $\kappa_a = 1.43606 \times 10^{-5}$.

Dimension check. $Z_0, \gamma_0, \beta_0, \kappa_f, \kappa_a$, and $\alpha_{\text{eff}}^{\text{gal}}$ are dimensionless; $[R_c] = L$; $[\hbar c/R_c] = E$; and $[c^2/R_c] = L/T^2$.

Status: Derived for the substitutions. The definitions of Z_0 and γ_0 are **Conditional normalizations**; R_c , f_a , and a_0 remain **Conditional calibrations**. No displayed equation derives L_* , κ_f , or κ_a .

Proposition 2.4 (Conditional logarithmic sensitivities). For sensitivity bookkeeping, denote the right-hand side of the existing equilibrium postulate by

$$\mathcal{C} = \beta_0(2 - \beta_0), \quad \mathcal{C}|_{\text{ATHENA}} = \frac{\pi}{12}.$$

On the selected branch $0 < \beta_0 < 1$, implicit differentiation gives

$$d \ln \beta_0 = s_\beta d \ln \mathcal{C}, \quad s_\beta = \frac{2 - \beta_0}{2(1 - \beta_0)} = 1.08194615598 \dots$$

Holding the exact constants c and $\hbar$ fixed, the declared parameter relations then imply

$$\begin{aligned} d \ln Z_0 &= d \ln \beta_0, \\ d \ln \gamma_0 &= 2 d \ln \beta_0, \\ d \ln R_c &= d \ln L_* - d \ln \beta_0, \\ d \ln f_a &= d \ln \kappa_f - d \ln L_* + d \ln \beta_0, \\ d \ln a_0 &= d \ln \kappa_a - d \ln L_* + 2 d \ln \beta_0 - d \ln \alpha_{\text{eff}}^{\text{gal}}. \end{aligned}$$

These are exact first-order identities for the displayed conditional definitions. They quantify propagation of assumed inputs; they do not derive the inputs or their uncertainties.

Status: Derived from the existing equilibrium relation and parameter definitions. The full sensitivity matrix and covariance formula are given in Appendix 14.3.

Remark 2.2 (Acceleration-Scale Audit). Without κ_a , the displayed values of R_c , β_0 , and $\alpha_{\text{eff}}^{\text{gal}}$ give $a_0 \simeq 7.66 \times 10^{-6} \text{ m/s}^2$. Thus the observed acceleration scale is not a consequence of those three quantities alone.

Conditional definition of Z_0 . ATHENA assigns five organizational stages and normalizes by three spatial dimensions, giving $Z_0 = 5\beta_0/3$. The counting rule is a model input; only the substitution $Z_0 \approx 0.2347$ is an algebraic consequence.

Conditional definition of γ_0 . The relation $\gamma_0 = \beta_0^2/2$ is adopted as a model normalization. No two-point function or Gauss–Bonnet calculation in the present source derives it. Substitution gives $\gamma_0 \approx 0.009914$.

Calibration of R_c . Dimensional consistency requires a reference length. The relation is $R_c = L_* 10/(3\beta_0)$; the value 23.67 kpc follows only after choosing $L_* = 1 \text{ kpc}$.

Compact-field scale. The periodicity scale is parameterized by $f_a = \kappa_f \hbar c/R_c$. It is not derived from topology without the additional coefficient κ_f and should not be identified automatically with a conventional QCD-axion decay constant.

Table 2.1: ATHENA V15.3 parameters and their declared origin.

Constant	Value	Origin
β_0	0.1408	Casimir/equilibrium model postulate
Z_0	0.2347	$5\beta_0/3$
γ_0	0.009914	$\beta_0^2/2$
R_c	23.67 kpc	Calibrated with $L_* = 1$ kpc
f_a	2.7×10^{-29} eV	$\kappa_f \hbar c / R_c$, $\kappa_f = 0.10$
a_0	1.1×10^{-10} m/s ²	Calibrated with $\kappa_a \simeq 1.44 \times 10^{-5}$

2.4 Dimensionless Relations

Remark 2.3. The numerical scale hierarchy is not fixed by β_0 alone. At minimum, L_* , κ_f , and the galactic screened-fit parameter $\alpha_{\text{eff}}^{\text{gal}}$ are phenomenological inputs. Predictions must report their sensitivity to these parameters.

2.5 Scaling Laws

Phenomenological Ansatz 2.1 (Discrete Cascade Scaling). ATHENA explores the cascade law

$$N_{k+1} = 2N_k, \quad E_{k+1} = \beta_0 E_k.$$

Iteration gives, by induction,

$$N_k = 2^k N_0, \quad E_k = \beta_0^k E_0.$$

Eliminating $k = \log(N_k/N_0)/\log 2$ gives the exact conditional power law

$$\frac{E_k}{E_0} = \left(\frac{N_k}{N_0} \right)^{\log \beta_0 / \log 2}, \quad \frac{\log \beta_0}{\log 2} = -2.828137 \dots$$

This exponent is a cascade exponent; it is not a Hausdorff, box-counting, or spectral dimension unless a self-similar metric space and a length-rescaling map are additionally specified.

If a hierarchy spans 31 decimal orders through binary doubling of N , the required number of doublings is

$$N_{\text{doublings}} = \frac{\log(10^{31})}{\log(2)} = \frac{31}{\log_{10}(2)} \approx 103.$$

By contrast, the declared energy recurrence gives $E_k = E_0 \beta_0^k$ and crosses 31 decimal orders after about $31/|\log_{10} \beta_0| = 36.4126 \dots$ steps. The two counts describe different scaling assumptions and must not be identified without an explicit map between N , length, and energy.

Status: Derived from the two recurrence assumptions. Scale invariance and a fractal interpretation are **Conditional**; the origin of the recurrences is an **Open Problem**. Full induction and dimensional details are in Appendix 14.7.

Chapter 3

Topological Tensor Ascent (TTA)

3.1 Introduction

The Topological Tensor Ascent is a proposed mechanism of the ATHENA framework. It seeks a map from winding sectors to particle representations; the existence and uniqueness of that map remain open problems.

3.2 Tensor Rank Ascent

Postulate 3.1 (Tensor-Rank Label). ATHENA assigns the organizational label

$$\text{Rank}(w) = w - 1.$$

This rule is restricted here to a selected positive winding $w \in \mathbb{Z}_{\geq 1}$; it does not yet define labels for $w = 0$ or negative winding. It is a model bookkeeping rule, not the tensor rank of a field implied by differential geometry. In the proposed correspondence:

- $w = 1$: Rank-0 (scalar)
- $w = 2$: Rank-1 candidate sector
- $w = 3$: Rank-2 (matrix)
- $w = 4$: Rank-3 (3-tensor)
- $w = 5$: Rank-4 (4-tensor)

Derivation 3.1 (Maximal formal construction supplied by the rank rule). Let

$$\mathcal{C}_w = \{\Phi : T^2 \rightarrow S^1 \mid w_C(\Phi) = w\}$$

be the selected-cycle configuration sector, and choose a vector space V . The assignment

$$\mathcal{T}_w := V^{\otimes(w-1)}$$

produces tensors of order $w - 1$ by definition. To turn this bookkeeping assignment into TTA one would still need maps

$$F_w : \mathcal{C}_w \longrightarrow \mathcal{T}_w$$

and rules for gauge transformations, coordinate changes, composition, and Lorentz/internal-symmetry actions. Neither the topology of $\mathcal{C}_w$ nor the integer w selects V , F_w , or a representation. In particular, many inequivalent choices of V and F_w have the same formal order.

Dimension and normalization check. The integer w and the tensor order are dimensionless. A norm or action on $\mathcal{T}_w$ requires additional dimensionful normalization data.

Status: Derived only as a tautological tensor-product construction after choosing V . The physical TTA map is an **Open Problem**. Detailed logical dependencies are listed in Appendix 14.8.

Remark 3.1. Winding number and tensor rank are mathematically independent data. Establishing this correspondence requires a functorial construction that maps each winding sector to a specified representation space.

Open Problem 3.1 (Topological Tensor Ascent). Construct an explicit map from the configuration space in winding sector $\mathbf{w}$ to Lorentz and internal-symmetry representations, prove that it is independent of gauge and coordinate choices, and determine whether the proposed rank rule follows.

3.3 Holonomy and Spinor Bundle

Proposition 3.1 (Levi-Civita holonomy of the declared metric). Along a curve $x^a(s)$, a parallel vector satisfies

$$\frac{dV^a}{ds} + \Gamma^a_{bc} \frac{dx^b}{ds} V^c = 0.$$

Since $\Gamma^a_{bc} = 0$ in the global product frame, $dV^a/ds = 0$ and a vector returns unchanged after either fundamental loop. Thus the Levi-Civita holonomy representation of the flat tangent bundle is trivial.

Status: Proved for the flat product metric. This does not imply a unique spin structure.

Proposition 3.2 (Spin structures and the intrinsic Dirac spectrum). Spin structures on T^2 are classified by $H^1(T^2; \mathbb{Z}_2) \simeq \mathbb{Z}_2^2$. Write $\varepsilon = (\varepsilon_\theta, \varepsilon_\phi)$ with each entry in $\{0, 1\}$. A spinor then obeys

$$\psi(\theta + 2\pi, \phi) = (-1)^{\varepsilon_\theta} \psi(\theta, \phi), \quad \psi(\theta, \phi + 2\pi) = (-1)^{\varepsilon_\phi} \psi(\theta, \phi).$$

Using the orthonormal coframe $e^1 = R d\theta$, $e^2 = r d\phi$ and Pauli matrices σ^1, σ^2 , the flat Euclidean Dirac operator is

$$\mathcal{D} = -i \left(\frac{\sigma^1}{R} \partial_\theta + \frac{\sigma^2}{r} \partial_\phi \right).$$

The allowed spinor modes are

$$\psi_{mn}(\theta, \phi) = \frac{1}{2\pi\sqrt{Rr}} e^{i[(m+\varepsilon_\theta/2)\theta + (n+\varepsilon_\phi/2)\phi]} \chi, \quad m, n \in \mathbb{Z},$$

where $\chi \in \mathbb{C}^2$. Acting on a mode gives the Hermitian matrix

$$\mathcal{D}_{mn} = \frac{m + \varepsilon_\theta/2}{R} \sigma^1 + \frac{n + \varepsilon_\phi/2}{r} \sigma^2.$$

Because $\{\sigma^a, \sigma^b\} = 2\delta^{ab}$,

$$\mathcal{D}_{mn}^2 = \left[\frac{(m + \varepsilon_\theta/2)^2}{R^2} + \frac{(n + \varepsilon_\phi/2)^2}{r^2} \right] \mathbf{1}_2,$$

and hence

$$\lambda_{mn}^{D,\pm} = \pm \left[\frac{(m + \varepsilon_\theta/2)^2}{R^2} + \frac{(n + \varepsilon_\phi/2)^2}{r^2} \right]^{1/2}.$$

Only the periodic-periodic structure has constant zero modes. The spin holonomy around $pC_\theta + qC_\phi$ is $(-1)^{p\varepsilon_\theta + q\varepsilon_\phi}$.

Dimension and normalization check. $[\mathcal{D}] = L^{-1}$, its eigenvalues have dimension L^{-1} , and the prefactor gives unit L^2 norm for a unit spinor χ .

Status: Proved for intrinsic two-dimensional spinors after a spin structure is chosen. This construction does not derive a four-dimensional Lorentz spinor, particle statistics, or a winding-spin identification. See Appendix 14.8.

Hypothesis 3.1 (Proposed Winding–Spin Correspondence). *ATHENA proposes that a selected $w = 2$ sector is associated with an antiperiodic spin structure, for which*

$$\psi(\theta + 2\pi) = -\psi(\theta).$$

The boundary condition is compatible with spinors but does not by itself derive four-dimensional spin-1/2 or identify $w = 2$ uniquely. The scalar winding pair lies in $H^1(T^2; \mathbb{Z})$, whereas a spin structure lies in $H^1(T^2; \mathbb{Z}_2)$. Reduction modulo two is a possible mathematical map, but no stated ATHENA axiom selects that map or singles out one cycle.

Status: *Conditional hypothesis.*

Open Problem 3.2 (Dirac Construction). Specify the spin structure, construct the Dirac operator and its spectrum, couple the resulting spinors to the Lorentzian four-dimensional EFT, and derive rather than assume the winding–spin map.

3.4 Pauli Exclusion and Anti-Commutation

Open Problem 3.3 (Spin–Statistics and Anti-Commutation). The desired fermionic sector must reproduce

$$\{a_i, a_j^\dagger\} = \delta_{ij}, \quad \{a_i, a_j\} = 0.$$

Equality of winding or linking numbers does not imply canonical anti-commutation. A valid derivation requires a local Lorentzian quantum field, microcausality, a positive Hilbert space, and an explicit quantization map from soliton configurations to fermionic operators.

Chapter 4

Renormalization and Gauge Structure

4.1 Liouville Theory

Definition 4.1 (Reference Liouville Action). The two-dimensional Liouville action used as a comparison model is

$$S_L = \frac{1}{4\pi} \int d^2x \sqrt{\hat{g}} \left[\hat{g}^{ab} \partial_a \phi \partial_b \phi + Q \hat{R} \phi + 4\pi \mu e^{2b\phi} \right],$$

with the tentative identifications

$$\phi = \frac{\Phi}{f_a}, \quad b = \sqrt{\beta_0}, \quad \mu = \frac{\Lambda^4}{f_a^2}.$$

Here ϕ , b , and Q are dimensionless, while $[\mu] = L^{-2}$ when x^a has dimension L . Since $[\hat{R}] = L^{-2}$, every term in S_L is dimensionless. On the declared flat torus, $\hat{R} = 0$ pointwise and $\int_{T^2} \sqrt{\hat{g}} \hat{R} d^2x = 4\pi \chi(T^2) = 0$.

Proposition 4.1 (What zero-mode integration establishes). Decompose a real representative as

$$\Phi(x) = \Phi_0 + \chi(x), \quad \Phi_0 = \frac{1}{A} \int_{T^2} \Phi dA, \quad \int_{T^2} \chi dA = 0.$$

For any specified microscopic Euclidean action $S[\Phi]$, integrating out the non-zero modes defines, rather than evaluates, the zero-mode action:

$$e^{-S_{\text{eff}}(\Phi_0)} = \int_{\chi=0} \mathcal{D}\chi e^{-S[\Phi_0+\chi]}.$$

If S is the free shift-symmetric Gaussian action, S depends only on $\partial\chi$; the Gaussian determinant is independent of Φ_0 and no $e^{2b\Phi_0/f_a}$ potential is generated. If a potential is already present, the resulting S_{eff} depends on that potential, the measure, regulator, and counterterms. Thus the Fourier decomposition alone neither produces the Liouville exponential nor fixes Q , b , or μ .

There is also a global compatibility condition. The compact identification $\phi \sim \phi + 2\pi$ sends $e^{2b\phi} \mapsto e^{4\pi b} e^{2b\phi}$, so a real Liouville exponential is not a single-valued potential for the compact scalar unless the compact shift is explicitly broken, a local branch is chosen, or the field is decompactified.

Status: Proved as an obstruction statement. A Liouville reduction is **Conjectural** and requires the missing microscopic data listed below. The mode-by-mode calculation is in Appendix 14.9.

Conjecture 4.1 (Liouville Reduction). ATHENA conjectures that a specified configuration-space reduction and functional integration over non-zero modes can produce the reference Liouville

action for the zero mode. A flat-torus mode expansion alone does not generate the exponential potential or fix the background charge Q . A proof requires the complete microscopic action, measure, regulator, counterterms, compactness prescription, and matching calculation.

Status: Conjectural.

4.2 RG Flow

Phenomenological Ansatz 4.1 (Truncated Beta Function for α). The following beta function is adopted as a two-term model ansatz:

$$\beta(\alpha) = \mu \frac{d\alpha}{d\mu} = \frac{\alpha^2}{4\pi}(2 - \gamma_{\text{str}}) - \frac{\alpha^3}{8\pi^2}\beta_0(2 - \beta_0),$$

where γ_{str} is dimensionless. The coefficients have not yet been derived from the declared four-dimensional EFT. Since α , β_0 , and γ_{str} are dimensionless and μ has mass dimension one, both sides are dimensionless.

Status: Conditional phenomenological ansatz.

Proposition 4.2 (Algebraic Fixed Point of the Truncated Ansatz). The zeros of the displayed beta function are $\alpha = 0$ and

$$\alpha_* = \frac{2\pi(2 - \gamma_{\text{str}})}{\beta_0(2 - \beta_0)}.$$

Proof. Setting $\beta(\alpha) = 0$:

$$\frac{\alpha^2}{4\pi}(2 - \gamma_{\text{str}}) - \frac{\alpha^3}{8\pi^2}\beta_0(2 - \beta_0) = 0.$$

Assuming $\alpha \neq 0$:

$$\frac{1}{4\pi}(2 - \gamma_{\text{str}}) - \frac{\alpha}{8\pi^2}\beta_0(2 - \beta_0) = 0.$$

Thus:

$$\alpha = \frac{2\pi(2 - \gamma_{\text{str}})}{\beta_0(2 - \beta_0)}.$$

Since $\beta_0(2 - \beta_0) = \pi/12$:

$$\alpha_* = \frac{2\pi(2 - \gamma_{\text{str}})}{\pi/12} = 24(2 - \gamma_{\text{str}}).$$

For $\gamma_{\text{str}} = 0$, the truncated ansatz therefore gives $\alpha_* = 48$, not 89.4. □

Derivation 4.1 (Factorization, local stability, and integrated RG curve). Define

$$A = \frac{2 - \gamma_{\text{str}}}{4\pi}, \quad B = \frac{\beta_0(2 - \beta_0)}{8\pi^2}.$$

Then, for $A \neq 0$ and $B \neq 0$,

$$\frac{d\alpha}{d\ln \mu} = A\alpha^2 - B\alpha^3 = A\alpha^2 \left(1 - \frac{\alpha}{\alpha_*}\right), \quad \alpha_* = \frac{A}{B}.$$

Differentiation gives

$$\beta'(\alpha) = 2A\alpha - 3B\alpha^2, \quad \beta'(\alpha_*) = -\frac{A^2}{B} = -A\alpha_*.$$

Thus, when $A > 0$ and $B > 0$, the non-zero fixed point is attractive as $\ln \mu \rightarrow +\infty$ within the truncated flow. The origin is a double zero; for small positive α , $\beta(\alpha) > 0$.

For $x = \alpha/\alpha_*$, separation of variables gives

$$d \ln \mu = \frac{1}{A\alpha_*} \frac{dx}{x^2(1-x)}.$$

Using

$$\frac{1}{x^2(1-x)} = \frac{1}{x} + \frac{1}{x^2} + \frac{1}{1-x},$$

integration between (μ_0, x_0) and (μ, x) yields

$$\ln \frac{\mu}{\mu_0} = \frac{1}{A\alpha_*} \left[\ln \left| \frac{x}{1-x} \right| - \frac{1}{x} - \ln \left| \frac{x_0}{1-x_0} \right| + \frac{1}{x_0} \right].$$

This is an implicit solution of the two-term ansatz away from its fixed points. With the equilibrium postulate and $\gamma_{\text{str}} = 0$, $A = 1/(2\pi)$, $B = 1/(96\pi)$, and $\alpha_* = 48$.

Status: Derived from the truncated beta-function ansatz. Fixed-point existence in the full EFT, scheme independence, and the physical identification of α remain **Open Problems**. The complete sign-dependent phase portrait, basin structure, exceptional cases, and Lambert- W inversion are given in Appendix 14.9.

For $\gamma_{\text{str}} = 0$, the explicit linear stiffness and its forward- $\ln \mu$ sign convention are evaluated in Proposition K.4.

Remark 4.1 (Status of the value 89.4). The value 89.4 does not follow from the displayed beta function with $\gamma_{\text{str}} = 0$. Within that two-term expression it would require $\gamma_{\text{str}} \simeq -1.725$. Unless independently derived higher-order terms, a scheme conversion, or a different normalization are supplied, 89.4 is retained only as a historical calibrated model input, denoted α_{cal} , and not as the fixed point α_* . The original calibration dataset, normalization convention, and uncertainty are not independently documented in the present archive.

4.3 Topological Phase Transition

Phenomenological Ansatz 4.2 (Coarse-Grained Winding Relaxation). For an ensemble-averaged variable $\bar{w}$, ATHENA considers

$$\frac{d\bar{w}}{d \ln \mu} = -\lambda(\bar{w} - 2), \quad \lambda > 0.$$

An individual winding number is an integer homotopy invariant and cannot obey this differential equation during a continuous nonsingular evolution. The ansatz can describe only a coarse-grained population with topology-changing events supplied by additional dynamics.

Conjecture 4.2 (Topological Phase Transition: $w = 1 \rightarrow w = 2$). ATHENA conjectures a topology-changing transition from $w = 1$ to $w = 2$ with:

- Order parameter: w
- Critical point: Driven by RG flow
- Conserved quantity: Topological charge (discrete jumps only)
- Type: First-order (discontinuous in w) with second-order features (energy and entropy continuous)

Open Problem 4.1 (Transition Dynamics). Provide a free-energy functional, nucleation or tunnelling mechanism, latent heat or critical scaling, and a treatment of the singular event that changes the homotopy class. Without these ingredients the thermodynamic order of the transition is undetermined.

4.4 Trefoil Knot

Definition 4.2 (Trefoil Knot). A $(2, 3)$ torus knot is an embedding $S^1 \hookrightarrow T^2$ whose image is then viewed in $\mathbb{R}^3$ through a chosen embedding of the torus. A generic planar projection has crossing number three. It is the simplest non-trivial knot up to crossing number.

4.5 Candidate SU(3) Correspondence

Hypothesis 4.1 (Trefoil–Color Correspondence). *ATHENA proposes that a finite-dimensional low-energy truncation of trefoil deformations may carry an $\mathfrak{su}(3)$ action. The number eight is a target dimension motivated by the color adjoint representation; it is not established by counting crossing-point displacements.*

Open Problem 4.2 (Gauge–Algebra Construction). Specify a knot energy functional, compute its Hessian and normalizable zero modes, define the Lie bracket on the reduced mode space, and prove closure, Jacobi identity, compact real form, gauge invariance, and coupling to matter representations. Confinement additionally requires a dynamical area law or an equivalent non-perturbative result.

4.6 Candidate U(1) Correspondence

Hypothesis 4.2 (Proposed U(1) Sector). *A continuous compact-field shift can generate a global Noether current,*

$$J^\mu = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \Phi)} \delta \Phi.$$

Identifying this global symmetry with electromagnetism requires introducing a local gauge redundancy, a gauge connection A_μ , a covariant derivative, charged representations, and a normalized charge operator. Winding alone does not complete these steps.

4.7 Candidate SU(2) Correspondence

Hypothesis 4.3 (Proposed Chiral SU(2) Sector). *ATHENA proposes that a metastable $w = 3$ resonance supports three reduced modes that may be organized as an $\mathfrak{su}(2)$ triplet, with a chiral energy splitting*

$$\Delta E = E_{\text{left}} - E_{\text{right}} \neq 0.$$

Because $w = 3$ is not in the postulated stable set, this sector must be treated as a finite-lifetime resonance rather than a stable vacuum configuration.

Open Problem 4.3 (Electroweak Completion). Construct generators with the $\mathfrak{su}(2)$ commutators, chiral fermion doublets, hypercharge assignments, symmetry breaking, gauge-boson masses, and complete gauge- and gravitational-anomaly cancellation.

Chapter 5

Mass Generation and Cosmological Models

5.1 Proposed Toroidal Internal Energy

Definition 5.1 (Dimensionally Normalized Internal Functional). Let $\hat{\Phi}$ and $\tilde{x}$ be dimensionless, and define

$$I_w^{(T^2)} = \int_{T^2} \left[\frac{1}{2} |\tilde{\nabla} \hat{\Phi}|^2 + \hat{V}(\hat{\Phi}) \right]_w d^2 \tilde{x}.$$

The dimensionless integral cannot by itself determine an energy. ATHENA's mass ansatz must therefore contain an explicit dimension-one scale E_* :

$$m_w c^2 = E_* I_w^{(T^2)}.$$

Remark 5.1. The boundary conditions, potential, domain metric, renormalization prescription, and value of E_* must be specified before a mass is predicted. Multiplying a dimensionless two-dimensional integral by f_a^2 would have mass dimension two in natural units and is not an energy.

Remark 5.2 (Scope of the compact-torus functional). $I_w^{(T^2)}$ is a two-dimensional internal configuration functional on a fixed compact torus. It is not the three-dimensional radial functional I_w^{rad} tested by Derrick scaling in Volume II, and the two quantities must not be interchanged. The compact functional may be retained as a proposed internal-mode contribution, but no dimensional-reduction or spacetime-embedding calculation currently converts it into a finite four-dimensional static particle mass.

Status: Conditional dimensional ansatz; physical particle mass Open Problem.

Conjecture 5.1 (Proposed Winding Dependence). For a specified family of classical solutions, ATHENA conjectures the leading scaling

$$m_w c^2 \sim E_* (A w^2 + B + \dots),$$

where A and B depend on the potential and geometry. The scaling is not a mass spectrum and does not identify particle species without further dynamics. It cannot be instantiated by the current three-dimensional radial functional, whose nontrivial finite static solutions are excluded by the Derrick test.

5.2 Particle Mass Integral

Phenomenological Ansatz 5.1 (Historical Particle-Mass Targets). The assignments $w = 1, 2, 4, 5$ to the proton, electron, muon, and tau are retained as ATHENA hypotheses. The

quoted values 938.272 MeV, 0.511 MeV, 105.66 MeV, and 1.776 GeV are fit targets, not outputs reproduced by the present source package. In particular, the earlier electron expression mixed an unspecified quantity R_e with dimensionless ratios and was not dimensionally closed. The three-dimensional radial functional later displayed in Volume II cannot produce a nontrivial finite static soliton under its current assumptions, so these numbers are not derivations from that functional.

Open Problem 5.1 (Non-Circular Mass Prediction). Declare the potential and boundary conditions, solve for each $I_w^{(T^2)}$, derive its relation to a four-dimensional energy, fix E_* from an observable not included in the mass table, propagate uncertainties, and predict at least one unused mass. The obstructed radial functional I_w^{rad} cannot supply this step. If measured particle masses enter the choice of E_* , the potential, or boundary data, agreement with those same masses is calibration rather than validation.

5.3 Topological Involution

Proposition 5.1 (Conditional Effective-Volume Identity). For the rank-one spatial ansatz $\tilde{\gamma}_{ij} = a^2 \delta_{ij} - \alpha M_*^{-4} \partial_i \Phi \partial_j \Phi$, define

$$S = \delta^{ij} \partial_i \Phi \partial_j \Phi, \quad Q_D = \frac{\alpha S}{a^2 M_*^4}, \quad a_{\text{vol}} = a(1 - Q_D)^{1/6}.$$

Assume $1 - Q_D > 0$ so that the spatial metric is positive definite. Then

$$H_{\text{vol}} = H - \frac{1}{6} \frac{\dot{Q}_D}{1 - Q_D}.$$

Proof. Set $A = a^2 I_3$, $v_i = \partial_i \Phi$, and $c = \alpha M_*^{-4}$. The matrix determinant lemma in the form

$$\det(A - cvv^T) = \det A (1 - cv^T A^{-1}v)$$

gives

$$\begin{aligned} \det \tilde{\gamma} &= a^6 (1 - \alpha M_*^{-4} v^T (a^{-2} I_3) v) \\ &= a^6 \left(1 - \frac{\alpha S}{a^2 M_*^4} \right) = a^6 (1 - Q_D). \end{aligned}$$

Therefore

$$\sqrt{\det \tilde{\gamma}} = a^3 \sqrt{1 - Q_D}.$$

Defining the local volume scale by $a_{\text{vol}}^3 = \sqrt{\det \tilde{\gamma}}$ gives

$$a_{\text{vol}} = a(1 - Q_D)^{1/6}.$$

Taking a logarithmic derivative,

$$H_{\text{vol}} = \frac{\dot{a}_{\text{vol}}}{a_{\text{vol}}} = \frac{\dot{a}}{a} - \frac{1}{6} \frac{\dot{Q}_D}{1 - Q_D} = H - \frac{1}{6} \frac{\dot{Q}_D}{1 - Q_D}.$$

If α and M_* are constant, then

$$\dot{Q}_D = \frac{\alpha}{a^2 M_*^4} (2\delta^{ij} \partial_i \Phi \partial_j \dot{\Phi} - 2HS).$$

Running α or M_* adds their logarithmic derivatives to $\dot{Q}_D/Q_D$. □

Dimension and normalization check. For spatial coordinates of length dimension L , $[\partial_i \Phi] = M^2$ in natural units, so S/M_*^4 and Q_D are dimensionless. Both H and $\dot{Q}_D$ have dimension T^{-1} .

Status: Conditional. The determinant and logarithmic-derivative identity are proved for the rank-one ansatz; their cosmological interpretation requires field equations and a coarse-graining prescription. See Appendix 14.10.

Corollary 5.1 (Inverse rank-one spatial metric). Let $\partial^i \Phi = \delta^{ij} \partial_j \Phi$. If $a \neq 0$ and $1 - Q_D \neq 0$, the inverse of the declared spatial metric is

$$(\tilde{\gamma}^{-1})^{ij} = \frac{1}{a^2} \delta^{ij} + \frac{\alpha}{a^4 M_*^4 (1 - Q_D)} \partial^i \Phi \partial^j \Phi.$$

Direct multiplication by $\tilde{\gamma}_{jk}$ gives δ^i_k . The metric is positive definite precisely when $a^2 > 0$ and $1 - Q_D > 0$; at $1 - Q_D = 0$ it is degenerate.

Status: Proved algebraically under the existing rank-one ansatz. The projector derivation, determinant, eigenvalues, and complete invertibility classification are given in Appendix 14.10.

Remark 5.3. Replacing the local Q_D by an averaged scalar before taking the determinant is an additional isotropic coarse-graining ansatz. The identity does not by itself replace a cosmological solution or establish “involution” as the physical cause of expansion.

Open Problem 5.2 (Mathematical involution). No map $\mathcal{I}$ satisfying $\mathcal{I} \circ \mathcal{I} = \text{id}$ is defined by the determinant calculation. If “Topological Involution” is meant literally as an involutive transformation, its domain, action, fixed points, and relation to $\tilde{\gamma}$ must be supplied. At present the proved content is the conditional pure-disformal volume identity only.

5.4 TEMP Postulate

Definition 5.2 (Topological Entropy).

$$H_{\text{topo}}(w) = - \sum_{n=1}^{\infty} P(w_n) \log_2 P(w_n).$$

Postulate 5.1 (Topological Entropy Minimization Principle (TEMP)). The disformal vacuum evolves toward configurations that minimize H_{topo} subject to winding number conservation.

Remark 5.4 (Variational Status). The exponential distribution obtained with normalization and fixed mean is the maximum-entropy distribution, because Shannon entropy is concave. It does not select powers of two. Selecting those sectors requires an independently specified energy or transition-rate functional whose stable minima occur there.

5.5 Modified Hubble Equation

Phenomenological Ansatz 5.2 (Modified Hubble Parameter). The observational phenomenological form is

$$H_{\text{obs}}(z) = H_{\Lambda}(z)[1 + \alpha_H \Omega_{\text{coll}}(z)], \quad \Omega_{\text{coll}}(z) = \frac{1}{1 + (z/z_c)^2}, \quad z_c = 0.35.$$

Here α_H denotes the Hubble-ansatz normalization and is distinct from the galactic screened-fit input $\alpha_{\text{eff}}^{\text{gal}}$. It is not derived from the EFT background equations in this monograph. The transition redshift z_c is a phenomenological parameter whose provenance is not independently documented in the present archive.

5.6 SPARC Screening Function Ansatz

Phenomenological Ansatz 5.3 (SPARC Screening Function). The screening function in the rotation velocity formula is:

$$\mathcal{F}(y) = \beta_0 [1 - e^{-y}(1 + y)], \quad y = r/R_c.$$

The scale R_c and any coupling entering the rotation curve must be fitted or fixed independently; no catalogue-level likelihood is included here.

5.7 Cosmic Dipole Misalignment Target

Phenomenological Ansatz 5.4 (Dipole Misalignment Target). The historical target for the residual misalignment is

$$\theta_{\text{res}} \approx 5.40^\circ, \quad p = 0.0022.$$

Its reported p value is not reproducible from the present package and is not a topological derivation.

5.8 Erdős–Straus Conjecture and Toroidal Resonance

Remark 5.5 (Mathematical Reflection). The Erdős–Straus conjecture asks whether for every integer $n > 1$ there exist positive integers x, y, z such that:

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}.$$

Within ATHENA, this finds a natural interpretation in terms of toroidal resonance modes. The three fractions can be viewed as the normalized energy contributions of three stable topological modes. The harmonic series $f_n = f_0/\beta_0^n$ ensures that for every energy level n there exists a resonant combination of modes.

Chapter 6

Generalized Kuramoto Dynamics in the Toroidal Vacuum

6.1 Introduction

The Kuramoto model is a paradigmatic framework for synchronization phenomena in coupled oscillator systems. The ATHENA extension postulates that the toroidal vacuum field $\Phi(x)$ may modify the phase dynamics of a coupled system.

6.2 Standard Kuramoto Model

The standard Kuramoto model is given by:

$$\frac{d\phi_i}{dt} = \omega_i + \sum_j K_{ij} \sin(\phi_j - \phi_i),$$

where ϕ_i is the phase of oscillator i , ω_i is its natural frequency, and K_{ij} is the coupling strength. The order parameter is:

$$re^{i\psi} = \frac{1}{N} \sum_{j=1}^N e^{i\phi_j}.$$

For identical coupling $K_{ij} = K/N$, the critical coupling threshold is:

$$K_c = \frac{2}{\pi g(\omega_0)}.$$

6.3 ATHENA Extension: Vacuum-Induced Phase Modulation

We introduce a periodic phenomenological term that couples the phase dynamics to the ATHENA vacuum field $\Phi(x)$:

$$\frac{d\phi_i}{dt} = \omega_i + \sum_j K_{ij} \sin(\phi_j - \phi_i) + \lambda_K \mathcal{P}\left(\frac{\Phi(\mathbf{x}, t)}{f_a}\right) R(\phi_i),$$

where λ_K has the units required to make the added term an angular frequency, $R(\phi)$ is a phase-dependent response function, and

$$\mathcal{P}(x + 2\pi) = \mathcal{P}(x).$$

No particular periodic function is selected by the present construction.

Remark 6.1 (Compact-shift status of the Kuramoto coupling). The periodic dependence is imposed solely so that the phenomenological ansatz is independent of the chosen real lift of the compact scalar. It is not a derivation of the coupling, of $\mathcal{P}$, or of additional dynamics.

Conjecture 6.1 (Modified Critical Coupling). After defining a dimensionless vacuum statistic Ξ_Φ and a coupling λ_K with units of angular frequency, a response-specific linearization may take the schematic form

$$K_c = \frac{2}{\pi g(\omega_0)} - \lambda_K \Xi_\Phi.$$

The sign and magnitude depend on $R(\phi)$, the oscillator distribution, and the field correlation function; they do not follow from the added term alone.

6.4 Proposed Observable Consequences

The extended ansatz motivates the following qualitative targets:

- Modified synchronization threshold depending on $\langle (\nabla \Phi)^2 \rangle$.
- Phase clustering induced by the vacuum field.
- Traveling waves produced by gradients in Φ .
- Hysteresis in the synchronization transition when λ is varied.

Chapter 7

Topological Information Theory

7.1 Introduction

The relationship between information, geometry, and gravity has been a central theme in theoretical physics since the work of Bekenstein and Hawking. This chapter asks whether an ATHENA encoding could be compatible with the covariant entropy bound; it does not derive that bound.

7.2 The Bousso Bound

Definition 7.1 (Reference Covariant Entropy Bound). For an appropriate light-sheet generated from a surface of area A , the covariant entropy conjecture states, in SI units,

$$S \leq \frac{k_B c^3 A}{4G\hbar}.$$

In units $k_B = G = \hbar = c = 1$, this becomes $S \leq A/4$. It is not a bound on an arbitrary spatial region without the light-sheet hypotheses.

7.3 ATHENA Interpretation

Hypothesis 7.1 (Topological Information Encoding). *If one independently postulates one bit per unit of winding and supplies a map from the torus sector to a gravitational light-sheet, then one may write the conditional estimate*

$$w_{\max} \leq \frac{A}{4}.$$

Here A must be measured in Planck-area units. Winding number is dimensionless, but it is not entropy without the declared encoding map.

Conjecture 7.1 (Black-Hole Information Channel). ATHENA conjectures that toroidal degrees of freedom may provide an information channel coupled to Hawking radiation. Unitary evolution, the Page curve, and an explicit recovery map remain open calculations; the information paradox is not resolved by the encoding hypothesis alone.

7.4 Holographic Principle and the 2D Toroidal Manifold

Hypothesis 7.2 (Candidate Toroidal Encoding). *ATHENA proposes a candidate relation*

$$I_{4D} \longleftrightarrow I_{T^2}.$$

The integral of Φ^2 is dimensionful and is not an information measure until normalized and connected to a Hilbert-space encoding. TTA does not yet supply an injective, surjective, or unitary map.

Proposition 7.1 (Conditional Entropy–Area Compatibility). If the covariant-bound hypotheses and the proposed encoding both hold, the normalized topological entropy must satisfy

$$H_{\text{topo}} \leq \frac{A}{4}.$$

This is a consistency requirement, not a derivation of either TEMP or the covariant entropy bound.

Chapter 8

Consistency and Uniqueness

8.1 Introduction

This chapter records the logical dependencies of ATHENA's organizational sector. The spectral assumption constrains the inputs to an organizational functional, but it does not select a unique formula. Rigidity, topological selection, and emergent time are therefore separated into an ansatz, conditional statements, and open problems.

8.2 The Spectral Determinism Axiom

The organizational axioms characterize Ω qualitatively but do not guarantee uniqueness. ATHENA introduces the following additional constraint.

Axiom 8.1 (Spectral Determinism). The organizational functional Ω is uniquely determined by the Laplacian spectrum $\{\lambda_i\}_{i=0}^N$ and the Betti numbers $\{\beta_k\}_{k=0}^{\dim K}$ of the relation complex K . That is, there exists a function

$$\Omega(K) = \Psi(\{\lambda_i\}, \{\beta_k\})$$

such that if K_1 and K_2 are isospectral and have identical Betti numbers, then $\Omega(K_1) = \Omega(K_2)$.

Remark 8.1. The axiom says that isospectral complexes with equal Betti numbers receive the same value. It does not specify the function Ψ , and it cannot distinguish isospectral non-isometric geometries.

Proposition 8.1 (Exact mathematical content of Spectral Determinism). Define the invariant map

$$\mathcal{I}(K) = (\text{Spec } \Delta_K, \beta_0(K), \dots, \beta_{\dim K}(K))$$

and the equivalence relation $K_1 \sim K_2$ when $\mathcal{I}(K_1) = \mathcal{I}(K_2)$. A functional Ω factors as $\Omega = \Psi \circ \mathcal{I}$ if and only if it is constant on every $\sim$ -equivalence class.

Proof. If $\Omega = \Psi \circ \mathcal{I}$ and $K_1 \sim K_2$, then $\mathcal{I}(K_1) = \mathcal{I}(K_2)$ and hence $\Omega(K_1) = \Omega(K_2)$. Conversely, if Ω is constant on equivalence classes, define $\Psi(y) = \Omega(K)$ for any K with $\mathcal{I}(K) = y$. Constancy on the fiber makes this definition independent of the representative. $\square$

The factorization is a restriction on an already chosen Ω ; it does not select one functional. For any two distinct functions F_1 and F_2 on the range of $\mathcal{I}$, both $F_1 \circ \mathcal{I}$ and $F_2 \circ \mathcal{I}$ obey the axiom but generally differ.

Status: Proved as a factorization statement. The assertion that one particular Ω is uniquely selected is an **Open Problem**. See Appendix 14.11.

8.3 Uniqueness of the Organizational Functional

Phenomenological Ansatz 8.1 (Candidate Organizational Functional). One function satisfying part of the intended qualitative behavior is

$$\Omega_{\text{cand}}(K) = C \frac{\beta_1(K)}{1 + \beta_1(K)^2} \exp \left(-\alpha_\Omega \sum_{i=1}^N \lambda_i^2 \right),$$

where $C > 0$ and $\alpha_\Omega > 0$. The subscript distinguishes the spectral penalty from the disformal coupling α . The cutoff N and eigenvalue normalization are part of the definition. Positivity and decay do not make this rational–Gaussian choice unique.

Dimension check. Since $[\lambda_i] = L^{-2}$, the exponent is dimensionless only if $[\alpha_\Omega] = L^4$, or if the eigenvalues are first replaced by dimensionless values $\hat{\lambda}_i = \ell_\Omega^2 \lambda_i$ using a declared reference length ℓ_Ω . The value of the candidate functional therefore depends on a scale convention in addition to the cutoff.

Status: Conditional phenomenological ansatz.

Open Problem 8.1 (Uniqueness of Organization). State a complete set of axioms and prove a representation theorem for Ω . Until then, $I(K) = -\log_2 \Omega(K)$ is candidate-dependent and must also be normalized before it is interpreted as information.

8.4 Rigidity of Emergence

Proposition 8.2 (What uniformization proves for genus one). Let Σ be a closed orientable surface of genus one and fix a conformal class and total area $A > 0$. Uniformization supplies a constant-curvature metric in that conformal class. Gauss–Bonnet gives

$$KA = \int_{\Sigma} K dA = 2\pi\chi(\Sigma) = 2\pi(2 - 2g) = 0,$$

so the constant curvature is $K = 0$. Thus the normalized representative is flat. However, the conformal class is parametrized by the torus modulus, and the area is an independent scale; neither is selected by Gauss–Bonnet or uniformization.

Status: Proved for existence of a flat representative after fixing conformal class and area. Selection of a unique modulus, rectangular metric, or area is **Open**. An explicit normalized representative, its uniqueness proof, and its Laplacian spectrum are derived in Appendix 14.11.

Corollary 8.1 (Normalized genus-one flat representative). Let $\tau = \tau_1 + i\tau_2$ with $\tau_2 > 0$, let $0 \leq x < 1$ and $0 \leq y < 1$, and fix total area $A > 0$. The metric

$$g_{\tau,A} = \frac{A}{\tau_2} [(dx + \tau_1 dy)^2 + \tau_2^2 dy^2]$$

is flat and has area A . Within the conformal class represented by τ , it is the unique flat metric of that area. Its scalar eigenvalues are

$$\lambda_{mn} = \frac{4\pi^2}{A\tau_2} |n - m\tau|^2, \quad (m, n) \in \mathbb{Z}^2.$$

The rectangular ATHENA metric is recovered with $\tau = i\beta$ and $A = 4\pi^2 Rr$.

Status: Proved for fixed (τ, A) . Neither the modulus nor the area is selected by this result.

Conjecture 8.1 (Rigidity of Emergence on the Torus). ATHENA conjectures that an emergence operator applied to a maximizing complex with $|K| \cong T^2$ selects

$$\mathcal{E}(K) = (T^2, g_{\text{flat}}),$$

but uniformization supplies only a flat representative in each conformal class; it does not select the modulus or area. The earlier relation $A = 4\pi^2/\Omega$ requires an independent area scale and a dimensionful normalization of Ω .

Remark 8.2. The flat metric remains non-unique over the torus moduli space. A combinatorial aspect ratio may be used to choose a modulus, but that is additional structure.

8.5 Organizational Uniformization

Proposition 8.3 (Topological factor over orientable surface genus). For a connected closed orientable surface Σ_g , $\beta_1(\Sigma_g) = 2g$. The Betti-number factor in Ω_{cand} is therefore

$$f(g) = \frac{2g}{1 + 4g^2}.$$

The sphere gives $f(0) = 0$. Treating $x = 2g$, differentiation gives

$$\frac{d}{dx} \frac{x}{1 + x^2} = \frac{1 - x^2}{(1 + x^2)^2} < 0 \quad \text{for } x \geq 2.$$

Hence among positive integer genera the topological factor alone is uniquely largest at $g = 1$. This calculation does not control the spectral exponential.

Moreover, under a metric rescaling $g_{ab} \mapsto c^2 g_{ab}$, $\lambda_i \mapsto c^{-2} \lambda_i$, so $\sum_{i=1}^N \lambda_i^2 \mapsto c^{-4} \sum_{i=1}^N \lambda_i^2$. Without fixed area or another scale constraint, the exponential tends to one as $c \rightarrow \infty$; a finite scale maximizer is therefore not selected by the displayed penalty.

The Betti-factor comparison and scale-rescaling audit are **Proved**. Maximization of the complete functional remains **Conjectural**.

Conjecture 8.2 (Organizational Uniformization). For closed orientable surfaces and the particular candidate functional above, ATHENA conjectures

$$\max_{g \geq 0} \Omega_{\text{cand}}(\Sigma_g) = \Omega_{\text{cand}}(T^2),$$

The Betti-number factor alone favors $g = 1$ among positive-genus closed orientable surfaces, but no result supplied here controls the regularized spectral factor over moduli space. A proof requires a fixed area, cutoff or spectral-zeta prescription, and a global extremization theorem.

8.6 Emergent Time

Definition 8.1 (Organizational Time). Let $\{K_\tau\}_{\tau \in \mathcal{I}}$ be a one-parameter family of relation complexes indexed by a totally ordered set $\mathcal{I}$. The parameter τ is called **organizational time** if:

1. $\Omega_{\text{cand}}(K_\tau)$ is strictly monotonically increasing in τ .
2. The map $\tau \mapsto K_\tau$ is continuous in the Gromov–Hausdorff topology on $|K_\tau|$.

Proposition 8.4 (Existence but Not Uniqueness of an Organizational Clock). If a family $\{K_\tau\}$ satisfies $\Omega_{\text{cand}}(K_{\tau_2}) > \Omega_{\text{cand}}(K_{\tau_1})$ for all $\tau_2 > \tau_1$, then $t = \Omega_{\text{cand}}(K_\tau) - \Omega_{\text{cand}}(K_{\tau_0})$ is an order coordinate on the image of the family. If in addition $\Omega_{\text{cand}}(K_\tau)$ is continuously differentiable and $d\Omega_{\text{cand}}/d\tau > 0$, then it has a differentiable local inverse and

$$\frac{d\Omega_{\text{cand}}}{dt} = 1.$$

Proof. Strict monotonicity makes t injective and order preserving. Under the stated differentiability condition, the chain rule gives

$$\frac{d\Omega_{\text{cand}}}{dt} = \frac{d\Omega_{\text{cand}}/d\tau}{dt/d\tau} = 1.$$

If f is any strictly increasing function, $f(t)$ has the same ordering, so the clock is not unique even up to affine maps. $\square$

Strict monotonicity alone does not imply that a derivative exists or is everywhere positive. Also, t is dimensionless when Ω_{cand} is dimensionless; a physical time requires an independent time scale.

Status: Proved as an order-parameter construction with the stated regularity assumptions. Emergence of physical time is an **Open Problem**. See Appendix 14.11.

Open Problem 8.2 (Physical Time). Derive a Lorentzian causal structure, clock observables, and the relation to proper time. A monotone organizational label is not yet physical time.

8.7 ATHENA Dependency Statement

Proposition 8.5 (Conditional Dependency Ledger). The ATHENA V15.3 relational sector has the following dependency structure:

1. Spectral Determinism is an axiom, not a uniqueness theorem.
2. Ω_{cand} is a non-unique phenomenological ansatz.
3. Geometric rigidity and torus selection are conjectures conditional on that ansatz and a specified spectral regularization.
4. Strict monotonicity supplies a non-unique organizational clock.
5. The EFT and phenomenology require additional postulates not entailed by the organizational sector.

This ledger prevents downstream physical assumptions from being used as proof of upstream mathematical claims. It is not an existence, uniqueness, or empirical-validity theorem.

Open Problem 8.3 (Higher-Dimensional Consistency). Construct a non-empty model satisfying a complete organizational axiom set, then study its extension to $d \geq 3$ and determine whether any topology is selected without inserting that preference into the functional.

Chapter 9

Emergent Field Theory: Volume I Prelude

9.1 From Geometry to Fields

The dependency ledger in Proposition 8.5 does not select a unique geometry. We now introduce a candidate EFT on the postulated flat torus as the bridge to Volume I.

9.2 The Organizational Functional Diagnostic

Definition 9.1 (Nondynamical organizational diagnostic). The candidate functional $\Omega_{\text{cand}}(K)$ remains a non-unique phenomenological functional on relation complexes. Its dimensionless diagnostic amplitude is

$$\mathcal{O}_\Omega(K) = \sqrt{\Omega_{\text{cand}}(K)}.$$

Neither Ω_{cand} nor $\mathcal{O}_\Omega$ is a spacetime field in V15.3. They have no kinetic term, stress tensor, equation of motion, or independent propagating degree of freedom.

Remark 9.1 (No global organizational-field identity). The V15.2 equality $\Phi = M_* \sqrt{\Omega_{\text{cand}}}$ is not part of V15.3. A globally single-valued real function of K_x would provide a global lift of Φ and force every winding integral to vanish. A future local, trivial-sector, vacuum-expectation, or phenomenological matching may be proposed only as an explicitly conditional additional input. No such matching is assumed here.

9.3 The Disformal Metric

Definition 9.2 (ATHENA Disformal Metric). The physical matter metric is related to the g -frame metric by the pure-disformal ansatz

$$\tilde{g}_{\mu\nu} = g_{\mu\nu} - \frac{\alpha}{M_*^4} \partial_\mu \Phi \partial_\nu \Phi,$$

where α is dimensionless and

$$X = -\frac{1}{2} g^{\mu\nu} \partial_\mu \Phi \partial_\nu \Phi, \quad \Delta_D = 1 + \frac{2\alpha X}{M_*^4}.$$

Because the ansatz depends only on $d\Phi$, it is invariant under $\Phi \mapsto \Phi + 2\pi f_a$ and is globally defined on the compact target.

Proposition 9.1 (Pure-disformal inverse, determinant, and signature). Let $u_\mu = \partial_\mu \Phi$ and assume that $g_{\mu\nu}$ is nondegenerate. Then

$$\tilde{g}^{\mu\nu} = g^{\mu\nu} + \frac{\alpha}{M_*^4 + 2\alpha X} \nabla^\mu \Phi \nabla^\nu \Phi,$$

$$\det \tilde{g} = \det g \Delta_D, \quad \sqrt{-\tilde{g}} = \sqrt{-g} \sqrt{\Delta_D} \quad (\Delta_D > 0).$$

The physical metric is invertible exactly when $\Delta_D \neq 0$. For a non-null gradient it has the same Lorentzian inertia as $g_{\mu\nu}$ exactly on the connected branch $\Delta_D > 0$.

Proof. Write $c = \alpha/M_*^4$. The mixed map is

$$A^\mu{}_\nu = g^{\mu\rho} \tilde{g}_{\rho\nu} = \delta^\mu{}_\nu - c u^\mu u_\nu.$$

Every vector orthogonal to u_μ is an eigenvector with eigenvalue one. If $u^2 = g^{\mu\nu} u_\mu u_\nu = -2X \neq 0$, the longitudinal vector u^μ has eigenvalue

$$1 - cu^2 = 1 + \frac{2\alpha X}{M_*^4} = \Delta_D.$$

Thus the determinant ratio is the product of the three transverse eigenvalues and the longitudinal one, giving $\det \tilde{g} / \det g = \Delta_D$.

For the inverse, take the rank-one ansatz

$$\tilde{g}^{\mu\nu} = g^{\mu\nu} + C u^\mu u^\nu.$$

Multiplication by $g_{\nu\rho} - cu_\nu u_\rho$ gives the identity precisely when $C(1 - cu^2) = c$. Hence $C = c/\Delta_D$, which yields the displayed inverse.

For a timelike gradient $X > 0$ and for a spacelike gradient $X < 0$, the only relative eigenvalue that can change sign is Δ_D . At $\Delta_D = 0$ the longitudinal direction is null and the metric is degenerate. For a null gradient $X = 0$, the rank-one part of $A - I$ is nilpotent: all eigenvalues are one, the determinant is unchanged, and the metric remains nondegenerate and Lorentzian although its null cone is generally shifted. $\square$

Corollary 9.1 (Matter causal cone). For a matter covector k_μ , the physical null condition is

$$\tilde{g}^{\mu\nu} k_\mu k_\nu = g^{\mu\nu} k_\mu k_\nu + \frac{\alpha}{M_*^4 + 2\alpha X} (\nabla^\mu \Phi k_\mu)^2 = 0.$$

The sign of the cone displacement depends on α , the type of scalar gradient, and the admissible branch. Invertibility alone proves neither subluminality nor hyperbolicity of the full coupled system.

Status: Proved for the declared algebraic metric ansatz. Coupled characteristic and stability questions are addressed conditionally below.

Remark 9.2 (V15.3.3 differential-geometric completion). The exact Levi-Civita connection difference, induced curvature identities, and the corresponding matter-geodesic equation are derived in Proposition K.7. These identities require no choice of matter Lagrangian, but they do not by themselves close the coupled matter principal symbol.

9.4 Minimal Covariant Effective Action

Postulate 9.1 (Minimal ATHENA EFT). A dimensionally consistent baseline action is

$$S_{\text{EFT}} = \int d^4x \sqrt{-g} \left[\frac{M_{\text{Pl}}^2}{2} \mathcal{R} - \frac{1}{2} g^{\mu\nu} \partial_\mu \Phi \partial_\nu \Phi - V(\Phi) \right] + S_m[\tilde{g}_{\mu\nu}, \Psi] + S_{\text{top}},$$

where S_{top} contains any explicitly specified topological terms. Higher-derivative operators must be organized as an EFT expansion and tested for ghost and gradient instabilities.

9.5 The Field Equation

Hypothesis 9.1 (ATHENA Field Equation). *The Euler–Lagrange equation of the scalar sector, before matter and topological source terms are added, is*

$$\square_g \Phi - V'(\Phi) = 0.$$

Matter coupled through $\tilde{g}_{\mu\nu}$ and terms in S_{top} generate additional source contributions that must be derived explicitly.

Definition 9.3 (Organizational Potential). Compatibility with the compact identification is maintained by the periodic potential

$$V(\Phi) = \Lambda^4 \left[1 - \cos \left(\frac{\Phi}{f_a} \right) \right],$$

whose minima occur at $\Phi_n = 2\pi n f_a$.

9.6 Static and Homogeneous Solutions

Theorem 9.1 (Vacuum Structure). The homogeneous minima of the periodic scalar potential are

$$\Phi_n = 2\pi n f_a, \quad n \in \mathbb{Z}.$$

These are local real representatives of one compact vacuum class, not physically distinct vacua of the quotient target.

Proof. $V'(\Phi) = (\Lambda^4/f_a) \sin(\Phi/f_a)$ vanishes at integer multiples of πf_a . The even multiples $2\pi n f_a$ have $V'' = \Lambda^4/f_a^2 > 0$ and are minima. $\square$

9.7 Perturbations and Dispersion

Proposition 9.2 (Linearized Spectrum). Let $\Phi = \Phi_n + \delta\Phi(x, t)$. The linearized field equation is

$$(\square + m_\Phi^2) \delta\Phi = 0, \quad m_\Phi^2 = \frac{\Lambda^4}{f_a^2}.$$

On the flat torus T^2 with periodic boundary conditions, the mode expansion yields the dispersion relation

$$\omega_{\mathbf{n}}^2 = \left(\frac{2\pi n_1}{L_1} \right)^2 + \left(\frac{2\pi n_2}{L_2} \right)^2 + m_\Phi^2,$$

where $\mathbf{n} = (n_1, n_2) \in \mathbb{Z}^2$ and L_1, L_2 are the torus periods.

Remark 9.3. The local mass gap is $m_\Phi^2 = \Lambda^4/f_a^2$. The integer-labelled flat-torus configuration-space spectrum is superimposed on this gap. This is not a Kaluza–Klein tower: the internal T^2 is not an additional spacetime dimension and no compactification mechanism is assumed.

9.8 Coupling to the Disformal Metric

Hypothesis 9.2 (Matter coupling and gravitational status). *Matter is postulated to couple universally through $S_m[\tilde{g}_{\mu\nu}, \Psi]$. The scalar kinetic term and Einstein–Hilbert term are written in the g frame. Consequently the canonical scalar stress tensor is obtained by varying the scalar action with respect to $g^{\mu\nu}$; it is not obtained by replacing $g_{\mu\nu}$ with $\tilde{g}_{\mu\nu}$ in the canonical formula.*

The V15.2 expression $G_{\text{eff}} = G_N/\varphi^2$ depended on the removed conformal factor and is not an active V15.3 equation. The baseline action has the constant g -frame normalization

$$G_N = \frac{1}{8\pi M_{\text{Pl}}^2}.$$

Any effective Newton constant measured by matter on a nontrivial scalar background requires a weak-field solution of the coupled equations and is currently open.

Remark 9.4. Frame transformations in a disformal theory require transforming the complete action, including matter. The existence of the physical metric does not derive Einstein dynamics, an effective Newton constant, or emergent gravity.

9.9 Summary of the Emergent Field Theory

This chapter established the field-theoretic content of the emergent torus geometry:

Concept	Symbol	Definition
Organizational data	$\mathcal{O}_\Omega$	Nondynamical $\sqrt{\Omega_{\text{cand}}(K)}$ diagnostic
Scalar field	$\Phi(x)$	One real compact scalar, $\Phi \sim \Phi + 2\pi f_a$
Disformal metric	$\tilde{g}_{\mu\nu}$	$g_{\mu\nu} - \alpha M_*^{-4} \partial\Phi \otimes \partial\Phi$
Field equation		$\square\Phi - V'(\Phi) = \text{sources}$
Vacua	Φ_n	$2\pi n f_a$
Dispersion	$\omega_{\mathbf{n}}$	$ \mathbf{k}_{\mathbf{n}} ^2 + m_\Phi^2$

Chapter 10

Topological Screening and $\alpha_* \rightarrow \alpha_{\text{eff}}$

10.1 Introduction

The truncated beta-function ansatz of Chapter 4 is

$$\beta(\alpha) = \mu \frac{d\alpha}{d\mu} = \frac{\alpha^2}{4\pi} (2 - \gamma_{\text{str}}) - \frac{\alpha^3}{8\pi^2} \beta_0 (2 - \beta_0).$$

For $\gamma_{\text{str}} = 0$ and the equilibrium postulate it has the non-zero algebraic root $\alpha_* = 48$. The historical value 89.4 is denoted α_{cal} and is not derived by this expression.

10.2 Topological Screening Mechanism

ATHENA considers the dimensionless screening ansatz

$$\alpha_{\text{eff}}(r) = \frac{\alpha_{\text{UV}}}{1 + \Xi_w(r)},$$

where α_{UV} must be declared as either α_* or the independent calibration α_{cal} , and

$$\Xi_w(r) = \frac{1}{A(r)} \int_{T^2(r)} \hat{\rho}_w dA$$

is dimensionless only if $\hat{\rho}_w$ has inverse-area units in the chosen coordinates.

Open Problem 10.1 (Dimensionally Complete Screening Model). The earlier expression $M_\Phi = \Phi_0 f_a R_c^2$ does not have the dimensions of mass under the conventions of this monograph and is therefore not used as a derivation. A valid screening calculation must define a soliton mass from the covariant energy functional, construct a dimensionally normalized winding density ρ_w , and then fit α_{eff} with a documented likelihood. The quoted galactic value $\alpha_{\text{eff}}^{\text{gal}} = 0.36 \pm 0.03$ remains a phenomenological fit target until that construction and its data products are supplied.

10.3 Comparison with Vainshtein Mechanism

In standard Horndeski theories, the Vainshtein mechanism screens the scalar field in high-density regions. ATHENA's topological screening is similar but relies on the winding number density rather than derivative self-interactions.

Chapter 11

Biological Resonance (mtDNA — 1420.405 MHz)

11.1 Introduction

The hydrogen 21-cm line is a hyperfine transition near 1420.405 MHz. ATHENA records a speculative biological-resonance analogy, but no coupling between Φ and a biological system at this frequency is derived.

11.2 Mitochondrial Membrane Potential

The inner mitochondrial membrane has a potential $\Delta\Psi \approx -180$ mV. The membrane thickness is $d \approx 5$ nm. The resulting electric field is:

$$E = \frac{\Delta\Psi}{d} = \frac{180 \times 10^{-3}}{5 \times 10^{-9}} = 3.6 \times 10^7 \text{ V/m.}$$

11.3 Magnetic-Field Audit

The relation $B = E/c$ applies to a propagating plane electromagnetic wave, not to a quasi-static transmembrane electric field. It therefore cannot be used to convert the membrane field into a local static magnetic field. Moreover, using the proton gyromagnetic ratio in frequency units, $\gamma_p/(2\pi) = 42.577$ MHz/T, a conventional proton NMR frequency of 1420.405 MHz would require

$$B_{\text{req}} = \frac{1420.405}{42.577} \text{ T} \simeq 33.36 \text{ T,}$$

not 0.033 T. The 21-cm line itself is a hydrogen hyperfine transition and should not be identified with membrane proton NMR without a microscopic model.

Open Problem 11.1 (Biological Resonance Test). Specify a Hamiltonian, coupling strength, linewidth, decoherence time, energy source, and measurable biological response. The membrane-potential estimate does not establish resonance or coupling to the ATHENA field.

Chapter 12

Topological Protection and Thermal Decoherence

12.1 Introduction

A key question for ATHENA is whether a proposed $w = 2$ sector could survive in thermal environments. No general protection theorem at 310 K is available from winding number alone.

12.2 Lindblad Decoherence

The density matrix evolution is:

$$\frac{d\rho}{dt} = -\frac{i}{\hbar}[H, \rho] + \sum_k \left(L_k \rho L_k^\dagger - \frac{1}{2} \{L_k^\dagger L_k, \rho\} \right).$$

The $w = 2$ projector is $P_{w=2} = |w = 2\rangle\langle w = 2|$. Topological protection requires $L_k P_{w=2} = 0$ for all k .

12.3 Phase-Flip Protection

The assumption that the environment couples only through a phase-flip operator is model-specific. A σ_z error generally changes relative phase and can decohere a superposition; it is not merely a global phase. Protection requires a code subspace on which every relevant error acts proportionally to the identity, or a verified energy-gap/error-correction mechanism.

Hypothesis 12.1 (Topological Protection). *ATHENA hypothesizes that a suitably engineered $w = 2$ sector may suppress some decoherence channels. The claimed lifetime at 310 K requires an explicit system–bath Hamiltonian, spectral density, material parameters, and an experimentally testable calculation.*

Chapter 13

Speculative Consciousness Analogy

13.1 Introduction

This chapter records a speculative analogy and is not part of the mathematically documented field-theory core of ATHENA. No accepted operational observable connecting the scalar field to consciousness is presently supplied.

13.2 Definition and Properties

Hypothesis 13.1 (Candidate Self-Reference Functional).

$$\hat{C} = \int d^3x \Phi(x) \cdot \hat{O}_{\text{self}}(x),$$

where $\hat{O}_{\text{self}}(x) = \Phi(x) \frac{\delta}{\delta \Phi(x)}$ is a formal functional operator. Its domain, inner product, self-adjoint extension, and physical measurement protocol remain unspecified.

Open Problem 13.1 (Operational Interpretation). Determine whether any normalized, experimentally accessible quantity derived from $\hat{C}$ correlates with independently defined cognitive observables. A nonzero expectation value alone is not a definition or test of consciousness.

Hypothesis 13.2 (Desired Mathematical Properties). *A viable operator construction would need to establish:*

1. $\hat{C}$ is Hermitian: $\hat{C} = \hat{C}^\dagger$.
2. $[\hat{C}, \hat{H}] \neq 0$ (non-conservation, conscious change).
3. $\langle \hat{C} \rangle$ is bounded by the Bousso Bound: $\langle \hat{C} \rangle \leq A/4$.

These properties are requirements for a future construction and are not proved by the formal expression above.

13.3 Implications for Artificial Intelligence

No claim about artificial consciousness follows from the present EFT. Such a claim would require an operational theory, quantitative thresholds, and blinded experimental tests independent of geometric analogy.

Chapter 14

Appendices

14.1 Numerical Convergence: Muon Mass

Table 14.1: Reported muon-mass convergence target; not reproduced in this package.

$N_\theta = N_\phi$	m_μ (MeV)
64	105.62
128	105.65
256	105.657
512	105.659
1024	105.660

The table has no accompanying solver, boundary conditions, input parameters, or reference output. It is therefore an archival target, not evidence of numerical convergence in the present V15.3 package.

14.2 Reproducibility Protocol

The present document package contains the LaTeX source but not the simulation scripts, input catalogues, covariance matrices, or environment lock file needed to reproduce the numerical tables. Consequently the quoted convergence and fit values are unverified within this package. A complete archival release must include exact data versions, checksums, executable scripts, dependency versions, random seeds, likelihood definitions, and machine-readable outputs.

14.3 Conditional Parameter-Sensitivity Matrix

14.3.1 Equilibrium-branch sensitivity

Write the existing equilibrium relation as

$$F(\beta_0, \mathcal{C}) = \beta_0(2 - \beta_0) - \mathcal{C} = 0, \quad \mathcal{C} = \frac{\pi}{12}.$$

Taking a differential gives

$$dF = 2(1 - \beta_0)d\beta_0 - d\mathcal{C} = 0,$$

and therefore

$$\frac{d\beta_0}{d\mathcal{C}} = \frac{1}{2(1 - \beta_0)}.$$

Multiplication by $\mathcal{C}/\beta_0$ gives the logarithmic sensitivity

$$\begin{aligned} s_\beta &\equiv \frac{d \ln \beta_0}{d \ln \mathcal{C}} = \frac{\mathcal{C}}{2\beta_0(1-\beta_0)} \\ &= \frac{\beta_0(2-\beta_0)}{2\beta_0(1-\beta_0)} = \frac{2-\beta_0}{2(1-\beta_0)}. \end{aligned}$$

At the selected ATHENA root this is $s_\beta = 1.08194615598\dots$. The symbol $\mathcal{C}$ is only a formal bookkeeping variable for perturbing the already declared normalization; no new physical parameter is introduced.

14.3.2 Jacobian of the conditional parameter map

Let

$$\mathbf{y} = (\beta_0, Z_0, \gamma_0, R_c, f_a, a_0)^T, \quad \mathbf{x} = (\mathcal{C}, L_*, \kappa_f, \kappa_a, \alpha_{\text{eff}}^{\text{gal}})^T.$$

For non-zero inputs, define the logarithmic Jacobian by $S_{ij} = \partial \ln y_i / \partial \ln x_j$. Direct differentiation of the declared relations gives

$$S = \begin{pmatrix} s_\beta & 0 & 0 & 0 & 0 \\ s_\beta & 0 & 0 & 0 & 0 \\ 2s_\beta & 0 & 0 & 0 & 0 \\ -s_\beta & 1 & 0 & 0 & 0 \\ s_\beta & -1 & 1 & 0 & 0 \\ 2s_\beta & -1 & 0 & 1 & -1 \end{pmatrix}.$$

Thus, for example, a fractional perturbation of the equilibrium normalization is amplified by $2s_\beta = 2.16389231197\dots$ in both γ_0 and a_0 when the other declared inputs are held fixed. The constants $5/3$, $1/2$, $10/3$, c , and $\hbar$ are treated as exact within the displayed definitions.

If a covariance matrix $C_{\ln x}$ for the logarithmic inputs is supplied, the linearized output covariance is determined algebraically by

$$C_{\ln y} = S C_{\ln x} S^T.$$

For statistically independent inputs this reduces to

$$\left(\frac{\sigma_{y_i}}{y_i} \right)^2 = \sum_j S_{ij}^2 \left(\frac{\sigma_{x_j}}{x_j} \right)^2.$$

No numerical uncertainty is assigned here because the source does not declare uncertainties or covariances for the calibrated inputs.

Final status: Derived from the existing conditional definitions. The normalizations and calibrations themselves retain their previous statuses.

14.4 Detailed Flat-Torus Geometry and Spectrum

14.4.1 Starting assumptions and quotient construction

Start with the Euclidean plane with arc-length coordinates (x, y) and impose the lattice identifications

$$(x, y) \sim (x + 2\pi R, y), \quad (x, y) \sim (x, y + 2\pi r).$$

The quotient is $S_R^1 \times S_r^1$. Introducing dimensionless angles $x = R\theta$ and $y = r\phi$ gives

$$dx = R d\theta, \quad dy = r d\phi, \quad ds^2 = dx^2 + dy^2 = R^2 d\theta^2 + r^2 d\phi^2.$$

The transition maps are translations, so the coordinate frame descends globally. Since all metric components are constant, all Christoffel symbols and all components of the Riemann tensor vanish. This proves intrinsic flatness; it does not describe the metric induced on a doughnut-shaped embedding in $\mathbb{R}^3$.

14.4.2 Volume form and scalar Laplacian

For

$$g_{ab} = \text{diag}(R^2, r^2), \quad g^{ab} = \text{diag}(R^{-2}, r^{-2}),$$

one obtains $\sqrt{\det g} = Rr$. Substitution into

$$\Delta f = \frac{1}{\sqrt{g}} \partial_a (\sqrt{g} g^{ab} \partial_b f)$$

gives, without omitted terms,

$$\begin{aligned} \Delta f &= \frac{1}{Rr} \partial_\theta \left(Rr \frac{1}{R^2} \partial_\theta f \right) + \frac{1}{Rr} \partial_\phi \left(Rr \frac{1}{r^2} \partial_\phi f \right) \\ &= \frac{1}{Rr} \left(\frac{r}{R} \partial_\theta^2 f + \frac{R}{r} \partial_\phi^2 f \right) \\ &= \frac{1}{R^2} \partial_\theta^2 f + \frac{1}{r^2} \partial_\phi^2 f. \end{aligned}$$

Periodic integration by parts shows

$$\int_{T^2} \bar{f} (-\Delta f) dA = \int_{T^2} g^{ab} \partial_a \bar{f} \partial_b f dA \geq 0,$$

so $-\Delta$ is non-negative on its periodic domain.

14.4.3 Mode normalization and spectrum

For

$$u_{mn} = C e^{i(m\theta + n\phi)},$$

periodicity requires $m, n \in \mathbb{Z}$. Normalization gives

$$1 = |C|^2 Rr \int_0^{2\pi} d\theta \int_0^{2\pi} d\phi = |C|^2 Rr (2\pi)^2,$$

so $C = (2\pi\sqrt{Rr})^{-1}$ up to a phase. Orthogonality follows from

$$\int_0^{2\pi} e^{i(m-p)\theta} d\theta = 2\pi \delta_{mp}$$

and the analogous ϕ integral. Finally,

$$\begin{aligned} -\Delta u_{mn} &= -\frac{1}{R^2} (-m^2) u_{mn} - \frac{1}{r^2} (-n^2) u_{mn} \\ &= \left(\frac{m^2}{R^2} + \frac{n^2}{r^2} \right) u_{mn}. \end{aligned}$$

Final result. $\lambda_{mn} = R^{-2}(m^2 + n^2/\beta^2)$ with a single scalar zero mode at $(0, 0)$. The result is **Proved**; $[\lambda_{mn}] = L^{-2}$.

14.5 Detailed Winding and Sector-Energy Derivations

14.5.1 Starting assumptions and lifts

Let the target be $\mathbb{R}/(2\pi f_a \mathbb{Z})$. A lift to the universal cover satisfies

$$\begin{aligned} \tilde{\Phi}(\theta + 2\pi, \phi) &= \tilde{\Phi}(\theta, \phi) + 2\pi f_a w_\theta, \\ \tilde{\Phi}(\theta, \phi + 2\pi) &= \tilde{\Phi}(\theta, \phi) + 2\pi f_a w_\phi. \end{aligned}$$

Subtracting $f_a(w_\theta\theta + w_\phi\phi)$ leaves a doubly periodic function φ . Direct integration gives the two integer periods. For a cycle $pC_\theta + qC_\phi$, linearity of integration gives $w_C = pw_\theta + qw_\phi$. This realizes $[T^2, S^1] \simeq H^1(T^2; \mathbb{Z}) \simeq \mathbb{Z}^2$.

14.5.2 Gradient-energy decomposition

For $\tilde{\Phi} = f_a(w_\theta\theta + w_\phi\phi) + \varphi$,

$$|\nabla\tilde{\Phi}|^2 = \frac{1}{R^2}(f_a w_\theta + \partial_\theta\varphi)^2 + \frac{1}{r^2}(f_a w_\phi + \partial_\phi\varphi)^2.$$

The cross terms vanish after integration because

$$\int_0^{2\pi} \partial_\theta\varphi d\theta = 0, \quad \int_0^{2\pi} \partial_\phi\varphi d\phi = 0.$$

The winding contribution is

$$\begin{aligned} \frac{1}{2M_*} \int_{T^2} f_a^2 \left(\frac{w_\theta^2}{R^2} + \frac{w_\phi^2}{r^2} \right) dA &= \frac{4\pi^2 Rr}{2M_*} f_a^2 \left(\frac{w_\theta^2}{R^2} + \frac{w_\phi^2}{r^2} \right) \\ &= \frac{2\pi^2 f_a^2}{M_*} (\beta w_\theta^2 + \beta^{-1} w_\phi^2). \end{aligned}$$

The remaining integral of $|\nabla\varphi|^2$ is non-negative, proving the sector minimum. It also proves that this quadratic functional does not select powers of two.

Final result. Integrality, homotopy conservation, and the quadratic sector minimum are **Proved**. Energetic stability and the power-of-two selection rule remain **Conditional/Open** because no suitable nonlinear energy and transition dynamics have been supplied.

14.6 Epstein-Zeta Continuation and Casimir Audit

14.6.1 Convergent domain and Mellin representation

Let Q be a positive-definite 2×2 matrix and define

$$Z_Q(u) = \sum'_{k \in \mathbb{Z}^2} (k^T Q k)^{-u}, \quad \text{Re } u > 1.$$

Using

$$x^{-u} = \frac{\pi^u}{\Gamma(u)} \int_0^\infty t^{u-1} e^{-\pi t x} dt,$$

one obtains

$$\pi^{-u} \Gamma(u) Z_Q(u) = \int_0^\infty t^{u-1} [\Theta_Q(t) - 1] dt,$$

where $\Theta_Q(t) = \sum_{k \in \mathbb{Z}^2} e^{-\pi t k^T Q k}$. Poisson summation gives

$$\Theta_Q(t) = (\det Q)^{-1/2} t^{-1} \Theta_{Q^{-1}}(1/t).$$

Splitting the Mellin integral at $t = 1$, applying this identity on $(0, 1)$, and changing variables $t \mapsto 1/t$ gives the meromorphic continuation and the functional equation

$$\pi^{-u} \Gamma(u) Z_Q(u) = (\det Q)^{-1/2} \pi^{-(1-u)} \Gamma(1-u) Z_{Q^{-1}}(1-u).$$

14.6.2 Evaluation at the zero-point argument

For $Q_\beta = \text{diag}(1, \beta^2)$, $\mathcal{Z}_\beta(s) = Z_{Q_\beta}(s/2)$. At $s = -1$, set $u = -1/2$:

$$\pi^{1/2} \Gamma(-1/2) Z_{Q_\beta}(-1/2) = \beta \pi^{-3/2} \Gamma(3/2) Z_{Q_\beta^{-1}}(3/2),$$

$$-2\pi Z_{Q_\beta}(-1/2) = \frac{\beta}{2\pi} Z_{Q_\beta^{-1}}(3/2),$$

$$Z_{Q_\beta}(-1/2) = -\frac{\beta}{4\pi^2} Z_{Q_\beta^{-1}}(3/2).$$

Since $Q_\beta^{-1} = \text{diag}(1, \beta^2)$, this is the convergent sum stated in the main text.

14.6.3 Separation from the reduced model ansatz

The exact continuation and $-(\pi/12)(\beta + \beta^{-1})$ are different functions. Consequently the latter cannot be used as the analytic continuation of the displayed Epstein zeta. At $\beta = 1$, the square-lattice identity

$$Z_{I_2}(u) = 4\zeta(u)L_{-4}(u), \quad L_{-4}(u) = \sum_{j=0}^{\infty} \frac{(-1)^j}{(2j+1)^u}$$

(under analytic continuation) gives $Z_{I_2}(-1/2) = -0.2288243104\dots$, while the reduced expression gives $-\pi/6 = -0.5235987756\dots$. It may therefore be retained only as a reduced model input. Differentiating its associated energy at fixed R gives the stationary point $\beta = 1$, not the postulated β_0 . Thus:

- Epstein continuation: **Proved**;
- physical Casimir subtraction: **Conditional**;
- reduced Casimir formula: **Conditional ansatz**;
- equilibrium equation for β_0 : **Postulate**;
- dynamical derivation of that equation: **Open Problem**.

14.7 Discrete Cascade and Fractal-Scaling Audit

Assume $N_{k+1} = 2N_k$ and $E_{k+1} = \beta_0 E_k$. For $k = 0$ the formulas $N_0 = 2^0 N_0$ and $E_0 = \beta_0^0 E_0$ hold. If they hold at k , then

$$N_{k+1} = 2(2^k N_0) = 2^{k+1} N_0, \quad E_{k+1} = \beta_0(\beta_0^k E_0) = \beta_0^{k+1} E_0.$$

Induction proves the closed forms. Eliminating k yields

$$E/E_0 = (N/N_0)^{\log \beta_0 / \log 2}.$$

N is a count and β_0 is dimensionless, so the recurrence preserves the units of E_0 . No fractal dimension follows until a contraction ratio for a metric length and a self-similar set are defined.

Final status: **Derived** from the recurrence assumptions; **Conditional** as a fractal law.

14.8 TTA, Holonomy, and Spinor Construction Audit

14.8.1 Configuration sectors versus representation spaces

The compact scalar supplies the disconnected configuration sectors $\mathcal{C}_{\mathbf{w}}$ indexed by $\mathbf{w} \in \mathbb{Z}^2$. A selected cycle supplies only the projected integer w . A tensor representation instead requires a vector space, a group action, and an equivariant map. Choosing $V^{\otimes(w-1)}$ verifies only that a tensor product with $w - 1$ factors can be written. It does not construct a canonical map from $\mathcal{C}_w$, because the same integer labels infinitely many choices of V , group action, and map. The proposed TTA correspondence therefore begins with an extra assignment; it is not a consequence of the homotopy classification.

The rule also has an undeclared domain issue: scalar winding permits $w = 0$ and $w < 0$, while $w - 1$ is not a non-negative tensor order there. Restricting the rule to $w \geq 1$ resolves the notation but is itself a model restriction.

14.8.2 Holonomy and spin structures

For the tangent bundle, parallel transport is trivial because the global frame has zero connection coefficients. For spinors, a spin structure is an independent lift of the oriented orthonormal frame bundle. The four lifts are the characters

$$\rho_\varepsilon : \pi_1(T^2) = \mathbb{Z}^2 \longrightarrow \{\pm 1\}, \quad \rho_\varepsilon(p, q) = (-1)^{p\varepsilon_\theta + q\varepsilon_\phi}.$$

This gives periodic or antiperiodic boundary conditions along each generator. The scalar winding class is integral, whereas the spin structure is mod two. Although reduction $\mathbb{Z}^2 \rightarrow \mathbb{Z}_2^2$ exists, topology alone does not require ATHENA to use it as the winding–spin map.

14.8.3 Dirac spectrum

With $\mathcal{D} = -i(\sigma^1 R^{-1} \partial_\theta + \sigma^2 r^{-1} \partial_\phi)$, define

$$k_\theta = \frac{m + \varepsilon_\theta/2}{R}, \quad k_\phi = \frac{n + \varepsilon_\phi/2}{r}.$$

On a plane-wave spinor, $\mathcal{D}$ reduces to $k_\theta \sigma^1 + k_\phi \sigma^2$. Its characteristic polynomial is

$$\det[(k_\theta \sigma^1 + k_\phi \sigma^2) - \lambda I_2] = \lambda^2 - k_\theta^2 - k_\phi^2,$$

so the two eigenvalues are $\lambda_\pm = \pm \sqrt{k_\theta^2 + k_\phi^2}$. This completes the intrinsic two-dimensional construction.

Final status. Holonomy, spin-structure classification, and the intrinsic Dirac spectrum are **Proved**. The TTA representation map, four-dimensional spinor lift, and spin–statistics construction are **Open Problems**.

14.9 Liouville Reduction and RG Details

14.9.1 Zero-mode calculation

For a topologically trivial representative, use the dimensionless field $\phi = \Phi/f_a$ and expand

$$\phi = \phi_0 + \sum_{(m,n) \neq (0,0)} q_{mn} u_{mn}, \quad q_{-m,-n} = \overline{q_{mn}}.$$

For the dimensionless free Gaussian internal action

$$S_0[\phi] = \frac{1}{2} \int_{T^2} |\nabla \phi|^2 dA,$$

orthonormality and $-\Delta u_{mn} = \lambda_{mn} u_{mn}$ give

$$S_0[\phi] = \frac{1}{2} \sum_{(m,n) \neq (0,0)} \lambda_{mn} |q_{mn}|^2.$$

Here $[q_{mn}] = L$, so every summand is dimensionless. The zero mode ϕ_0 is absent. Gaussian integration over all q_{mn} gives a determinant proportional to $(\det'[-\Delta])^{-1/2}$, independent of ϕ_0 . Therefore a real exponential potential for ϕ_0 cannot follow from the free mode integration. It must originate in a declared interaction, measure anomaly, boundary term, or other microscopic input, followed by an explicit matching calculation.

For the compact scalar, the Liouville factor $e^{2b\Phi/f_a}$ is not invariant under $\Phi \mapsto \Phi + 2\pi f_a$. Thus the proposed reduction must additionally state whether compactness is broken, restricted to a branch, or abandoned in the Liouville sector.

14.9.2 RG integration and fixed-point audit

With $A = (2 - \gamma_{\text{str}})/(4\pi)$ and $B = \beta_0(2 - \beta_0)/(8\pi^2)$,

$$\beta(\alpha) = \alpha^2(A - B\alpha).$$

The roots are $\alpha = 0$ and $\alpha_* = A/B$. When the equilibrium postulate is inserted,

$$B = \frac{\pi/12}{8\pi^2} = \frac{1}{96\pi}, \quad \alpha_* = 24(2 - \gamma_{\text{str}}).$$

Hence $\gamma_{\text{str}} = 0$ gives 48. Requiring 89.4 in this same two-term formula would instead give

$$89.4 = 24(2 - \gamma_{\text{str}}) \implies \gamma_{\text{str}} = -1.725.$$

No displayed equation supplies that value. The partial-fraction integration in the main text gives the complete implicit trajectory of the truncated ODE.

14.9.3 Complete autonomous phase portrait

Set $t = \ln(\mu/\mu_0)$. For constant A and B , the truncated flow is the autonomous equation

$$\dot{\alpha} = \alpha^2(A - B\alpha), \quad \dot{\alpha} \equiv \frac{d\alpha}{dt}.$$

The origin is always a double fixed point. If $B \neq 0$, the second fixed point is $\alpha_* = A/B$. Since $\alpha^2 \geq 0$, away from the fixed points the direction of flow is determined entirely by the sign of $A - B\alpha$. Moreover,

$$\beta'(\alpha_*) = -\frac{A^2}{B},$$

so the non-zero fixed point is forward- t attractive for $B > 0$ and repulsive for $B < 0$, independently of the sign of A . At the non-hyperbolic origin, $\dot{\alpha} \sim A\alpha^2$: for $A > 0$ the origin attracts from the negative side and repels from the positive side, while for $A < 0$ these one-sided roles are reversed.

On the physically relevant half-line $\alpha \geq 0$, the complete sign classification is:

- (i) $A > 0, B > 0$: $\alpha_* = A/B > 0$ is globally attractive within $\alpha > 0$. Couplings in $(0, \alpha_*)$ increase and couplings above α_* decrease. The boundary point $\alpha = 0$ is forward unstable.
- (ii) $A < 0, B < 0$: $\alpha_* = A/B > 0$ is a separatrix and is forward unstable. Initial data in $(0, \alpha_*)$ decrease to zero, while data above α_* grow without bound in finite forward time.
- (iii) $A > 0, B < 0$: the non-zero fixed point is negative and $\dot{\alpha} > 0$ for every $\alpha > 0$. Every positive trajectory grows without bound in finite forward time.
- (iv) $A < 0, B > 0$: the non-zero fixed point is negative and $\dot{\alpha} < 0$ for every $\alpha > 0$. Every positive trajectory approaches zero as $t \rightarrow +\infty$.

These statements also give the full real-line portrait when combined with the one-sided behavior at zero and the stability rule for α_* . A divergent trajectory reaches infinity in finite RG time because the integral $\int^\infty d\alpha/\alpha^3$ converges.

The exceptional coefficient cases are elementary. If $B = 0$ and $A \neq 0$,

$$\alpha(t) = \frac{\alpha_0}{1 - A\alpha_0(t - t_0)}.$$

Thus positive trajectories blow up for $A > 0$ and approach zero for $A < 0$. If $A = 0$ and $B \neq 0$,

$$\alpha(t) = \frac{\alpha_0}{\sqrt{1 + 2B\alpha_0^2(t - t_0)}},$$

with the sign of α inherited from α_0 . Positive trajectories approach zero for $B > 0$ and blow up for $B < 0$. If $A = B = 0$, every constant α is a fixed trajectory.

For the ATHENA values $A = 1/(2\pi) > 0$ and $B = 1/(96\pi) > 0$, the applicable case is (i): the conditional fixed point $\alpha_* = 48$ attracts every initial $\alpha_0 > 0$ as $\mu \rightarrow \infty$ within the two-term ansatz.

14.9.4 Lambert-W inversion

Assume $A \neq 0$, $B \neq 0$, and define

$$x = \frac{\alpha}{\alpha_*}, \quad F(x) = \ln \left| \frac{x}{1-x} \right| - \frac{1}{x}.$$

The implicit solution is

$$F(x(t)) = K(t), \quad K(t) = F(x_0) + A\alpha_*(t - t_0).$$

Set

$$z = \frac{1-x}{x} = \frac{1}{x} - 1.$$

Then $F(x) = -\ln|z| - z - 1$, so

$$\ln|z| + z = -K(t) - 1.$$

Consequently the real solution on each invariant interval can be written

$$\boxed{x(t) = \frac{1}{1 + W_k(\sigma e^{-K(t)-1})}, \quad \alpha(t) = \alpha_* x(t)},$$

where $\sigma = \text{sgn } z$. The real branch is fixed by the initial interval:

$$\begin{aligned} 0 < x < 1 : \quad \sigma &= +1, \quad k = 0, \\ x > 1 : \quad \sigma &= -1, \quad k = 0, \\ x < 0 : \quad \sigma &= -1, \quad k = -1. \end{aligned}$$

The fixed solutions $x = 0$ and $x = 1$ are treated separately. If a trajectory runs to $|\alpha| = \infty$, then $F(x) \rightarrow 0$ and its finite blow-up time is

$$t_\infty = t_0 - \frac{F(x_0)}{A\alpha_*}.$$

This inversion introduces no new dynamics; it is the explicit solution of the already declared two-term ansatz.

Final status. The zero-mode obstruction is **Proved**. The truncated ODE results are **Derived** from the ansatz. Liouville matching and derivation of the beta-function coefficients from the EFT remain **Open Problems**; the value 89.4 is not the fixed point of the displayed ansatz at $\gamma_{\text{str}} = 0$.

14.10 Topological-Involution Determinant Calculation

Let $v_i = \partial_i \Phi$. Relative to the Euclidean spatial background, the rank-one matrix vv^T has eigenvalue $S = v^T v$ along v and eigenvalue zero on the two-dimensional orthogonal complement. Therefore

$$\tilde{\gamma} = a^2 I_3 - \alpha M_*^{-4} vv^T$$

has two eigenvalues a^2 and one eigenvalue $a^2 - \alpha M_*^{-4} S = a^2(1 - Q_D)$. Their product gives

$$\det \tilde{\gamma} = a^6(1 - Q_D),$$

in agreement with the matrix determinant lemma. Positive definiteness requires $a^2 > 0$ and $1 - Q_D > 0$. Taking the positive square root and cubic root gives $a_{\text{vol}} = a(1 - Q_D)^{1/6}$, and the logarithmic derivative gives the displayed H_{vol} identity.

For $S > 0$, introduce the orthogonal projectors

$$P_{\parallel} = \frac{vv^T}{S}, \quad P_{\perp} = I_3 - P_{\parallel}.$$

They obey $P_{\parallel}^2 = P_{\parallel}$, $P_{\perp}^2 = P_{\perp}$, and $P_{\parallel}P_{\perp} = 0$. The metric therefore has the spectral decomposition

$$\tilde{\gamma} = a^2 [P_{\perp} + (1 - Q_D)P_{\parallel}].$$

Its eigenvalues are

$$\lambda_{\perp,1} = a^2, \quad \lambda_{\perp,2} = a^2, \quad \lambda_{\parallel} = a^2(1 - Q_D).$$

When $a \neq 0$ and $1 - Q_D \neq 0$, inversion of the eigenvalues gives

$$(\tilde{\gamma})^{-1} = \frac{1}{a^2} \left[P_{\perp} + \frac{1}{1 - Q_D} P_{\parallel} \right].$$

Equivalently, the Sherman–Morrison identity gives

$$(\tilde{\gamma}^{-1})^{ij} = \frac{1}{a^2} \delta^{ij} + \frac{\alpha}{a^4 M_*^4 (1 - Q_D)} \partial^i \Phi \partial^j \Phi.$$

Multiplication by $\tilde{\gamma}_{jk} = a^2 \delta_{jk} - \alpha M_*^{-4} \partial_j \Phi \partial_k \Phi$ cancels the rank-one correction and yields δ^i_k . If $S = 0$, then $Q_D = 0$ and the same formula reduces continuously to $(\tilde{\gamma})^{-1} = a^{-2} I_3$.

Assuming the declared scale $M_* \neq 0$, the complete algebraic classification is:

- $a = 0$: the background matrix is singular and the displayed rank-one formula does not define an invertible three-metric;
- $a \neq 0$ and $1 - Q_D = 0$: the parallel eigenvalue vanishes, the matrix has rank two, and $\det \tilde{\gamma} = 0$;
- $a \neq 0$ and $1 - Q_D > 0$: all three eigenvalues are positive and the spatial metric is positive definite;
- $a \neq 0$ and $1 - Q_D < 0$: the metric remains invertible but has two positive and one negative eigenvalue, so it is not a positive-definite spatial metric.

For $\alpha \leq 0$, $1 - Q_D \geq 1$ and the spatial metric is positive whenever $a \neq 0$. For $\alpha > 0$, positivity requires

$$S < \frac{a^2 M_*^4}{\alpha}.$$

All determinant, inverse, and signature statements follow from the existing rank-one ansatz and introduce no coarse-graining or cosmological dynamics.

Final status. The local determinant, inverse, eigenvalue, positivity, and invertibility identities are **Proved under the existing assumptions**. Spatial averaging, dynamical cosmology, and a literal involutive map remain **Open Problems**.

14.11 Spectral Determinism, Rigidity, and Emergent Time

14.11.1 Non-uniqueness after spectral factorization

Spectral Determinism says that Ω is constant on fibers of the invariant map $\mathcal{I}$. This guarantees that an already selected Ω descends to the quotient by isospectral/equal-Betti equivalence. It does not choose the function on that quotient. For example, after fixing dimensionless normalized eigenvalues $\widehat{\lambda}_i$, every member of the family

$$\Omega_{p,c}(K) = C \frac{\beta_1(K)}{1 + \beta_1(K)^p} \exp \left[-c \sum_{i=1}^N \widehat{\lambda}_i^2 \right], \quad p > 1, \quad c > 0,$$

depends only on the declared invariants and has positive spectral decay, but different (p, c) give different functionals. Additional precise axioms are needed for a representation or uniqueness theorem.

14.11.2 Uniformization and missing rigidity data

For genus one, Gauss–Bonnet forces the constant-curvature uniformizing metric to be flat. The remaining statement can be made explicit. A marked conformal torus is represented by

$$T_\tau = \mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z}), \quad \tau = \tau_1 + i\tau_2, \quad \tau_2 > 0.$$

For $0 \leq x < 1$, $0 \leq y < 1$, and fixed area $A > 0$, define

$$\begin{aligned} g_{\tau,A} &= \frac{A}{\tau_2} |dx + \tau dy|^2 \\ &= \frac{A}{\tau_2} [dx^2 + 2\tau_1 dx dy + |\tau|^2 dy^2]. \end{aligned}$$

Its matrix and determinant are

$$(g_{ij}) = \frac{A}{\tau_2} \begin{pmatrix} 1 & \tau_1 \\ \tau_1 & |\tau|^2 \end{pmatrix}, \quad \det g = A^2.$$

Thus $\sqrt{\det g} = A$ on the unit square and the total area is exactly A . The coefficients are constant, so the metric is flat.

This representative is unique within its conformal class after the area is fixed. Indeed, if $\widetilde{g} = e^{2u}g$ and both metrics are flat, the two-dimensional curvature-transformation law gives

$$K_{\widetilde{g}} = e^{-2u}(K_g - \Delta_g u) = 0.$$

Since $K_g = 0$, one has $\Delta_g u = 0$. Every harmonic function on a compact connected torus is constant, so $u = u_0$. Equality of the two total areas then requires $e^{2u_0} = 1$, hence $u_0 = 0$ and $\widetilde{g} = g$.

The inverse matrix is

$$(g^{ij}) = \frac{1}{A\tau_2} \begin{pmatrix} |\tau|^2 & -\tau_1 \\ -\tau_1 & 1 \end{pmatrix}.$$

For Fourier modes $e^{2\pi i(mx+ny)}$, the scalar Laplacian therefore has eigenvalues

$$\begin{aligned} \lambda_{mn} &= 4\pi^2 g^{ij}(m, n)_i(m, n)_j \\ &= \frac{4\pi^2}{A\tau_2} (|\tau|^2 m^2 - 2\tau_1 mn + n^2) \\ &= \frac{4\pi^2}{A\tau_2} |n - m\tau|^2. \end{aligned}$$

For the rectangular product torus, take $\tau = i\beta$ and $A = 4\pi^2 Rr$. The formula reduces to

$$\lambda_{mn} = \frac{m^2}{R^2} + \frac{n^2}{r^2},$$

so the earlier spectrum is recovered exactly.

Unmarked conformal tori identify τ under the modular transformation

$$\tau \mapsto \frac{a\tau + b}{c\tau + d}, \quad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{PSL}(2, \mathbb{Z}).$$

Consequently the fixed-area moduli space remains $\mathbb{H}/\text{PSL}(2, \mathbb{Z})$. Uniformization proves uniqueness only after a point in this moduli space and the area have been fixed; it cannot select either datum.

For the candidate Betti factor, $\beta_1 = 2g$ proves that genus one maximizes that factor among closed orientable surfaces. The spectral factor can reverse or erase this ordering unless its normalization, area, cutoff, and global extremization are controlled. Metric rescaling already shows that no finite area is selected by the displayed exponential alone.

14.11.3 Order parameter versus physical time

If $\Omega(\tau)$ is strictly increasing, it supplies an order coordinate. If it is also C^1 with $\Omega'(\tau) > 0$, the inverse-function theorem justifies the differentiable reparametrization $t = \Omega - \Omega_0$ and gives $d\Omega/dt = 1$. Strict monotonicity without regularity does not guarantee a derivative. Any strictly increasing $f(t)$ preserves the same order, so no unique clock normalization follows. Finally, a dimensionless order coordinate does not supply Lorentzian signature, causal cones, proper time, or a physical unit of seconds.

Final status. Spectral factorization, the genus-one Betti-factor maximum, uniformization within a fixed conformal class, and the regular monotone-clock construction are **Proved**. Uniqueness of Ω , full rigidity, topology selection by the complete functional, and physical emergent time remain **Conjectural/Open**.

14.12 Status Notes on Key Claims

14.12.1 Status of the β_0 Relation

The relation $\beta_0(2 - \beta_0) = \pi/12$ is the equilibrium postulate of Chapter 2. Its selected root is derived correctly, but the postulate is not the stationary condition of the displayed reduced Casimir ansatz. A derivation requires a dimensionally normalized effective action, declared variation constraint, and complete spectral Casimir calculation.

14.12.2 Status of TTA

The winding-spin correspondence remains an open construction. An antiperiodic spin structure is compatible with spinors, but it is not selected uniquely by $w = 2$ and does not by itself prove four-dimensional spin-statistics. The intrinsic two-dimensional Dirac spectrum is proved in Appendix 14.8.

14.12.3 Conditional Topological-Involution Identity

For the rank-one spatial ansatz, define $Q_D = \alpha|\nabla\Phi|^2/(a^2M_*^4)$. The matrix determinant lemma then gives $H_{\text{vol}} = H - \dot{Q}_D/[6(1 - Q_D)]$. Replacing Q_D by an isotropic ensemble average and interpreting the correction dynamically are additional assumptions.

14.13 Stage 1 Mathematical Status Ledger

Table 14.2: Stage 1 mathematical status ledger.

Topic	Final status	Mathematical conclusion
T^2 metric and geometry	Proved	Product metric, inverse, determinant, area, vanishing connection coefficients, and intrinsic flatness derived.
Laplace–Beltrami operator	Proved	Coordinate formula reduces to $R^{-2}\partial_\theta^2 + r^{-2}\partial_\phi^2$.
Scalar spectrum	Proved	Normalized Fourier basis and $\lambda_{mn} = R^{-2}(m^2 + n^2/\beta^2)$.
Winding numbers	Proved	Lift, integrality, $H^1(T^2; \mathbb{Z})$ classification, conservation, and harmonic sector representative.
Power-of-two stability	Conditional/Open	Topology protects every winding sector; neither topology nor the quadratic energy selects powers of two.
Epstein regularization	Proved mathematically	Functional equation gives the convergent dual-lattice expression at $s = -1$. Physical subtraction remains scheme-dependent.
Reduced Casimir formula	Conditional	The expression $-(\pi/12)(\beta + \beta^{-1})$ is not the continuation of the defined Epstein zeta and is retained only as a model ansatz.

Table 14.3: Stage 1 mathematical status ledger (continued).

Topic	Final status	Mathematical conclusion
β_0	Derived from postulate	Both quadratic roots and the selected $0 < \beta_0 < 1$ branch are explicit; the equilibrium equation is not derived by Casimir minimization.
Z_0, γ_0	Conditional/Derived	Numerical values follow from declared normalizations; no independent correlation or curvature derivation is given.
R_c, f_a, a_0	Conditional calibration	Units and substitutions close, but L_* , κ_f , and κ_a are independent inputs.
Cascade/fractal scaling	Derived/Conditional	Recurrences imply a power-law cascade exponent; no mathematical fractal dimension is defined.
TTA	Open Problem	Tensor order can be assigned formally, but no canonical equivariant map from winding sectors to particle representations exists.
Holonomy	Proved	Tangent holonomy is trivial for the declared flat metric; four independent spin structures remain.
Spinor construction	Proved intrinsically/Open physically	The full 2D Dirac spectrum is derived; the 4D winding–spin and spin–statistics maps are open.

Table 14.4: Stage 1 mathematical status ledger (continued).

Topic	Final status	Mathematical conclusion
Liouville reduction	Conjectural	Free non-zero modes do not generate a zero-mode exponential; compactness is also incompatible with a real single-valued Liouville exponential without extra assumptions.
RG equations and α_*	Derived from ansatz	Fixed points, stability, and implicit flow are derived. The coefficients are not derived from the EFT; $\alpha_* = 48$ for $\gamma_{\text{str}} = 0$.
Topological Involution	Conditional/Open	Rank-one determinant identity is proved; cosmological averaging and a literal involutive map are absent.
Spectral Determinism	Axiom with proved consequence	It implies factorization through spectral/Betti data, not uniqueness of the function.
Uniqueness and rigidity	Partial/Conjectural	Genus one maximizes the Betti factor and uniformizes to a flat metric in each fixed conformal class; modulus, area, and full spectral extremization remain open.
Emergent time	Proved order parameter/Open physically	A regular monotone functional defines a non-unique order coordinate; Lorentzian proper time is not derived.

Part II

Volume I: Physical Foundations

Chapter 15

Introduction

15.1 Purpose

This volume builds the physical interpretation of the ATHENA framework, using the mathematical foundations established in Volume 0. The goal is to translate the topological and mathematical structures — the toroidal manifold T^2 , the scalar field $\Phi(x)$, the winding number w , the Topological Tensor Ascent (TTA), and the renormalization group flow — into a coherent physical picture of the universe.

The physical discussion distinguishes:

- conditional consequences of declared assumptions,
- physical interpretations and analogies, and
- phenomenological targets and conditional predictions requiring independent tests.

Notation 15.1 (Stage 2 physical-audit convention). For every physical construction below, the audit separates (i) mathematical input inherited from Volume 0, (ii) an additional physical postulate or ansatz, and (iii) consequences that follow after that input is accepted. A calculation labelled *derived from a stated postulate* verifies the consequence but not the physical postulate. Whenever the required spacetime embedding, action, operator matching, state definition, or observational likelihood is absent, the status is stated explicitly as: *Open — requires new physical input*.

15.2 Relation to Volume 0

The physical content of ATHENA is proposed on the basis of the mathematical structures and additional assumptions of Volume 0:

- The toroidal manifold T^2 and winding number w are proposed as labels for candidate particle and force sectors; the mathematical classification does not establish that physical identification.
- The dimensionless relations among β_0 , Z_0 , and γ_0 are supplemented by the calibrated scales L_* and κ_f defining R_c and f_a .
- TTA proposes, but does not yet derive, spin-1/2 and fermionic statistics.
- The Liouville and RG constructions are ansatzes to be matched to the 4D EFT.
- The dependency ledger in Chapter 8 identifies which conclusions remain conjectural.

15.3 Physical Axioms

Hypothesis 15.1 (Coarse-grained EM diagnostic). *A possible electromagnetic-vacuum diagnostic is the dimensionless composite quantity*

$$\mathcal{O}_{\text{EM}}(x) \equiv \frac{1}{\Lambda^4} \langle F_{\mu\nu} F^{\mu\nu} \rangle_{\text{cg}},$$

where Λ is the ultraviolet cutoff, $F_{\mu\nu}$ is the electromagnetic field tensor, and $\langle \cdot \rangle_{\text{cg}}$ denotes coarse-graining over scales below Λ^{-1} . This composite is not identified globally with the compact scalar Φ , carries no separately postulated dynamics, and does not define a new physical field. Any matching to a local expectation of $U = \exp(i\Phi/f_a)$ or to a trivial-sector representative is *Open* — requires additional physical input.

Proposition 15.1 (What follows from the EM diagnostic). In four spacetime dimensions, $[F_{\mu\nu}] = 2$ in mass units, so $[F_{\mu\nu} F^{\mu\nu}] = 4$. Therefore

$$[\Lambda^{-4} \langle F_{\mu\nu} F^{\mu\nu} \rangle_{\text{cg}}] = 0, \quad [\mathcal{O}_{\text{EM}}] = 0.$$

The contraction $F_{\mu\nu} F^{\mu\nu}$ is a Lorentz scalar and is invariant under the electromagnetic gauge transformation $A_\mu \mapsto A_\mu + \partial_\mu \chi$. These statements establish dimensional and symmetry consistency only. They do not define the coarse-graining map, derive the compact scalar, or show that the composite operator has the periodic potential used later.

Status: Derived for the dimensional and symmetry statements. A composite-to-compact-scalar matching is *Open* — requires new physical input, namely a declared microscopic state, coarse-graining prescription, renormalized composite operator, and matching calculation to the scalar EFT.

Definition 15.1 (Kinetic Term).

$$X = -\frac{1}{2} g^{\mu\nu} \partial_\mu \Phi \partial_\nu \Phi.$$

15.3.1 Three Layers of the Vacuum

The following layer notation is a schematic filtering of the one compact scalar into three energy-band components; it does not introduce independent fundamental or propagating fields:

$$\Phi = \Phi_\gamma + \Phi_{\text{vis}} + \Phi_{\text{cmb}}, \quad \rho_{\text{vac}} = \rho_\gamma + \rho_{\text{vis}} + \rho_{\text{cmb}},$$

where:

- Φ_γ : Gamma/Nuclear layer ($E > 100$ keV). Its gravitational shadow is interpreted as dark matter.
- Φ_{vis} : Visible layer (1.7–3.1 eV). Our observational layer.
- Φ_{cmb} : CMB/Microwave layer ($E < 0.001$ eV). Its expansive pressure is interpreted as dark energy.

Proposition 15.2 (Audit of the layer decomposition). Even if a specified linear filter defines $\Phi = \sum_i \Phi_i$, the filtered components are not new degrees of freedom. Moreover, that field identity does not, for the canonical action, imply the energy identity $\rho = \sum_i \rho_i$. For example,

$$\frac{1}{2} \partial_\mu \Phi \partial^\mu \Phi = \frac{1}{2} \sum_i \partial_\mu \Phi_i \partial^\mu \Phi_i + \sum_{i < j} \partial_\mu \Phi_i \partial^\mu \Phi_j,$$

and in general

$$V\left(\sum_i \Phi_i\right) \neq \sum_i V(\Phi_i).$$

Thus the displayed additive energy split is valid only if the filtering map is defined and the cross terms vanish under a specified projection or averaging operation, or if the layers are promoted to independent fields with a separable action.

Status: Conditional consequence. The energy-layer interpretation is *Open* — *requires new physical input*: an orthogonality theorem, decoherence/coarse-graining rule, or a multi-field action that makes the split dynamically meaningful.

Chapter 16

Electromagnetic Layered Vacuum

Postulate 16.1 (EM Layered Vacuum). The physical vacuum is not empty. It is a multi-layered electromagnetic continuum with distinct energy densities. These layers are different energy regimes of a single scalar field.

Table 16.1: The three principal energy layers of the vacuum.

Layer	Energy Range	Physical Interpretation
Gamma/Nuclear	$> 100 \text{ keV}$	“Dark matter” shadow
Visible	$1.7\text{--}3.1 \text{ eV}$	Observational layer
CMB/Microwave	$< 0.001 \text{ eV}$	“Dark energy” pressure

Remark 16.1. ATHENA hypothesizes that the “dark matter” shadow and “dark energy” pressure are effective manifestations of different regimes of one vacuum field. This interpretation does not constitute a solution until the covariant equations reproduce lensing, clustering, background expansion, and perturbations.

Postulate 16.2 (Information Conservation). Total information in the universe is conserved:

$$C_{\text{record}} + C_{\text{collapse}} \approx 1,$$

where C_{record} is potential information encoded in the EM filament (the vacuum), and C_{collapse} is realized classical matter.

Remark 16.2 (Information-balance audit). The displayed quantities are dimensionless but no entropy, information measure, Hilbert-space partition, evolution law, or error term for the approximate equality is defined. The equation is therefore a physical postulate rather than a conservation law derived from Noether symmetry or unitary dynamics. *Open — requires new physical input.*

Hypothesis 16.1 (Compact scalar and EM-vacuum interpretation). *The physical scalar remains the single compact field $\Phi \sim \Phi + 2\pi f_a$. The reported dimensionless number 0.782 ± 0.015 is retained only as a historical organizational/EM diagnostic target. It is not a value of Φ/M_* , does not select a scalar vacuum, and does not enter the V15.3 metric. Its calculation and uncertainty provenance are not independently documented in the present archive.*

Status: the EM-condensate interpretation is **Conditional/Open**. No global composite-field identity is assumed.

Chapter 17

Field Theory

Postulate 17.1 (Covariant EFT Action).

$$S = \int d^4x \sqrt{-g} \left[\frac{M_{\text{Pl}}^2}{2} R - \frac{1}{2} (\partial\Phi)^2 - V(\Phi) + \mathcal{L}_{\text{EFT}}^{(>4)} + \frac{c_{\Phi F}}{4M_*} \Phi F_{\mu\nu} \tilde{F}^{\mu\nu} \right] + S_m[\tilde{g}_{\mu\nu}, \psi_m] + S_{\text{top}}.$$

Every operator in $\mathcal{L}_{\text{EFT}}^{(>4)}$ must be accompanied by its suppression scale. This action defines the baseline model; additional Horndeski-like terms below are optional EFT deformations, not already-derived consequences of topology.

Proposition 17.1 (Variation of the declared minimal scalar sector). Let

$$S_\Phi = \int d^4x \sqrt{-g} \left[-\frac{1}{2} g^{\mu\nu} \partial_\mu \Phi \partial_\nu \Phi - V(\Phi) \right].$$

For a compactly supported variation $\Phi \mapsto \Phi + \delta\Phi$,

$$\begin{aligned} \delta S_\Phi &= \int d^4x \sqrt{-g} \left[-\nabla^\mu \Phi \nabla_\mu \delta\Phi - V'(\Phi) \delta\Phi \right] \\ &= \int d^4x \sqrt{-g} \left[\square_g \Phi - V'(\Phi) \right] \delta\Phi, \end{aligned}$$

where the boundary term has been discarded. Hence the source-free equation is

$$\square_g \Phi - V'(\Phi) = 0.$$

With

$$T_{\mu\nu}^{(\Phi)} = -\frac{2}{\sqrt{-g}} \frac{\delta S_\Phi}{\delta g^{\mu\nu}},$$

metric variation gives

$$T_{\mu\nu}^{(\Phi)} = \partial_\mu \Phi \partial_\nu \Phi - g_{\mu\nu} \left(\frac{1}{2} \partial_\rho \Phi \partial^\rho \Phi + V(\Phi) \right).$$

Direct covariant differentiation then yields

$$\nabla^\mu T_{\mu\nu}^{(\Phi)} = (\square_g \Phi - V'(\Phi)) \partial_\nu \Phi,$$

so the minimal scalar stress tensor is conserved on its source-free equation of motion. Since $[\Phi] = 1$, each term in the scalar Lagrangian has mass dimension four and $[T_{\mu\nu}] = 4$.

Status: Derived from the stated EFT postulate. This calculation does not determine the higher-order operators, the disformal matter source, or S_{top} .

Remark 17.1 (Electromagnetic-sector audit). The displayed action contains the pseudoscalar operator $\Phi F_{\mu\nu} \tilde{F}^{\mu\nu}$ but does not explicitly display a Maxwell kinetic term $-Z_F(\Phi) F_{\mu\nu} F^{\mu\nu}/4$. Unless such a term and its gauge field are declared inside S_m , the electromagnetic tensor acts only as an external or incompletely specified field. An EM-medium interpretation requires constitutive relations, propagation equations, and a stress tensor; these are *Open — require new physical input*. In addition, because Φ is compact, the shift $\Phi \mapsto \Phi + 2\pi f_a$ changes the action by

$$\Delta S = 2\pi \frac{c_{\Phi F} f_a}{M_*} \mathcal{Q}_F, \quad \mathcal{Q}_F = \frac{1}{4} \int d^4x \sqrt{-g} F_{\mu\nu} \tilde{F}^{\mu\nu}.$$

Global quantum consistency therefore requires

$$\frac{c_{\Phi F} f_a}{M_*} \mathcal{Q}_F \in \mathbb{Z}$$

for every allowed topological sector, subject to the normalization chosen for $\mathcal{Q}_F$. V15.3.1 does not assert that this condition is satisfied. The gauge normalization, allowed topological sectors, and boundary conditions are not specified, so the displayed linear operator remains an optional local EFT term rather than a completed global compact-scalar coupling. **Status: Open — requires additional physical input.**

Proposition 17.2 (Sources in the baseline scalar equation). Define the matter-frame tensor by

$$\tilde{T}^{\mu\nu} = \frac{2}{\sqrt{-\tilde{g}}} \frac{\delta S_m}{\delta \tilde{g}_{\mu\nu}}$$

and set $u_\mu = \partial_\mu \Phi$. Varying the declared disformal metric gives

$$\delta \tilde{g}_{\mu\nu} = -\frac{\alpha}{M_*^4} (u_\mu \partial_\nu \delta \Phi + u_\nu \partial_\mu \delta \Phi).$$

Indeed,

$$\begin{aligned} \delta_\Phi S_m &= \frac{1}{2} \int d^4x \sqrt{-\tilde{g}} \tilde{T}^{\mu\nu} \delta \tilde{g}_{\mu\nu} \\ &= -\frac{\alpha}{M_*^4} \int d^4x \sqrt{-\tilde{g}} \tilde{T}^{\mu\nu} u_\mu \partial_\nu \delta \Phi \\ &= \int d^4x \sqrt{-g} \mathcal{J}_m^{(D)} \delta \Phi, \end{aligned}$$

after a compactly supported integration by parts. Consequently the exact matter contribution is

$$\mathcal{J}_m^{(D)} = \frac{\alpha}{M_*^4 \sqrt{-g}} \partial_\nu \left(\sqrt{-\tilde{g}} \tilde{T}^{\mu\nu} \partial_\mu \Phi \right).$$

For the displayed pseudoscalar coupling, the lower-order scalar equation is therefore

$$\square_g \Phi - V'(\Phi) + \frac{c_{\Phi F}}{4M_*} F_{\mu\nu} \tilde{F}^{\mu\nu} + \mathcal{J}_m^{(D)} + \mathcal{J}_{\text{top}} = 0,$$

where $\mathcal{J}_{\text{top}} = (\sqrt{-g})^{-1} \delta S_{\text{top}} / \delta \Phi$. The formula fixes the chain-rule source but not the matter state or the unspecified topological action.

Status: Derived from the displayed baseline couplings; conditional on the stated stress-tensor convention.

Definition 17.1 (V15.3 pure-disformal metric).

$$\tilde{g}_{\mu\nu} = g_{\mu\nu} - \frac{\alpha}{M_*^4} \partial_\mu \Phi \partial_\nu \Phi, \quad X = -\frac{1}{2} g^{\mu\nu} \partial_\mu \Phi \partial_\nu \Phi.$$

The reported galactic screened-fit target $\alpha_{\text{eff}}^{\text{gal}} = 0.36 \pm 0.03$ is model- and dataset-dependent and must be reported together with the likelihood and nuisance-parameter treatment. It does not by itself determine the bare disformal coefficient α .

Proposition 17.3 (Inverse, determinant, and Lorentzian signature). Let $u_\mu = \partial_\mu \Phi$ and

$$\Delta_D \equiv 1 + \frac{2\alpha X}{M_*^4}.$$

The direct rank-one derivation in Definition 9.2 gives

$$\tilde{g}^{\mu\nu} = g^{\mu\nu} + \frac{\alpha}{M_*^4 + 2\alpha X} \nabla^\mu \Phi \nabla^\nu \Phi,$$

and, in four dimensions,

$$\det \tilde{g} = \det g \Delta_D, \quad \sqrt{-\tilde{g}} = \sqrt{-g} \sqrt{\Delta_D} \quad (\Delta_D > 0).$$

The mixed rank-one map $g^{-1}\tilde{g}$ has three transverse eigenvalues equal to one and one non-null longitudinal eigenvalue Δ_D . Thus invertibility requires $\Delta_D \neq 0$. If g is Lorentzian, the sufficient and, for a non-null gradient, necessary condition for retaining the same inertia is

$$\Delta_D > 0.$$

For a null gradient, $X = 0$, the determinant is unchanged and the metric remains nondegenerate for finite α/M_*^4 .

These algebraic conditions do not prove hyperbolicity, absence of ghosts, or that the g - and $\tilde{g}$ -causal cones satisfy a desired nesting relation. Those properties depend on the background and the kinetic operators of every propagating field.

Status: Proved for the declared rank-one metric ansatz. A viable causal EFT remains *Open* — *requires new physical input* in the form of a complete operator basis and background-dependent characteristic analysis.

Hypothesis 17.1 (Optional Higher-Order EFT Functions).

$$\begin{aligned} G_2(\Phi, X) &= X - V(\Phi) + \frac{\beta(\Phi)}{M_*^4} X^2, \\ G_3(\Phi, X) &= \frac{\gamma(\Phi)}{M_*^3} X, \\ V(\Phi) &= \Lambda^4 \left(1 - \cos \frac{\Phi}{f_a} \right), \\ \beta(\Phi) &= \beta_0 \cos \frac{\Phi}{f_a}, \quad \gamma(\Phi) = \gamma_0 \sin \frac{\Phi}{f_a}. \end{aligned}$$

Proposition 17.4 (Local stability check for the G_2 -only truncation). If the optional G_3 term, metric mixing, and matter mixing are temporarily omitted, the scalar pressure function is

$$P(\Phi, X) = X - V(\Phi) + \frac{\beta(\Phi)}{M_*^4} X^2.$$

On a homogeneous timelike background, its derivatives are

$$P_X = 1 + \frac{2\beta X}{M_*^4}, \quad P_{XX} = \frac{2\beta}{M_*^4}, \quad P_X + 2XP_{XX} = 1 + \frac{6\beta X}{M_*^4}.$$

The standard quadratic $P(X)$ test therefore requires

$$1 + \frac{2\beta X}{M_*^4} > 0, \quad 1 + \frac{6\beta X}{M_*^4} > 0,$$

and gives

$$c_s^2 = \frac{1 + 2\beta X/M_*^4}{1 + 6\beta X/M_*^4}.$$

These are necessary local conditions only for the stated truncation. The G_3 interaction, disformal matter coupling, gravitational constraints, and spacetime dependence of $\beta(\Phi)$ change the complete perturbation system.

Status: Conditional consequence of the optional G_2 ansatz. Full stability is *Open* — requires new physical input through a single fixed EFT basis and its quadratic action.

Proposition 17.5 (Matter-frame and g -frame stress tensors). Keep the matter-frame convention

$$\tilde{T}^{\mu\nu} = \frac{2}{\sqrt{-\tilde{g}}} \frac{\delta S_m}{\delta \tilde{g}_{\mu\nu}}$$

and define the g -frame matter tensor by

$$T_{\rho\sigma}^{(m,g)} = -\frac{2}{\sqrt{-g}} \frac{\delta S_m}{\delta g^{\rho\sigma}}.$$

At fixed Φ , the covariant metric variation satisfies

$$\delta \tilde{g}_{\mu\nu} = \delta g_{\mu\nu} = -g_{\mu\rho} g_{\nu\sigma} \delta g^{\rho\sigma}.$$

Therefore

$$T_{\rho\sigma}^{(m,g)} = \frac{\sqrt{-\tilde{g}}}{\sqrt{-g}} g_{\rho\mu} g_{\sigma\nu} \tilde{T}^{\mu\nu} = \sqrt{\Delta_D} g_{\rho\mu} g_{\sigma\nu} \tilde{T}^{\mu\nu} \quad (\Delta_D > 0).$$

There is no independent tensor $T_{\mu\nu}^{(\text{dis})}$ in this baseline bookkeeping: the pure-disformal effect is already contained in the relation between the two matter tensors and in $\mathcal{J}_m^{(D)}$.

Status: Derived from the displayed matter coupling and stress-tensor conventions.

Proposition 17.6 (Baseline metric field equation). For the lower-order action consisting of the Einstein–Hilbert term, the canonical compact scalar, matter coupled through $\tilde{g}_{\mu\nu}$, and the explicitly declared source sectors, variation with respect to $g^{\mu\nu}$ gives

$$M_{\text{Pl}}^2 G_{\mu\nu} = T_{\mu\nu}^{(\Phi)} + T_{\mu\nu}^{(m,g)} + T_{\mu\nu}^{(\text{top},g)} + T_{\mu\nu}^{(\Phi F)} + T_{\mu\nu}^{(>4)}.$$

The pure-disformal metric contains lower-index derivatives $\partial_\mu \Phi$ and hence produces no additional metric-derivative term at fixed Φ beyond $T_{\mu\nu}^{(m,g)}$ above. Explicit expressions for $T_{\mu\nu}^{(\text{top},g)}$ and $T_{\mu\nu}^{(>4)}$ require the unspecified actions and cannot be manufactured from the baseline.

The equation is Einstein-like because the Einstein–Hilbert term is an explicit postulate. It is not a derivation of gravity from topology or organization.

Status: Derived for the declared lower-order action; **Open** for the unspecified operator sectors and for emergent-gravity interpretations.

Proposition 17.7 (Energy exchange and Bianchi consistency). On the matter equations of motion, diffeomorphism invariance of $S_m[\tilde{g}, \psi_m]$ gives

$$\tilde{\nabla}_\mu \tilde{T}^\mu{}_\nu = 0$$

in the physical matter frame. The g -frame matter tensor is not separately conserved on a nontrivial scalar background. For the scalar–matter subsystem, the diffeomorphism identity and the derived scalar source give

$$\nabla^\mu T_{\mu\nu}^{(m,g)} = \mathcal{J}_m^{(D)} \partial_\nu \Phi,$$

while

$$\nabla^\mu T_{\mu\nu}^{(\Phi)} = (\Box_g \Phi - V'(\Phi)) \partial_\nu \Phi.$$

After the complete scalar equation and the source-sector equations are used, the total right-hand side of the metric equation is covariantly conserved, as required by $\nabla^\mu G_{\mu\nu} = 0$. Neither the scalar tensor nor the g -frame matter tensor is generally conserved in isolation.

Status: Derived for the scalar–matter exchange, conditional on the matter equations and on diffeomorphism invariance of every included source sector.

Proposition 17.8 (Stability scope of the V15.3 baseline). The isolated canonical scalar has a positive two-derivative kinetic coefficient and propagates on the g -cone with unit sound speed. Matter propagates on the $\tilde{g}$ -cone when its action is hyperbolic with respect to that metric. The necessary algebraic metric condition is $\Delta_D > 0$; the inverse and matter volume element become singular as $\Delta_D \rightarrow 0$.

These statements do not prove hyperbolicity or absence of ghosts for the full coupled EFT. The optional G_2 result above applies only to its isolated truncation. A complete quadratic action including gravity, matter, G_3 , higher operators, and background mixing is not specified.

Status: Derived for the isolated canonical sector and metric invertibility; full perturbative stability is **Open** — **additional physical input required**.

Chapter 18

Spacetime Embedding, Spin, and Physical Time Audit

18.1 Winding sectors as physical configurations

Volume 0 proves that a compact map $T^2 \rightarrow S^1$ has a conserved winding pair under continuous nonsingular deformations and identifies the harmonic representative of each sector. The frozen ontology takes T^2 to be internal configuration space, not physical space and not an extra spacetime dimension. The four-dimensional action in Chapter 17, however, does not specify the bundle or dynamical map relating that configuration space to the spacetime field $\Phi(x)$.

Open Problem 18.1 (Physical realization of winding). To turn (w_θ, w_ϕ) into a four-dimensional conserved particle label, one must provide a map or bundle construction relating spacetime configurations to the internal torus, finite-energy boundary conditions, and a covariant current whose charge equals the selected winding. Without that construction, the mathematical winding sectors are not yet particle states.

Status: Open — requires new physical input.

18.2 Intrinsic spin structures versus four-dimensional fermions

The intrinsic result of Chapter 3 and Appendix 14.8 is that the flat two-torus has four spin structures and the corresponding two-dimensional Dirac spectra can be written explicitly. This does not supply a representation of $\text{Spin}(1,3)$, a four-component Lorentzian spinor, or a map from winding number to four-dimensional helicity.

Open Problem 18.2 (Winding–spin and spin–statistics maps). A completed construction requires: (i) a declared four-dimensional spin bundle and Clifford action; (ii) a representation map from torus modes or TTA data to four-dimensional spinor states; (iii) a Lorentz-covariant local quantum field algebra; and (iv) positivity, locality/microcausality, and a stable vacuum so that a spin–statistics theorem can be applied. Antiperiodicity on one torus cycle alone establishes none of these four items.

Status: Open — requires new physical input.

18.3 Emergent order parameter versus proper time

Volume 0 constructs monotone order coordinates from an assumed monotone functional. If H_{topo} is differentiable along a trajectory and $dH_{\text{topo}}/d\lambda < 0$, then

$$t_H(\lambda) = -\ln \frac{H_{\text{topo}}(\lambda)}{H_{\text{ref}}}$$

is monotone because

$$\frac{dt_H}{d\lambda} = -\frac{1}{H_{\text{topo}}} \frac{dH_{\text{topo}}}{d\lambda} > 0.$$

Any strictly increasing function of t_H is equally admissible, so this calculation supplies an ordering but no unique duration, clock rate, metric proper time, or causal cone.

Status: Conditional consequence of the TEMP monotonicity hypothesis. The emergence of physical Lorentzian time is *Open* — *requires new physical input*: a Lorentzian state space, causal dynamics, and an operational clock prescription.

18.4 Lorentzian causal structure

The algebraic signature conditions in Chapter 17 ensure only that the disformal tensor can remain a Lorentzian metric. Matter characteristics are defined by $\tilde{g}_{\mu\nu}$, gravitational characteristics by the metric sector of the chosen action, and scalar characteristics by the principal symbol of its perturbation equation. Equality or nesting of these cones is not a consequence of the determinant condition.

Status: Conditional at the metric level; Open — **requires new physical input at the dynamical level.**

18.5 Perturbative content of the baseline

Proposition 18.1 (Scalar and tensor perturbations in the minimal truncation). On a specified solution of the source-free minimal scalar theory, a scalar perturbation obeys the principal equation

$$(\Box_g - V''(\Phi_{\text{bg}})) \delta\Phi = 0$$

up to background-curvature and metric-constraint mixing. Its principal part is the g -frame wave operator, so the isolated two-derivative scalar has $c_{s,g}^2 = 1$ and no ghost when its canonical kinetic coefficient is positive.

Because the gravitational action is the Einstein–Hilbert action with constant M_{Pl} , the transverse-traceless quadratic tensor action on a flat FLRW background is

$$S_T^{(2)} = \frac{M_{\text{Pl}}^2}{8} \int dt d^3x a^3 \left[\dot{h}_{ij} \dot{h}^{ij} - \frac{\partial_k h_{ij} \partial^k h^{ij}}{a^2} \right],$$

when anisotropic source terms are omitted. Thus

$$\ddot{h}_{ij} + 3H\dot{h}_{ij} - \frac{\nabla^2}{a^2} h_{ij} = 0,$$

with two tensor polarizations and $c_{T,g} = 1$ in the baseline g frame.

The disformal matter coupling can alter source terms and the operational speed measured with matter clocks. For the homogeneous lapse $\tilde{dt} = N_D dt$, a g -null tensor ray has matter-frame speed $c_{T,\text{matter}} = 1/N_D$, whereas matter null rays have unit speed in the $\tilde{g}$ frame. On a constant scalar background, $N_D = 1$ and the two cones coincide.

Status: Derived for the minimal source-free principal operators. Scalar–metric mixing, matter perturbations, lensing potentials, gravitational slip, and numerical wave phases on nontrivial backgrounds are **Open** — **additional physical input required**.

Proposition 18.2 (Conditional Newtonian and equivalence-principle limit). Consider a constant background $\partial_\mu \Phi_0 = 0$ and neglect scalar stress, optional higher operators, and unspecified source sectors. Then

$$\tilde{g}_{\mu\nu} = g_{\mu\nu}$$

through linear order, because the pure-disformal correction is quadratic in scalar perturbation gradients. The linear weak-field metric equation is then the general-relativistic one with Newton constant G_N , and the baseline gives the conditional values $\gamma_{\text{PPN}} = \beta_{\text{PPN}} = 1$ in this restricted limit.

Universal coupling of all test matter to one $\tilde{g}_{\mu\nu}$ implements the metric weak equivalence principle on any admissible background. The strong equivalence principle, fifth-force bounds, screening, and PPN parameters for a nonzero scalar gradient require an explicit background solution and compact source boundary conditions; they are not fixed by the rank-one identity.

Status: **Derived conditionally** for the constant-background minimal limit; otherwise **Open**.

Chapter 19

Emergent Gravity

Hypothesis 19.1 (Emergent Gravity). *ATHENA proposes that the stress tensor of the scalar and its disformal matter coupling may produce a background-dependent gravitational response. The V15.3 action nevertheless contains the Einstein–Hilbert term as a postulate; it does not generate it from topology, Ω_{cand} , or an EM condensate. No scalar-dependent Newton constant is postulated in the active V15.3 theory.*

Proposition 19.1 (What the baseline action establishes about gravity). The action in Chapter 17 contains the Einstein–Hilbert term $M_{\text{Pl}}^2 R/2$ as an explicit physical postulate. Varying the specified lower-order part with respect to $g^{\mu\nu}$ gives the frame- g equation

$$M_{\text{Pl}}^2 G_{\mu\nu} = T_{\mu\nu}^{(\Phi)} + T_{\mu\nu}^{(m,g)} + T_{\mu\nu}^{(\text{top},g)} + T_{\mu\nu}^{(\Phi F)},$$

where each non-scalar tensor is defined by variation of its displayed action with respect to $g^{\mu\nu}$. Diffeomorphism invariance gives conservation of the total tensor on the complete equations of motion; the scalar tensor alone exchanges energy-momentum with matter whenever $\mathcal{J}_m^{(D)} \neq 0$.

For the conventional four-dimensional definition of $F_{\mu\nu}\tilde{F}^{\mu\nu}$, the density $\sqrt{-g}\Phi F\tilde{F}$ has no direct metric stress tensor; its physical effects enter through the gauge and scalar equations. A Maxwell stress tensor requires the missing or separately declared Maxwell kinetic sector.

Therefore the baseline EFT is a postulated gravitational scalar–matter theory; it does not derive the Einstein–Hilbert term from the vacuum condensate. A Sakharov calculation could instead generate such a term only after specifying the fluctuating field content, regulator, counterterms, and matching scale.

Status: Derived from the stated action. Gravity as an emergent phenomenon is *Open* — requires new physical input.

Proposition 19.2 (Newton-coupling audit). The coefficient of the active Einstein–Hilbert term fixes the constant g -frame coupling

$$G_N = \frac{1}{8\pi M_{\text{Pl}}^2}.$$

On a constant scalar background, $\partial_\mu \Phi_0 = 0$, one has $\tilde{g}_{\mu\nu} = g_{\mu\nu}$ and $\mathcal{J}_m^{(D)} = 0$ at linear order. If scalar stress and all optional operators are negligible, the weak-field equations reduce conditionally to the general-relativistic normalization G_N .

For a nonzero scalar gradient, the relation between sources measured with $\tilde{g}_{\mu\nu}$ and the g -frame potentials is background dependent. No unique $G_{\text{eff}}(\Phi, X)$ follows from the algebraic metric definition alone; it requires a solved weak-field boundary-value problem.

Status: Derived for the constant-background limit; **Open** for a general effective Newton constant.

Phenomenological Ansatz 19.1 (Topological Screening).

$$\alpha_{\text{eff}}(r) = \frac{\alpha_{\text{UV}}}{1 + \Xi_w(r)}.$$

Here Ξ_w is a dimensionless environmental statistic. The two-term beta function gives $\alpha_* = 48$ for $\gamma_{\text{str}} = 0$; the historical 89.4 is the separate calibration α_{cal} . The reported galactic value $\alpha_{\text{eff}}^{\text{gal}} = 0.36 \pm 0.03$ remains a fit target pending reproduction.

Hypothesis 19.2 (Sakharov-Type Induced Gravity). *In natural units, an induced-gravity calculation would have the scaling*

$$G_{\text{ind}}^{-1} = c_S M_{\text{UV}}^2,$$

where the dimensionless coefficient c_S depends on field content, regulator, and counterterms. ATHENA has not calculated c_S .

Remark 19.1. Emergent gravity is an ATHENA interpretation. It must reproduce precision tests of general relativity and provide a consistent quantum effective theory before claims about gravitons or the quantum-gravity problem can be decided.

Chapter 20

Cosmological Framework

Proposition 20.1 (Homogeneous pure-disformal cosmology). Let the g -frame background be

$$ds^2 = -dt^2 + a^2(t)d\mathbf{x}^2, \quad \Phi = \Phi(t).$$

Then $X = \dot{\Phi}^2/2$ and the physical metric is

$$d\tilde{s}^2 = -N_D^2 dt^2 + a^2(t)d\mathbf{x}^2, \quad N_D^2 = \Delta_D = 1 + \frac{\alpha \dot{\Phi}^2}{M_*^4}.$$

Physical proper time is $d\tilde{t} = N_D dt$. The physical scale factor is still a ; there is no conformal a_{eff} in a homogeneous background. The physical Hubble and acceleration rates are

$$\begin{aligned} \tilde{H} &= \frac{1}{a} \frac{da}{d\tilde{t}} = \frac{H}{N_D}, \\ \frac{1}{a} \frac{d^2 a}{d\tilde{t}^2} &= \frac{1}{N_D^2} \left(\dot{H} + H^2 - H \frac{\dot{N}_D}{N_D} \right). \end{aligned}$$

The scalar background equation obtained from the complete sourced equation is

$$\ddot{\Phi} + 3H\dot{\Phi} + V'(\Phi) = \mathcal{J}_m^{(D)} + \mathcal{J}_{\text{top}} + \frac{c_{\Phi F}}{4M_*} F_{\mu\nu} \tilde{F}^{\mu\nu},$$

with the right-hand side absent in the isolated minimal truncation.

Status: Derived kinematically and from the stated scalar equation. A solution for $a(t)$ and $\Phi(t)$ requires specified matter, source sectors, and initial data.

Proposition 20.2 (Spatial-gradient volume identity). For a configuration with spatial gradients, define $Q_D = \alpha |\nabla \Phi|^2 / (a^2 M_*^4)$. The induced spatial determinant and volume rate are

$$\det \tilde{\gamma} = a^6 (1 - Q_D), \quad \frac{1}{3} \frac{d}{dt} \ln \sqrt{\det \tilde{\gamma}} = H - \frac{1}{6} \frac{\dot{Q}_D}{1 - Q_D}.$$

This is distinct from the homogeneous lapse correction above. It establishes a kinematic relation for the pure-disformal rank-one spatial metric. It does not supply a Friedmann equation, an equation of state, an energy density, or a solution for $Q_D(t)$. In particular, the sign of the correction is not fixed without dynamics for Φ .

Status: Conditional consequence of the metric ansatz. Its use as a cosmological mechanism is *Open* — *requires new physical input* through background and perturbation solutions of one complete EFT.

Remark 20.1. “Involution” is the proposed interpretation of the correction term; it is not an alternative cosmological solution established by the determinant identity.

Hypothesis 20.1 (Phenomenological Fractional-Dimension Ansatz). *The model introduces the dimensionless parametrization*

$$n(z) = 2 + \frac{(z/z_n)^2}{1 + (z/z_n)^2}, \quad z_n = 0.80.$$

Here z_n is a phenomenological shape parameter whose provenance is not independently documented in the present archive. Deriving this expression from the declared EFT requires a covariant definition of effective dimension and a solution of the background field equations. The earlier expression involving $\dot{\Phi}^2/\Phi_0^2$ is not dimensionless without an additional time or mass scale.

20.1 Rapid early structure formation (JWST)

JWST observations motivate tests of early structure formation. Demonstrating an ATHENA enhancement requires solving the linear and nonlinear perturbation-growth equations and comparing the resulting mass function with survey selection functions and covariance. The previously quoted factor 10^{25} is therefore treated as an unverified estimate rather than a derived prediction.

Hypothesis 20.2 (Vacuum Evolution via TEMP).

$$\frac{dH_{\text{topo}}}{dt} \leq 0.$$

The powers-of-two sectors are postulated asymptotic states, while sectors such as $w = 3$ may appear as metastable resonances. Connecting this monotonicity to a thermodynamic arrow of time requires a microscopic dynamics and coarse-graining theorem.

Chapter 21

Observational Horizon and the Toroidal Equator Hypothesis

21.1 The Toroidal Equator and the Milky Way

In ATHENA, the toroidal field profile is written in dimensionally explicit form as

$$B_\phi(r, \theta) = B_0 \sin 2\theta e^{-r/L_B} (1 + a \sin^2 \theta).$$

Here B_0 is a phenomenological field-amplitude parameter, L_B is a phenomenological radial length scale, and a is a dimensionless shape parameter. The historical denominator 20 is the numerical representation of L_B in the radial unit chosen for that profile; the physical unit, normalization, and provenance of B_0 , L_B , and a are not independently documented in the present archive. For the displayed profile the factor $\sin 2\theta$ vanishes at $\theta = \pi/2$ and is extremal near $\theta = \pi/4$ and $3\pi/4$ before the additional factor is included. The location and width of any “equatorial band” must therefore be obtained by differentiating the complete profile and fitting its parameter a ; the Milky Way placement is a hypothesis.

For the angular factor $f(\theta) = \sin 2\theta(1 + a \sin^2 \theta)$,

$$f'(\theta) = 2 \cos 2\theta(1 + a \sin^2 \theta) + a \sin^2 2\theta.$$

Writing $x = \cos 2\theta$, the non-boundary stationary points satisfy

$$2ax^2 - (2 + a)x - a = 0, \quad x = \frac{2 + a \pm \sqrt{(2 + a)^2 + 8a^2}}{4a} \quad (a \neq 0),$$

with only roots in $[-1, 1]$ physically admissible. For $a = 0$, the extrema are at $\theta = \pi/4$ and $3\pi/4$. The exponential is dimensionless because r/L_B is a ratio of lengths. Retaining the historical numerical value 20 requires an explicit choice of radial unit, which is not supplied here.

Status: Derived from the proposed profile. The profile itself, its normalization, radial scale, and relation to Maxwell or magnetohydrodynamic equations remain phenomenological inputs.

21.2 The Eridanus Supervoid

ATHENA interprets the Eridanus underdensity as a possible polar-region feature. This is a qualitative correspondence until a sky map, selection function, and likelihood are compared with standard large-scale-structure explanations.

21.3 The Cosmic Dipole Misalignment

The residual dipole misalignment ($\theta_{\text{res}} \approx 5.40^\circ$, $p = 0.0022$) is a reported phenomenological target of the toroidal-equator hypothesis, pending independent reproduction and comparison with non-toroidal models.

21.4 M87* Toroidal Magnetic Field Observations

EHT polarization observations reveal ordered magnetic-field structure around M87*. Such observations are qualitatively compatible with toroidal-field models, but they do not uniquely identify the ATHENA mechanism.

21.5 Biological Resonance and Consciousness

Mitochondrial electromagnetic scales may motivate laboratory resonance tests, but no demonstrated coupling to the ATHENA vacuum field is established here.

Chapter 22

Summary

22.1 Proposed Physical Correspondences

This volume has formulated:

- The postulated EM layered vacuum and its hypothesized observational shadows (dark matter and dark energy).
- The conditional interpretation of the compact scalar as an effective vacuum degree of freedom; no condensate-density identity is assumed.
- A baseline covariant EFT action and optional higher-order operators.
- A hypothesis for emergent gravity from vacuum pressure.
- A phenomenological cosmological framework with a proposed expansion correction.
- The Toroidal Equator Hypothesis and its proposed observational consequences.
- The topological screening ansatz relating α_* to α_{eff} .
- Speculative biological analogies, separated from the mathematically audited core.

Part III

Volume II: Particle Physics

Chapter 23

Introduction and Mathematical Roadmap

This volume develops proposed correspondences with Standard Model gauge groups, fermionic structure, and particle masses. The missing algebraic, anomaly, and dynamical steps are stated explicitly rather than treated as completed derivations.

The proposed research chain is:

$$\begin{aligned} \text{Topology } (T^2, w) &\longrightarrow \text{Trefoil Geometry} \longrightarrow \text{Configuration Space} \\ &\longrightarrow \text{Lie Algebra} \longrightarrow \text{Gauge Group} \end{aligned}$$

No completed implication from knot geometry to a Standard Model gauge theory is claimed. Volume I supplies a candidate EFT rather than a finished dynamical completion.

For this audit, three logically distinct levels are used. A *mathematical representation* is an action of a group or Lie algebra on a declared state space. A *candidate correspondence* is a proposed matching between such mathematical data and a physical sector. A *physical gauge theory* additionally requires a local principal-bundle connection, covariant derivatives, charged matter representations, an invariant action, constraints, quantization, and anomaly consistency. No passage between these levels is assumed automatically.

Chapter 24

Physical Audit of Topological Tensor Ascent

The mathematically established inputs are the integer winding sectors, the tautological assignment $\mathcal{T}_w = V^{\otimes(w-1)}$ after a vector space V is chosen, the four intrinsic torus spin structures, and the corresponding two-dimensional Dirac spectra. The physical TTA postulate adds the proposed sector-to-particle interpretation.

No functor $F_w : \mathcal{C}_w \rightarrow \mathcal{T}_w$, Lorentz action on $\mathcal{T}_w$, internal-symmetry action, invariant inner product, or four-dimensional kinetic operator is specified. Consequently the rank label $w - 1$ has no demonstrated relation to spacetime spin, gauge representation, or particle statistics.

Status: the tensor-product order is derived after choosing V ; TTA as a physical construction is *Open* — *requires new physical input*. The minimum missing input is an explicit representation map with covariance, dynamics, and quantization, not an additional algebraic relabelling.

Chapter 25

Trefoil Geometry

The trefoil knot is the simplest non-trivial torus knot, embedded in T^2 with winding numbers $(2, 3)$. Its explicit parametrization is:

$$\mathbf{r}(t) = \begin{pmatrix} (2 + \cos 3t) \cos 2t \\ (2 + \cos 3t) \sin 2t \\ \sin 3t \end{pmatrix}, \quad 0 \leq t < 2\pi.$$

The induced metric on the knot is $ds^2 = g_{ij}dx^i dx^j$. The curvature and torsion are:

$$\kappa = \frac{|\mathbf{r}' \times \mathbf{r}''|}{|\mathbf{r}'|^3}, \quad \tau = \frac{(\mathbf{r}' \times \mathbf{r}'') \cdot \mathbf{r}'''}{|\mathbf{r}' \times \mathbf{r}''|^2}.$$

Proposition 25.1 (Regularity and induced line element). For the displayed curve, direct differentiation gives

$$|\mathbf{r}'(t)|^2 = 9 \sin^2 3t + 4(2 + \cos 3t)^2 + 9 \cos^2 3t = 25 + 16 \cos 3t + 4 \cos^2 3t.$$

The quadratic in $c = \cos 3t$ is increasing on $[-1, 1]$ and has minimum 13 at $c = -1$. Therefore $\mathbf{r}'(t) \neq 0$ and the parametrization is regular, with induced metric

$$ds^2 = (25 + 16 \cos 3t + 4 \cos^2 3t) dt^2.$$

This verifies the differential-geometric knot input but produces no internal color space or gauge connection.

Status: Proved for the displayed embedding.

Chapter 26

Candidate $SU(3)$ Correspondence

The trefoil–color correspondence is a proposed truncation, not a completed derivation. Volume 0 lists the required construction: a declared knot energy functional, its Hessian and zero modes, a closed compact Lie algebra, matter representations, anomaly cancellation, and a confinement mechanism. Until those steps are supplied, the eight Gell–Mann generators serve only as a target algebra for the proposed reduced sector.

Remark 26.1 ($SU(3)$ dependency audit). The trefoil supplies a knot type, a one-dimensional induced metric, and a knot group. None of these data canonically selects eight dynamical zero modes or a bracket isomorphic to $\mathfrak{su}(3)$. Writing the Gell–Mann matrices therefore supplies a mathematical target representation only. A physical $SU(3)_c$ theory remains *Open* — *requires new physical input*: a declared state space, eight generators obtained from the Hessian or constraints, closure and normalization of their bracket, local gauge fields, quark representations, anomaly consistency, and confinement dynamics.

Chapter 27

Candidate U(1) Correspondence

A compact scalar shift provides a candidate global U(1) current. Electromagnetism requires the additional local gauge connection, covariant derivative, charge normalization, and charged representations described in Volume 0. The $w = 2$ label is therefore a proposed sector assignment rather than a derivation of electric charge.

Proposition 27.1 (Continuous-shift audit). For the kinetic scalar theory with $V = 0$ and without explicit sources, the constant transformation $\delta\Phi = -f_a\epsilon$ has Noether current

$$j^\mu = f_a \nabla^\mu \Phi.$$

For the actual periodic potential and displayed sources,

$$\nabla_\mu j^\mu = f_a \square \Phi = f_a V'(\Phi) - f_a \left(\frac{c_{\Phi F}}{4M_*} F \tilde{F} + \mathcal{J}_m^{(D)} + \mathcal{J}_{\text{top}} \right),$$

using the V15.3 Volume I pure-disformal equation of motion. Hence the continuous shift current is not conserved in the full displayed model; the cosine potential retains only the compact identification by $2\pi f_a$.

Status: Derived from the stated action. The compact target and a global shift of the kinetic truncation are mathematical structures, not a derivation of local hypercharge or electromagnetism. A physical U(1) theory is *Open* — *requires new physical input* through a connection, representations, charge assignments, kinetic normalization, and anomaly matching.

Chapter 28

Candidate $SU(2)$ Correspondence

Hypothesis 28.1 (Metastable $w = 3$ Electroweak Sector). *The $w = 3$ sector is treated as a metastable resonance whose reduced dynamics may admit an $\mathfrak{su}(2)$ action and a chiral energy splitting*

$$\Delta E = E_{\text{left}} - E_{\text{right}} \neq 0.$$

The lifetime, generators, chiral representations, electroweak symmetry breaking, and anomaly cancellation remain to be derived.

Remark 28.1 ($SU(2)$, chirality, and parity audit). A two-level splitting $\Delta E \neq 0$ does not by itself define three generators with the commutators of $\mathfrak{su}(2)$. It establishes parity violation only after a parity operator is defined with $P|L\rangle = e^{i\delta}|R\rangle$ and the Hamiltonian is shown not to commute with P . Weak chirality further requires left-handed $\text{Spin}(1, 3)$ representations, a local $SU(2)$ connection, the correct doublets and singlets, symmetry breaking, and cancellation of gauge and global anomalies.

Status: Candidate correspondence. A physical $SU(2)_L$ theory and parity-violating dynamics are *Open* — requires new physical input.

Chapter 29

Spinors and Dirac

The flat torus admits four spin structures. ATHENA selects an antiperiodic boundary condition for its candidate fermionic sector, but the identification of that choice with $w = 2$ remains a hypothesis. A complete derivation must construct the torus Dirac spectrum and its four-dimensional Lorentzian embedding before using the action below.

The Dirac action is:

$$S_D = \int d^4x \sqrt{-g} \bar{\psi} (i\gamma^\mu \nabla_\mu - m) \psi.$$

Varying with respect to $\bar{\psi}$ yields the Dirac equation:

$$(i\gamma^\mu \nabla_\mu - m)\psi = 0.$$

Proposition 29.1 (What follows from the postulated four-dimensional Dirac action). The four-dimensional measure and first-order kinetic term require $[\psi] = 3/2$ and $[m] = 1$. Variation with respect to ψ gives the adjoint equation, and together the two equations imply conservation of the global-phase current

$$j^\mu = \bar{\psi} \gamma^\mu \psi, \quad \nabla_\mu j^\mu = 0.$$

This is the standard consequence of a postulated Lorentzian spinor action. The action is not obtained from the two-dimensional torus spectrum, and its mass parameter is not fixed by winding.

Status: Derived from the stated Dirac action. The TTA-to-spinor map, winding–spin relation, and spin–statistics correspondence are *Open — require new physical input* as specified in Chapter 18.

Chapter 30

Proposed Radial Mass Functional and Derrick Obstruction

Phenomenological Ansatz 30.1 (Dimensionally Closed Mass Formula). The proposed dimensionless three-dimensional radial functional is

$$I_w^{\text{rad}} \equiv 4\pi \int_0^\infty \left[\frac{1}{2} \left(\frac{d\hat{\Phi}}{d\tilde{r}} \right)^2 + \hat{V}(\hat{\Phi}) \right] \tilde{r}^2 d\tilde{r},$$

and the retained dimensional research parametrization is

$$m_w c^2 \stackrel{\text{ansatz}}{=} E_* I_w^{\text{rad}}.$$

This equation states only how a dimensionless integral would be converted into an energy if a nontrivial admissible solution existed. The earlier prefactor $f_a^2 R_0^2$ is dimensionless in natural units and cannot supply an energy.

Status: Conditional dimensional ansatz. It is not an established finite static soliton mass.

Proposition 30.1 (Complete Derrick-scaling test). For the exact radial functional above, write

$$T = 4\pi \int_0^\infty \frac{1}{2} (\hat{\Phi}')^2 \tilde{r}^2 d\tilde{r}, \quad U = 4\pi \int_0^\infty \hat{V}(\hat{\Phi}) \tilde{r}^2 d\tilde{r}.$$

Thus $I_w^{\text{rad}} = T + U$. For ordinary variations whose boundary term $4\pi \tilde{r}^2 \hat{\Phi}' \delta \hat{\Phi}|_0^\infty$ vanishes,

$$\delta I_w^{\text{rad}} = 4\pi \int_0^\infty \left[-\frac{d}{d\tilde{r}} (\tilde{r}^2 \hat{\Phi}') + \tilde{r}^2 \frac{d\hat{V}}{d\hat{\Phi}} \right] \delta \hat{\Phi} d\tilde{r}.$$

Thus any stationary radial profile must obey

$$\hat{\Phi}'' + \frac{2}{\tilde{r}} \hat{\Phi}' = \frac{d\hat{V}}{d\hat{\Phi}}.$$

For the Derrick variation, scale every candidate profile by

$$\hat{\Phi}_\lambda(\tilde{r}) = \hat{\Phi}(\lambda \tilde{r}), \quad \lambda > 0.$$

With $\rho = \lambda \tilde{r}$, one has $d\tilde{r} = d\rho/\lambda$ and $\tilde{r}^2 = \rho^2/\lambda^2$. The kinetic term scales as

$$\begin{aligned} T(\lambda) &= 4\pi \int_0^\infty \frac{1}{2} \left[\lambda \hat{\Phi}'(\lambda \tilde{r}) \right]^2 \tilde{r}^2 d\tilde{r} \\ &= 4\pi \lambda^2 \int_0^\infty \frac{1}{2} [\hat{\Phi}'(\rho)]^2 \frac{\rho^2}{\lambda^2} \frac{d\rho}{\lambda} = \lambda^{-1} T. \end{aligned}$$

The potential term scales as

$$\begin{aligned} U(\lambda) &= 4\pi \int_0^\infty \hat{V}(\hat{\Phi}(\lambda\tilde{r})) \tilde{r}^2 d\tilde{r} \\ &= 4\pi \int_0^\infty \hat{V}(\hat{\Phi}(\rho)) \frac{\rho^2}{\lambda^2} \frac{d\rho}{\lambda} = \lambda^{-3}U. \end{aligned}$$

Hence

$$I_w^{\text{rad}}(\lambda) = \lambda^{-1}T + \lambda^{-3}U,$$

and

$$\frac{dI_w^{\text{rad}}}{d\lambda} = -\lambda^{-2}T - 3\lambda^{-4}U, \quad \left. \frac{dI_w^{\text{rad}}}{d\lambda} \right|_{\lambda=1} = -T - 3U.$$

A stationary finite-energy configuration must therefore satisfy

$$T + 3U = 0.$$

For the canonical kinetic term, $T \geq 0$. For the nonnegative ATHENA cosine-type potential normalized to vanish at its vacuum, $U \geq 0$. Thus $T = U = 0$, so $\hat{\Phi}$ is constant almost everywhere and lies in a vacuum of $\hat{V}$. There is no nontrivial finite-energy static radial soliton.

The obstruction holds precisely when: (i) the configuration is static and sufficiently regular in three spatial dimensions; (ii) the energy contains only the displayed canonical quadratic gradient term and nonnegative local potential; (iii) the energy and boundary terms are finite; (iv) the scaled profiles remain admissible; and (v) no additional constraint or fixed charge forbids the scaling variation. It does not address a different functional, a time-dependent state, or the compact internal functional $I_w^{(T^2)}$. None of those alternatives is adopted here as a repair. The label w is also absent from the radial integrand and can enter only through unstated boundary or profile data.

Status: Proved obstruction under the stated assumptions. The claim that this current functional yields a nontrivial finite static particle soliton is **Invalid under the current functional**. A particle-mass derivation is an **Open Problem**.

Proposition 30.2 (Dependency of a numerical mass). For a future admissible mass functional with a nontrivial solution, the retained dimensional parametrization would give

$$m_w = \frac{E_*}{c^2} I_w^{\text{admissible}}, \quad \frac{\delta m_w}{m_w} = \frac{\delta E_*}{E_*} + \frac{\delta I_w^{\text{admissible}}}{I_w^{\text{admissible}}}$$

at first order. A numerical mass therefore depends on the chosen potential, stabilizing dynamics, winding boundary data, numerical solution, and the scale E_* . Fixing E_* or potential parameters with a measured particle mass makes that mass calibrated rather than predicted; only masses computed with no additional tuning could be out-of-sample predictions.

Status: Conditional algebraic consequence for a future admissible functional; not applicable to a nontrivial static solution of I_w^{rad} .

The historical numerical targets are:

The present archive does not state which masses, if any, fixed E_* , the potential, or boundary data. More fundamentally, the displayed radial functional has no nontrivial finite static solution under its stated assumptions. The table is therefore a record of proposed targets, not a mass derivation, reproduced fit, or out-of-sample agreement.

Table 30.1: Historical ATHENA mass targets; not derived from the obstructed radial functional.

Particle	w	Model target	Reference value	Status
Proton	1	938.272 MeV	938.272 MeV	Historical target only
Electron	2	0.511 MeV	0.511 MeV	Historical target only
Muon	4	105.66 ± 0.02 MeV	105.658 MeV	Historical target only
Tau	5	1.776 ± 0.005 GeV	1.777 GeV	Historical target only

Chapter 31

The Higgs Sector and Lepton Generations

Conjecture 31.1 (Generation Truncation). *ATHENA conjectures that strain growing approximately as w^2 eventually destabilizes higher winding sectors. Scaling alone does not locate a threshold at $w = 6$: a normalized binding energy, full energy functional, decay channels, and stability analysis are required. Exactly three lepton generations are therefore a target of the mechanism, not a theorem.*

Remark 31.1 (Generation-count dependency). The scaling statement $E_{\text{strain}} \propto w^2$ orders sectors once a positive proportionality coefficient is supplied, but it contains no decay threshold and no reason to terminate after three fermion families. A completed count requires the binding energy in every sector, allowed decay products, selection rules, and a stability eigenvalue calculation. The generation count is therefore *Open — requires new physical input*.

Hypothesis 31.1 (Collective Higgs Interpretation). *ATHENA proposes a collective breathing-mode interpretation. The historical relation*

$$m_H c^2 \propto \frac{\sqrt{\beta_0} \hbar c}{R_c}$$

is dimensionally correct but, for $R_c = 23.67$ kpc, gives an energy of order 10^{-28} eV rather than 125 GeV. Recovering the observed Higgs scale requires a separate microscopic length or an explicit dimensionless hierarchy; that new parameter must be calibrated and cannot be hidden in the proportionality symbol.

Proposition 31.1 (Higgs-scale dependency audit). Write the proportionality explicitly as

$$m_H c^2 = C_H \frac{\sqrt{\beta_0} \hbar c}{R_c}.$$

Using $\beta_0 = 0.1408139828 \dots$ and $R_c = 23.67189158 \dots$ kpc gives

$$\frac{\sqrt{\beta_0} \hbar c}{R_c} \simeq 1.01 \times 10^{-28} \text{ eV}.$$

Matching $m_H c^2 \simeq 125$ GeV would require $C_H \simeq 1.2 \times 10^{39}$. The formula is dimensionally correct, but the observed scale is not derived unless this hierarchy follows from an independently specified microscopic theory.

Status: Conditional dimensional relation; numerical Higgs mass *Open — requires new physical input*.

Chapter 32

Effective Field Theory and Renormalization

Proposition 32.1 (Fixed Point of the Truncated Ansatz). For the displayed two-term beta function, the non-zero algebraic root is

$$\alpha_* = \frac{2\pi(2 - \gamma_{\text{str}})}{\beta_0(2 - \beta_0)}.$$

Thus $\alpha_* = 48$ for $\gamma_{\text{str}} = 0$ and the equilibrium postulate. The value 89.4 is not produced by this beta function.

The beta function is:

$$\beta(\alpha) = \mu \frac{d\alpha}{d\mu} = \frac{\alpha^2}{4\pi}(2 - \gamma_{\text{str}}) - \frac{\alpha^3}{8\pi^2}\beta_0(2 - \beta_0).$$

Proposition 32.2 (RG dependency and coupling-identification audit). Let α_{RG} denote the variable in the displayed beta function and α_{dis} the coefficient in the disformal metric when their roles must be distinguished. The algebra of Chapter 4 and Appendix 14.9 proves the fixed points, phase portrait, implicit flow, and Lambert- W inversion of the two-term function once its coefficients are accepted. It does not derive those coefficients from the four-dimensional EFT.

Moreover, setting $\alpha_{\text{RG}} = \alpha_{\text{dis}}$ is a matching hypothesis: the source contains no loop calculation showing that the disformal operator renormalizes with the displayed beta function, no renormalization scheme, and no operator-mixing matrix. The reported galactic fit target $\alpha_{\text{eff}}^{\text{gal}} = 0.36 \pm 0.03$ is a third, environment-dependent quantity.

Status: $\alpha_* = 48$ is analytically derived from the truncated beta ansatz and its parameter postulates. Identification with a physical EFT coupling and the full-EFT RG coefficients are *Open — require new physical input*.

Chapter 33

Experimental Targets and Tests

Table 33.1: Summary of ATHENA particle-physics targets with experimental counterparts.

Target or benchmark	Equation	Value	Experiment	Timeline
Muon mass	m_μ	$105.66 \pm 0.02 \text{ MeV}$	PDG	Current
Muon $g - 2$	Δa_μ	$\sim 8 \times 10^{-10}$	Fermilab	Current
Higgs coupling	g_{hWW}	$g_{\text{SM}}(1 - O(Z_0))$	HL-LHC	2029
Electron EDM	d_e	$< 10^{-34} e \cdot \text{cm}$	ACME	Ongoing
Axion coupling	$g_{a\gamma\gamma}$	$\sim 10^{-14} \text{ GeV}^{-1}$	ADMX	Ongoing

Proposition 33.1 (Dependency audit of the particle-physics targets). The numerical entries in the table have the following status.

- The muon mass depends on an unarchived profile, potential, stabilizing dynamics, boundary data, and E_* ; it is a reported target pending reproduction.
- No displayed loop integral or Wilson coefficient yields $\Delta a_\mu \sim 8 \times 10^{-10}$; the value is proposed.
- $g_{hWW} = g_{\text{SM}}[1 - O(Z_0)]$ fixes neither the coefficient nor its sign and is therefore a scaling ansatz, not a numerical prediction.
- The electron-EDM bound requires a CP-violating operator basis and loop matching that are not supplied; it is a proposed threshold.
- Comparing $c_{\Phi F} \Phi F \tilde{F} / (4M_*)$ with the conventional normalization $g_{\Phi\gamma\gamma} \Phi F \tilde{F} / 4$ gives the algebraic relation $g_{\Phi\gamma\gamma} = c_{\Phi F} / M_*$. The value $10^{-14} \text{ GeV}^{-1}$ is not fixed because neither $c_{\Phi F}$ nor M_* is determined by the preceding construction.

Status: the normalization relation for $g_{\Phi\gamma\gamma}$ is derived from the postulated operator; all listed numerical particle targets remain conditional, calibrated, or proposed. Their calculations are *Open* — require new physical input.

Chapter 34

Summary

Volume II has formulated the following research targets:

- a candidate trefoil–color correspondence targeting $SU(3)_c$;
- a candidate compact-shift sector targeting $U(1)_Y$;
- a metastable $w = 3$ resonance targeting chiral $SU(2)_L$;
- a winding–spin correspondence requiring a Dirac and spin–statistics construction;
- conditional particle-mass integrals requiring calibrated scales and uncertainty propagation;
- an emergent collective scalar-mode interpretation of the Higgs sector; and
- a conjectured generation-count mechanism requiring a dynamical stability proof.

Part IV

Volume III: Cosmology and Observations

Chapter 35

Introduction

This volume compares ATHENA hypotheses with observations and states a falsification roadmap. Compatibility, parameter fitting, and model-discriminating evidence are kept distinct.

For every numerical statement, the dependency chain is recorded in the order

observable $\longleftarrow$ equation $\longleftarrow$ model parameters $\longleftarrow$ calibrations and independent inputs.

A value is called *derived* only when all arrows are supplied without using that value as an input. Values copied from data are *inherited*; parameters optimized on data are *fitted*; normalizations fixed to a reference datum are *calibrated*; and values without an archived calculation are *proposed*.

Chapter 36

Galactic Dynamics

Postulate 36.1 (Rotation Velocity). The rotation velocity of galaxies in ATHENA is given by:

$$v^2(r) = \frac{GM(r)}{r} + \sqrt{GM(r)a_0\mathcal{F}(r/R_c)},$$

with:

$$\mathcal{F}(y) = \beta_0 [1 - e^{-y}(1 + y)], \quad y = r/R_c.$$

Proposition 36.1 (Analytical consequences of the rotation postulate). For $y \geq 0$ and the selected positive branch $\beta_0 > 0$,

$$\mathcal{F}(0) = 0, \quad \mathcal{F}'(y) = \beta_0 y e^{-y} \geq 0, \quad \lim_{y \rightarrow \infty} \mathcal{F}(y) = \beta_0.$$

Expanding at the origin gives

$$\mathcal{F}(y) = \beta_0 \left(\frac{y^2}{2} - \frac{y^3}{3} + O(y^4) \right).$$

The second term in v^2 is dimensionally consistent because $[GMa_0] = L^4 T^{-4}$ and $\mathcal{F}$ is dimensionless. If the enclosed baryonic mass approaches a finite value M_b at large radius, the postulate implies

$$v^2 \longrightarrow \sqrt{GM_b a_0 \beta_0}, \quad v^4 \longrightarrow GM_b a_0 \beta_0.$$

This is a conditional baryonic mass–velocity scaling derived from the rotation law, not from the covariant EFT.

Status: Derived from the stated postulate.

Proposition 36.2 (Parameter and conservation-law audit of the rotation law). The dependency chain is

$$v(r) \leftarrow \{M(r), \mathcal{F}(r/R_c), a_0\} \leftarrow \{\beta_0, R_c, a_0\} \leftarrow \{L_*, \kappa_a, \alpha_{\text{eff}}^{\text{gal}}\}.$$

Here β_0 follows from the equilibrium postulate, R_c is calibrated with L_* , and a_0 requires both κ_a and $\alpha_{\text{eff}}^{\text{gal}}$. The baryonic profile $M(r)$ is an independent galaxy-by-galaxy input. Thus the dimensional form and asymptotics are derived after the rotation postulate, while the normalization is calibrated or supplied by the reported fit claim.

The algebraic circular-speed law does not specify a covariant stress tensor or a relativistic equation of motion. It therefore cannot by itself establish energy-momentum conservation, gravitational-wave propagation, or lensing. Those consequences are *Open* — *require new physical input* through solutions of the complete metric and scalar equations.

Proposition 36.3 (Status of R_c , f_a , and a_0). The Stage 1 definitions are

$$R_c = L_* \frac{10}{3\beta_0}, \quad f_a = \kappa_f \frac{\hbar c}{R_c}, \quad a_0 = \kappa_a \frac{c^2}{R_c} \frac{\beta_0}{2\pi\alpha_{\text{eff}}^{\text{gal}}}.$$

Their dimensions are respectively length, energy, and acceleration. Their numerical dependency chain is

$$\begin{aligned} &\beta_0 \text{ (equilibrium postulate),} & L_* \text{ (length calibration),} \\ &\kappa_f, \kappa_a \text{ (normalizations),} & \alpha_{\text{eff}}^{\text{gal}} \text{ (galactic fit target).} \end{aligned}$$

Accordingly R_c is calibrated, f_a is calibrated from R_c and κ_f , and a_0 is calibrated from R_c , κ_a , and $\alpha_{\text{eff}}^{\text{gal}}$. None is an independent parameter-free prediction of topology. The exact logarithmic sensitivities are given in Appendix 14.3.

Status: Conditional definitions with derived substitutions and dimensions.

This formula has two components: the Newtonian contribution from baryonic matter and the toroidal vacuum contribution.

Table 36.1: Reported SPARC fit targets; pending independent reproduction.

Metric	Value
χ^2	731.1
$\alpha_{\text{eff}}^{\text{gal}}$	0.36 ± 0.03
χ^2/dof	Not computable without the declared data vector and parameter count

The quoted SPARC values refer to a proposed 175-galaxy analysis. The current source package does not contain the catalogue, covariance, likelihood, or code, so these numbers are not independently reproduced here. Moreover R_c depends on L_* and f_a depends on κ_f ; they are not fixed by topology alone. No statement is made that all 175 galaxies were successfully reanalyzed in the present audit.

Chapter 37

Cosmological Observations

37.1 Phenomenological Hubble-Tension Ansatz

ATHENA proposes the following phenomenological parametrization of the Hubble tension:

$$H_0^{\text{early}} \simeq 67.4 \text{ km/s/Mpc}, \quad H_0^{\text{local}} \simeq 73.2 \text{ km/s/Mpc}.$$

ATHENA hypothesizes that some or all of the discrepancy is represented by the redshift-dependent phenomenological form

$$H_{\text{obs}}(z) = H_{\Lambda}(z) [1 + \alpha_H \Omega_{\text{coll}}(z)], \quad \Omega_{\text{coll}}(z) = \frac{1}{1 + (z/z_c)^2}, \quad z_c = 0.35.$$

Here z_c is the phenomenological transition-redshift parameter declared in Volume 0; its provenance is not independently documented in the present archive.

Proposition 37.1 (Hubble-ansatz normalization audit). Let $z_c = 0.35$. The ansatz implies

$$\frac{H_{\text{obs}}(0)}{H_{\Lambda}(0)} = 1 + \alpha_H, \quad \frac{H_{\text{obs}}(z_c)}{H_{\Lambda}(z_c)} = 1 + \frac{\alpha_H}{2}, \quad \lim_{z \rightarrow \infty} \frac{H_{\text{obs}}}{H_{\Lambda}} = 1.$$

For $z > 0$,

$$\frac{d\Omega_{\text{coll}}}{dz} = -\frac{2z/z_c^2}{[1 + (z/z_c)^2]^2} < 0.$$

If the displayed values are identified as

$$H_{\Lambda}(0) = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad H_{\text{obs}}(0) = 73.2 \text{ km s}^{-1} \text{ Mpc}^{-1},$$

consistency requires

$$\alpha_H = \frac{73.2}{67.4} - 1 = 0.086053412 \dots \simeq 0.0861.$$

By contrast, inserting the distinct reported galactic fit target $\alpha_{\text{eff}}^{\text{gal}} = 0.36$ gives $H_{\text{obs}}(0) = 1.36(67.4) \simeq 91.7 \text{ km s}^{-1} \text{ Mpc}^{-1}$. Therefore the same numerical value cannot simultaneously reproduce the two displayed H_0 values under this ansatz without an additional mapping or a redefinition of $H_{\Lambda}(0)$.

Status: Derived algebraic normalization within a phenomenological ansatz. The value $\alpha_H = 0.086053412 \dots$ is a conditional calibration from the two displayed H_0 inputs, not a derivation from the EFT. An EFT derivation remains *Open* — *requires new physical input*.

37.2 Fractional Dimension Evolution

$$n(z) = 2 + \frac{(z/z_n)^2}{1 + (z/z_n)^2}, \quad z_n = 0.80.$$

The transition redshift z_n is a phenomenological shape parameter. Its provenance is not independently documented in the present archive. At $z \gtrsim 10$, the ansatz approaches $n(z) \simeq 3$. The value of $n(z)$ follows algebraically from the ansatz, but a structure-growth factor does not. The previously quoted enhancement of 10^{25} has no archived perturbation calculation and is not used as a prediction in V15.3.

37.3 Recent Model-Compatible Observations (2026)

Table 37.1: Observation-to-test audit for ATHENA.

Observation	ATHENA hypothesis or target	Competing explanation	Discriminating test
Hubble discrepancy	Conditional $H_{\text{obs}}(z)/H_{\Lambda}(z)$ ansatz	Calibration systematics or other early/late-time models	Joint pre-registered fit to BAO, SNe, CMB, lenses, and distance ladder
DESI DR2 trends	Vacuum-layer hypothesis; no specific $w_{\text{DE}}(z)$ derived	Flexible dark-energy or modified-gravity models	ATHENA-specific $w(z)$ and growth/lensing covariance
High-redshift galaxies	Proposed growth interpretation of $n(z)$	Selection, stellar populations, feedback, or alternative cosmology	End-to-end luminosity/mass function with survey completeness
Ordered magnetic fields	Declared $B_{\phi}(r, \theta)$ profile	Plasma dynamos and MHD accretion/jet models	Bayesian image/polarization fit against MHD alternatives
Synthetic crystal rotation	Analogy to rotational wave exchange; ATHENA dependence open	Standard time-modulated electrodynamics	Dependence unique to the ATHENA disformal coupling
IXPE pulsar polarimetry	Candidate correspondence with toroidal/ordered morphology	Standard synchrotron transport in pulsar-wind nebulae	Quantitative polarization map and evidence ratio
SPARC rotation curves	Declared screening function $\mathcal{F}$	Dark halos, MOND-family laws, baryonic systematics	Public likelihood and out-of-sample predictive comparison
Cosmic dipole	Residual 5.40° directional target	Survey masks, local structure, kinematic or intrinsic dipoles	Common-mask multi-survey estimator fixed in advance

Chapter 38

Background, Lensing, and Magnetic-Field Audit

38.1 Homogeneous scalar cosmology

For the source-free minimal scalar on a flat FLRW background, the stress tensor derived in Volume I gives

$$\rho_\Phi = \frac{1}{2}\dot{\Phi}^2 + V(\Phi), \quad p_\Phi = \frac{1}{2}\dot{\Phi}^2 - V(\Phi), \quad w_\Phi = \frac{\dot{\Phi}^2/2 - V}{\dot{\Phi}^2/2 + V}.$$

The homogeneous equation and its conservation identity are

$$\ddot{\Phi} + 3H\dot{\Phi} + V'(\Phi) = 0, \quad \dot{\rho}_\Phi + 3H(\rho_\Phi + p_\Phi) = 0.$$

Thus negative pressure requires potential domination; it is not implied by the label “CMB layer.” In the complete model, the disformal matter source and optional operators modify both equations and permit energy exchange between sectors. No solution $H(z)$, $w(z)$, or perturbation transfer functions follows until those terms and initial data are fixed.

The physical matter metric is instead

$$ds^2 = - \left(1 + \frac{\alpha\dot{\Phi}^2}{M_*^4} \right) dt^2 + a^2(t)d\mathbf{x}^2.$$

Thus $\tilde{dt} = N_D dt$ and $\tilde{H} = H/N_D$, with $N_D^2 = 1 + \alpha\dot{\Phi}^2/M_*^4$. This lapse relation does not reproduce the phenomenological factor $1 + \alpha_H\Omega_{\text{coll}}(z)$ without a background solution and an independently derived parameter matching.

Status: Derived for the source-free minimal truncation; full cosmological evolution Open — requires new physical input.

The homogeneous same-epoch DM/DE obstruction and the instability of the positive cosine-potential maximum are proved respectively in Propositions K.5 and K.6.

38.2 Gravitational dynamics and lensing

In a scalar-perturbed cosmological metric one may write schematically

$$ds^2 = -(1 + 2\Psi)dt^2 + a^2(t)(1 - 2\Phi_N)d\mathbf{x}^2.$$

Slow circular motion primarily constrains the potential Ψ , whereas weak lensing constrains a line-of-sight combination proportional to $\Psi + \Phi_N$. The galactic rotation postulate supplies neither

Φ_N nor a gravitational-slip relation between the two potentials. Consequently rotation-curve agreement cannot be converted into a lensing, cluster, or relativistic-gravity prediction without solving the covariant field equations.

Status: Open — requires new physical input.

38.3 Toroidal magnetic profile

Treat the proposed profile as an axisymmetric, purely azimuthal field in spherical coordinates, $\mathbf{B} = B_\phi(r, \theta)\mathbf{e}_\phi$. Then

$$\nabla \cdot \mathbf{B} = \frac{1}{r \sin \theta} \frac{\partial B_\phi}{\partial \phi} = 0,$$

so the magnetic Gauss constraint is satisfied by axisymmetry. Its curl is

$$(\nabla \times \mathbf{B})_r = \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta B_\phi), \quad (\nabla \times \mathbf{B})_\theta = -\frac{1}{r} \frac{\partial}{\partial r} (r B_\phi), \quad (\nabla \times \mathbf{B})_\phi = 0.$$

Thus Maxwell's equations would require a corresponding current and, for a time-dependent configuration, an electric field. The ATHENA profile does not specify those sources, its amplitude, radial unit, boundary conditions, or an MHD evolution law.

Status: The divergence constraint is proved for the axisymmetric profile; the magnetic-field profile is a phenomenological ansatz. Its dynamics are Open — require new physical input.

Chapter 39

Observational Status: August 2026

39.1 Assessment Framework

This chapter surveys the observational landscape as of August 2026 and classifies each development according to its relation to ATHENA. We distinguish three epistemic categories:

- Definition 39.1** (Evidence Classification). 1. **Model-Compatible Observation.** An observation whose qualitative or quantitative features are compatible with an ATHENA hypothesis, conditional prediction, or proposed target, but may also be explained by competing models.
2. **Convergent Anomaly.** An observation reported as being in tension with a standard interpretation in a direction discussed by ATHENA, but which does not constitute a positive detection of the ATHENA mechanism.
3. **Pending Precision Test.** A forthcoming measurement for which ATHENA records a proposed threshold or benchmark (Chapter 41); its completeness must be assessed separately.

Remark 39.1 (Honesty of Status). Convergent anomalies and model-compatible observations do not, individually, confirm ATHENA. Decisive evidence requires a pre-specified quantitative test that discriminates ATHENA from competing models. This chapter therefore avoids interpreting compatibility as validation.

39.2 Model-Compatible Observations (2026)

39.2.1 Synthetic crystal rotation in spacetime metamaterials

Liberal and Manjavacas developed synthetic crystal rotation using spatiotemporal modulation, obtaining frequency/spin-angular-momentum locking and rotational wave-exchange effects without mechanically rotating a bulk crystal¹. This is a metamaterial/electrodynamics result. It is not a laboratory confirmation of the black-hole Penrose process and does not test ATHENA unless an independently derived disformal dependence is measured.

39.2.2 Vacuum birefringence in a magnetar

Phase- and energy-resolved X-ray polarimetry of the radio magnetar 1E 1547.0–5408 has been interpreted as evidence for magnetospheric vacuum birefringence². An independent analysis of

¹I. Liberal and A. Manjavacas, “Synthetic Crystal Rotation with Spacetime Metamaterials,” *Phys. Rev. Lett.* **136**, 146903 (2026), arXiv:2506.02495.

²R.E. Stewart *et al.*, “Vacuum birefringence in the polarized X-ray emission of a radio magnetar,” arXiv:2509.19446 (2025, revised 2026).

Table 39.1: Observations that are compatible with, but do not uniquely confirm, ATHENA mechanisms.

Observation	ATHENA mechanism	Status
Synthetic crystal rotation (metamaterial)	Rotational wave-exchange analogy	Context only; not an ATHENA test
Vacuum-birefringence interpretation (magnetar)	Vacuum as EM medium Φ	Compatible with QED; interpretation debated
IXPE pulsar field (PSR J1101–6101)	Toroidal magnetic structure	Relevant observation
EHT M87* / OJ 287 polarization	Toroidal B_ϕ profile	Qualitatively compatible
Cluster magnetic-field reconstruction	μG large-scale fields	Necessary, not sufficient
Quantum entanglement from sunlight	Polarization as organizational key	Contextual analogy

the same source found the data compatible with QED effects but not a compelling stand-alone smoking gun under its geometry model³. Either interpretation concerns standard strong-field QED; a separate discriminating observable is required for ATHENA.

39.2.3 Direct pulsar magnetic-field measurement (IXPE)

The Imaging X-ray Polarimetry Explorer mapped the projected magnetic-field geometry of the Lighthouse pulsar complex PSR J1101–6101 through X-ray polarimetry⁴. This constrains synchrotron transport and magnetic ordering, but comparison with ATHENA requires a quantitative fit against pulsar-wind nebula models.

39.2.4 Toroidal fields at event horizons (EHT)

Polarization observations of M87* and OJ 287 constrain ordered magnetic structure near compact objects⁵. Toroidal field components are also generic outcomes of magnetized accretion and jet models; the specific ATHENA profile must be fitted rather than inferred from morphology.

39.2.5 Large-scale cluster magnetism

The MeerKAT Fornax Survey has characterized intracluster magnetic structure in Fornax⁶. Such measurements supply inputs for a future comparison, but the existence of cluster magnetism is not specific to ATHENA.

39.2.6 Quantum entanglement from sunlight

Experiments using sunlight in quantum-information protocols provide a contextual example of polarization-based state preparation. They do not test the ATHENA vacuum postulate and are excluded from the evidentiary ledger.

³R. Taverna *et al.*, “The long quest for vacuum birefringence in magnetars: 1E 1547.0–5408 and the elusive smoking gun,” arXiv:2601.15452 (2026).

⁴J.T. Dinsmore *et al.*, “IXPE Polarizations of the Lighthouse Pulsar, Trail, and Filament,” arXiv:2604.22914; *Astrophysical Journal* (2026).

⁵J.L. Gómez *et al.*, “Spatially resolved polarization swings in the supermassive binary black hole candidate OJ 287,” *Astronomy & Astrophysics* (2026).

⁶F. Loi *et al.*, “The MeerKAT Fornax Survey VII: Characterisation of the Fornax cluster’s magnetic field,” arXiv:2604.18338 (2026).

39.3 Convergent Anomalies and Open Cosmological Tensions

Remark 39.2. The following observations do not confirm ATHENA. Some may place pressure on particular Λ CDM parameterizations or astrophysical assumptions, while remaining open to conventional and alternative explanations. They are included as motivation for explicit comparative tests.

39.3.1 Dynamical dark energy (DESI)

DESI DR2 combined with CMB, supernova, and weak-lensing data strengthens hints that a time-dependent dark-energy parametrization can fit the joint data better than rigid Λ CDM⁷. The significance and inferred evolution depend on the data combination and parametrization. ATHENA must predict a specific $w(z)$ and correlated growth history before this becomes a test of its vacuum layer.

39.3.2 A check on one Hubble-tension systematic

JWST observations provide a strong check on Cepheid crowding in the distance ladder⁸. Excluding one systematic does not exclude every calibration or model explanation. The result motivates, but does not support uniquely, ATHENA’s redshift-dependent $H_{\text{obs}}(z)$.

39.3.3 Rapid early structure formation (JWST)

JWST has motivated renewed study of massive galaxies and black holes at high redshift. Stellar-population assumptions, selection functions, feedback, seed models, and cosmology all affect the inferred tension. ATHENA’s $n(z)$ ansatz has not yet been propagated through perturbation growth and survey selection; no numerical enhancement factor is claimed in V15.3.

Proposition 39.1 (Mathematical content of the fractional-dimension ansatz). With $z_n = 0.80$,

$$n(0) = 2, \quad \lim_{z \rightarrow \infty} n(z) = 3, \quad \frac{dn}{dz} = \frac{2z/z_n^2}{[1 + (z/z_n)^2]^2} \geq 0 \quad (z \geq 0).$$

At $z = 10$, the formula gives $n(10) \simeq 2.9936$. These bounds and monotonicity are derived from the phenomenological form. No covariant observable called dimension, no modified measure or field equation, and no map from $n(z)$ to a growth factor is defined.

Status: Derived properties of a phenomenological ansatz. Its physical interpretation and structure-growth consequences are *Open* — *require new physical input*.

39.3.4 Cosmic dipole anomaly

Reported number-count dipoles motivate tests of isotropy, but survey masks, flux calibration, local structure, and estimator choices remain competing explanations. The ATHENA value $\theta_{\text{res}} \approx 5.40^\circ$ is a reported analysis target pending a common-mask multi-survey analysis.

39.3.5 Cosmic birefringence

Analyses of CMB polarization have reported rotation estimators that can be modeled with pseudoscalar couplings. ATHENA contains the optional operator $c_{\Phi F} \Phi F_{\mu\nu} \tilde{F}^{\mu\nu} / (4M_*)$, but it has not yet predicted a calibrated rotation spectrum distinct from instrumental-angle systematics or other axion-like models.

⁷DESI Collaboration, “DESI DR2 Results: Extended Dark Energy Analysis,” arXiv:2503.14743 (2025), together with the DESI DR2 BAO papers.

⁸A.G. Riess *et al.*, arXiv:2401.04773 (2024).

39.3.6 MOND-type regularities

Galaxy regularities such as the radial-acceleration relation motivate models with a characteristic acceleration scale⁹. In ATHENA, a_0 depends on calibrated R_c and $\alpha_{\text{eff}}^{\text{gal}}$ as well as β_0 ; it is not derived from topology alone.

39.3.7 Geometric tension models

An anonymous 2026 Zenodo deposit proposed a geometric-tension fit to SPARC. Because authorship, catalogue choices, covariance, likelihood, and an archival reproduction are not established in this monograph, it is not used as evidence for ATHENA or against particle dark matter.

39.4 Pending Precision Tests

Table 39.2: Precision frontiers with ATHENA proposed benchmarks (August 2026 status).

Probe	2026 status	ATHENA proposed benchmark
Muon $g - 2$	Final measurement, 127 ppb	$\Delta a_\mu \sim 8 \times 10^{-10}$
Electron EDM (ACME III)	$d_e < 1.1 \times 10^{-29} \text{ e}\cdot\text{cm}$; toward 10^{-31}	$d_e < 10^{-34} \text{ e}\cdot\text{cm}$
Axion (ADMX)	No detection in searched bands	$g_{a\gamma\gamma} \sim 10^{-14} \text{ GeV}^{-1}$
Sterile neutrinos	No evidence (IceCube)	3σ PMNS deviation by 2030
Bullet Cluster	Residual-mass debate ongoing	$\Delta r \simeq 117 \text{ kpc}$

Remark 39.3 (Interpretation of Null Results). Null results in other new-physics searches do not by themselves favor ATHENA. Only a pre-specified ATHENA signal or exclusion region can update its status.

39.5 Synthesis

Remark 39.4 (What the 2026 Landscape Says). The August 2026 landscape motivates five comparison programs:

1. strong-field QED versus an additional ATHENA vacuum response;
2. rigid Λ CDM versus specified evolving-dark-energy models;
3. standard and alternative high-redshift growth histories;
4. MHD-generated versus ATHENA-specific magnetic profiles; and
5. survey/systematic versus cosmological explanations of dipoles.

None currently confirms ATHENA. The framework becomes evidentially competitive only after it supplies quantitative likelihoods and outperforms the stated alternatives on pre-registered tests.

⁹A. Hees *et al.*, *MNRAS* 455, 449 (2016); S.S. McGaugh *et al.*, *Phys. Rev. Lett.* 117, 201101 (2016).

Chapter 40

Astrophysical Tests

Open Problem 40.1 (V15.3 gravitational-wave phase prediction). The V15.3 baseline derives the g -frame tensor equation and the homogeneous matter lapse but supplies no binary solution, waveform map, propagation history, detector response, or likelihood. The historical V15.2 values 3×10^{-6} rad, 4.07×10^{-6} rad, and the five-event 3.5σ statement are therefore removed as active predictions and retained only as unreproduced historical targets. A V15.3 phase prediction requires a specified scalar background, source solution, and waveform calculation.

Phenomenological Ansatz 40.1 (Local-Lift Aharonov–Bohm Phase Benchmark).

$$\Delta\phi_{\text{total}} = \frac{q}{\hbar}\Phi_B + w \frac{\Delta L}{\lambda_C} \frac{\alpha\beta_0}{2\pi} \frac{\Phi}{f_a}, \quad \delta\phi \sim 10^{-9} \text{ rad/m}.$$

Here α denotes the bare disformal coefficient of the metric ansatz, not the Hubble normalization α_H or the galactic screened-fit target $\alpha_{\text{eff}}^{\text{gal}}$. In this expression Φ/f_a denotes a chosen real lift on a local patch or in the topologically trivial sector. The source does not derive a lift-independent phase difference or path observable, so the displayed expression is not a global observable on the compact field space. **Status: Phenomenological and local-lift only; a global compact completion is Open.**

Phenomenological Ansatz 40.2 (Proposed Dipole-Misalignment Benchmark).

$$\theta_{\text{res}} \approx 5.40^\circ, \quad p = 0.0022 \text{ (} 3\sigma \text{)}.$$

Phenomenological Ansatz 40.3 (Proposed Bullet-Cluster Offset Benchmark).

$$\tau/\tau_{\text{gas}} = Z_0\beta_0 = 0.033, \quad \Delta r \simeq 117 \text{ kpc}.$$

Proposition 40.1 (Dependency audit of the astrophysical targets). The four proposed benchmarks above have distinct derivational status.

1. The historical GW phase is dimensionless, but no modified waveform, source parameters, propagation equation, detector noise curve, or likelihood maps the V15.3 EFT to 4.07×10^{-6} rad. A five-event stack alone does not determine a 3.5σ significance without the individual signal-to-noise ratios and correlations. The numbers are historical unreproduced targets, not an active V15.3 prediction.
2. In the Aharonov–Bohm expression, both $q\Phi_B/\hbar$ and

$$w \frac{\Delta L}{\lambda_C} \frac{\alpha\beta_0}{2\pi} \frac{\Phi}{f_a}$$

are dimensionless. This verifies dimensional closure, but the second term is lift dependent and is valid only under the local/trivial-sector qualification stated in the ansatz. It is not derived from a particle Hamiltonian or path integral. The quoted 10^{-9} rad/m is a phase gradient and requires a specified path, field profile, particle, and parameter normalization.

3. The dipole angle and p -value are reported analysis outputs. No sky mask, estimator, covariance, null distribution, or forward model is archived, so they are inherited targets rather than derived predictions.
4. The Stage 1 values give the algebraic product

$$Z_0\beta_0 = \frac{5}{3}\beta_0^2 \simeq 0.03305,$$

so the dimensionless optical-depth ratio follows from the two normalization rules if the physical identification is accepted. No equation converts that ratio into $\Delta r \simeq 117$ kpc; the offset is proposed.

Status: dimensional checks and the product $Z_0\beta_0$ are derived from stated inputs. The numerical GW, interferometric, dipole, and cluster offset forecasts are *Open — require new physical input* and archived calculations.

Chapter 41

Falsification

Table 41.1: Proposed falsification tests and incomplete benchmarks for ATHENA V15.3.

Test	Platform	Threshold	Year
DESI DR3 α_H	DESI	$\alpha_H < 0.05$ (1σ)	2027
GW propaga- tion	LISA	Open: waveform-dependent V15.3 threshold re- quired	2035
Weak lensing	Euclid	Open: signed $\Delta\kappa$ prediction required	2028
Intracuster B-field	SKA	μG toroidal fields	2030
CMB seasonal dipole	Planck TOI	Proposed $\theta > 134^\circ$	2026–27
EHT toroidal shift	EHT + ngEHT	π -phase shift	2027–28
Neutrino anomaly	IceCube + KM3NeT	3σ deviation from PMNS	2030
Muon $g - 2$	Fermilab	Δa_μ deviation	Current
Higgs coupling	HL-LHC	g_{hWW} deviation	2029
Electron EDM	ACME	d_e detection	Ongoing

The thresholds and dates in this table are historical ATHENA test targets. Where no likelihood, derivation, or analysis configuration is archived, the number is a proposed benchmark rather than a validated forecast. A future calendar date alone does not define a falsification criterion.

In particular, the 134° dipole entry has no archived derivation, unambiguous observable definition, estimator, covariance, or decision rule in V15.3. It is therefore a proposed benchmark with an incomplete falsification criterion, not a validated observational threshold.

Proposition 41.1 (Which Stage 2 claims are presently falsifiable). A claim is operationally falsifiable only after its parameters and analysis rule are fixed independently of the test data. Under that standard:

- the rotation law and its screening function define a falsifiable functional form once $M(r)$, parameter priors, galaxy selection, and likelihood are frozen and tested out of sample;
- the Hubble-ratio ansatz defines a falsifiable redshift shape after its cosmological coupling is separated from the galactic one and recalibrated;
- the toroidal magnetic angular profile is falsifiable after its amplitude, radial scale, parameter a , source geometry, and comparison statistic are specified;

- the Aharonov–Bohm correction is in principle falsifiable after a Hamiltonian derivation fixes the coupling and experimental path dependence;
- the current GW, weak-lensing, dipole, EHT, neutrino, muon, Higgs, EDM, and cluster numbers are proposed benchmarks, not yet closed forecasts, because at least one equation-to-observable map or archived likelihood is missing.

The DESI threshold uses the explicitly cosmological parameter α_H ; no matching to either the bare disformal or galactic-fit parameter is assumed. The weak-lensing row does not state a predicted sign and interval for $\Delta\kappa$, so the inequality alone is not yet a model-specific exclusion region.

Status: Derived audit classification.

Chapter 42

Scope, Assumptions, and Limitations

42.1 Status of the Framework

ATHENA is presented as a mathematically controlled physical framework and research program with a specified lower-order theoretical core. It is not an experimentally established replacement for the Standard Model or general relativity. Its mathematical statements fall into four distinct classes: standard results imported from established mathematics, conditional results derived from stated assumptions, phenomenological ansatzes, and proposed empirical tests or conditional predictions. These classes must not be conflated.

42.2 Core Assumptions

The principal model assumptions are:

1. the relevance of a compact scalar field and toroidal configuration space to fundamental physics;
2. the proposed effective action and disformal coupling;
3. the Casimir and equilibrium ansatzes used to define β_0 ;
4. the identification of winding sectors with particle and gauge degrees of freedom; and
5. the mapping from the effective fields to astrophysical, cosmological, and biological observables.

All subsequent conclusions are conditional on some or all of these assumptions.

42.3 Outstanding Mathematical and Physical Tests

Before the framework can be regarded as a complete physical theory, it must provide:

- a full spectral calculation of the toroidal Casimir energy for specified fields, boundary conditions, and renormalization conventions;
- a proof of stability and absence of ghost or gradient instabilities in the effective action;
- explicit recovery of tested Standard Model and general-relativistic limits;
- derivations of gauge representations, interaction strengths, and masses without hidden parameter fitting; and
- a consistent treatment of uncertainties, selection effects, and nuisance parameters in every observational comparison.

42.4 Evidence and Model Comparison

Qualitative agreement is not sufficient for confirmation because many observations admit multiple explanations. Quantitative ATHENA predictions should therefore be specified before comparison with new data and evaluated with a documented likelihood. The relevant comparison is not only goodness of fit, but also performance relative to competing models after accounting for parameter count and prior assumptions.

42.5 Falsifiability Standard

A proposed test is genuinely discriminating only if it specifies the observable, uncertainty budget, analysis procedure, ATHENA prediction, and competing-model prediction in advance. Failure of a robust, model-specific threshold counts against the assumptions responsible for that prediction; successful agreement counts as evidence only when the same result is not generically expected from other models.

Final Conclusions

ATHENA Ultimate V15.3 presents a pure-disformal reformulation of a unified topo-geometric research framework connecting gauge structure, fermionic sectors, proposed particle-mass targets, gravitation, and cosmology. The present monograph defines conjectured mechanisms and a set of proposed empirical tests and conditional predictions. Establishing the framework as a physical theory will require completion of the mathematical checks in Chapter 42 and successful model-discriminating tests.

Chapter 43

Reproducibility Checklist

The repository address identifies the intended external code archive, but the current monograph package does not contain the referenced simulation scripts or data products. The numerical results below must therefore be treated as targets for independent reproduction rather than as reproduced results:

- SPARC likelihood and the reported fit target for $\alpha_{\text{eff}}^{\text{gal}}$;
- historical LISA phase-shift targets, which are not active V15.3 predictions;
- GRB 221009A time-profile fit — not implemented in the present version; no corresponding equation, dataset, or reproducible analysis is supplied; and
- cosmic-dipole inversion and its reported significance.

A citable release must complete the following manifest. No missing value has been inferred or invented in this audit.

Until these fields are supplied, $\chi^2 = 731.1$, $\alpha_{\text{eff}}^{\text{gal}} = 0.36 \pm 0.03$, mass tables, phase shifts, and dipole significances are reported targets pending independent reproduction.

Required item	V15.3 archive status
Repository URL	Author-supplied GitHub address; reachability is not proof of the archived calculation
Repository commit hash	Not yet archived
Release tag or DOI	Not yet archived
SPARC catalogue/data version	Not yet archived in this package
Data licences and SHA-256 checksums	Not yet archived
Covariance matrices and likelihood	Not yet archived
Priors and nuisance parameters	Not yet archived
Python version	Not yet archived
Dependency names and exact versions	Not yet archived
Random seed and stochastic configuration	Not yet archived
Exact commands/workflow	Not yet archived
Reference outputs and numerical tolerances	Not yet archived
Hardware-sensitive settings	Not yet archived

Chapter 44

Stage 2 Physical Derivation Ledger

The ledger below records the physical dependency structure after the Stage 2 audit. “Postulate-derived” means that the calculation follows after the listed physical postulate is accepted; it does not validate that postulate.

Table 44.1: Stage 2 physical derivation ledger: Volume I.

Claim	Mathematical Input	Physical Input	Derivation Status	Missing Input	Testability
EM condensate scalar	Dimensions and Lorentz/gauge contraction of F^2	Coarse-grained vacuum postulate	Dimension and symmetry checks derived	Microscopic state, coarse graining, operator matching	Composite correlators and EFT matching
Three vacuum layers	Linear field decomposition	Energy-band and dark-sector interpretation	Hypothesis; additive energy is conditional	Orthogonality or decoherence rule, or multi-field action	Layer-specific stress, clustering, and pressure
Minimal scalar equation	Covariant variation and integration by parts	Baseline scalar action	Postulate-derived	Matter state and S_{top} for a solution	Background and perturbation solutions
Scalar stress tensor	Metric variation and Bianchi identity	Baseline scalar action	Postulate-derived; conserved on shell when isolated	Complete exchange terms for disformal matter	Energy conditions and cosmological evolution
Disformal metric	Rank-one inverse and determinant identities	Choice of matter metric and bare disformal α	Algebraically proved for the ansatz	Background and full characteristic operators	Signature, cone nesting, and propagation tests
G_2 stability	$P(X)$ quadratic-action criteria	Optional X^2 truncation	Conditional local result	Fixed full EFT including G_3 , gravity, and matter	Ghost, gradient, and sound-speed bounds
Physical winding sector	$H^1(T^2; \mathbb{Z})$ and conserved winding pair	Identification with a particle configuration	Open — requires new physical input	Spacetime/internal bundle map, finite-energy charge	Soliton spectrum and charge conservation
Four-dimensional spin	Torus spin structures and 2D Dirac spectrum	Winding-spin/TTA hypothesis	Open — requires new physical input	$\text{Spin}(1, 3)$ representation map and dynamics	Lorentz-covariant fermion spectrum
Spin-statistics	Intrinsic antiperiodic boundary data	Fermionic interpretation	Open — requires new physical input	Local QFT, positivity, micro-causality, quantization map	Exchange and exclusion observables
Physical time	Monotone order coordinate	TEMP monotonicity hypothesis	Conditional ordering; physical time open	Lorentzian causal dynamics and operational clock	Clock-rate and causal predictions

Table 44.2: Stage 2 physical derivation ledger: Volume I (continued).

Claim	Mathematical Input	Physical Input	Derivation Status	Missing Input	Testability
Einstein gravity	Variation of $M_{\text{Pl}}^2 R/2$	Einstein–Hilbert term in baseline action	Postulate-derived, not emergent	Induced-gravity loop calculation if emergence is claimed	GR limit, PPN, waves, and stability
Effective Newton response	Constant G_N in the active action	Pure-disformal matter coupling	GR normalization derived only for constant scalar background; general response open	Weak-field background solution and source boundary data	PPN, fifth force, and equivalence-principle tests
Topological screening	Dimensionless quotient	Environmental statistic Ξ_w	Phenomenological ansatz	Covariant winding density and likelihood	Radial and environmental variation of coupling
Topological Involution	Pure-disformal spatial determinant	Cosmological interpretation of $Q_D(t)$	Re-derived conditional identity with $1 - Q_D$	Background and perturbation dynamics for Q_D	Expansion, growth, and lensing covariance
Dark-matter shadow	Scalar stress is covariantly definable	Gamma-layer interpretation	Hypothesis	Clustering solutions and relativistic potentials	Joint dynamics, lensing, and clusters
Dark-energy pressure	Scalar ρ_Φ, p_Φ formulas	CMB-layer interpretation	Hypothesis	Potential-dominated solution and specific $w(z)$	Expansion and growth history
Information balance	Dimensionless sum	Conservation postulate	Unspecified postulate, not Noether-derived	Information measure, dynamics, and error definition	Operational entropy/information protocol
Toroidal equator	Differentiable angular profile	Milky-Way placement and phenomenological a	Profile extrema derived; interpretation phenomenological	Units, normalization, sources, sky likelihood	Polarization and field-map fit

Table 44.3: Stage 2 physical derivation ledger: Volume II.

Claim	Mathematical Input	Physical Input	Derivation Status	Missing Input	Testability
TTA	Winding sectors and tautological tensor products	Sector-to-particle rank rule	Bookkeeping derived; physical construction open	Functor, representation space, invariant action, quantization	Particle representation spectrum
Trefoil geometry	Smooth embedded curve	Trefoil as physical organizing object	Regularity and line metric proved	Knot energy and physical state map	Geometry-dependent mode spectrum
Candidate $U(1)$	Compact scalar and kinetic shift current	Identification with hypercharge or electromagnetism	Kinetic global current derived; broken by displayed potential and sources	Local connection, charges, representations, anomalies	Gauge interactions and charge quantization
Candidate $SU(2)$	Possible two-state labels	Metastable $w = 3$ sector	Candidate correspondence	Three closing generators, chiral reps, breaking, anomalies	Weak amplitudes and parity observables
Candidate $SU(3)$	Trefoil geometry and target matrices	Trefoil-color identification	Candidate correspondence	Eight derived modes, bracket, quarks, confinement, anomalies	Color representations and QCD dynamics
4D Dirac action	Standard Lorentzian spin geometry	Independent Dirac action postulate	Equation and current derived from action	Derivation from TTA/winding and mass mechanism	Fermion spectrum and currents
Soliton mass functional	Radial variational calculus	Static scalar-particle interpretation	Euler equation derived; Derrick obstruction proved	Stabilizing operator/dynamics and winding boundary data	Existence and stability of finite-energy states
Particle masses	$m_w c^2 = E_* I_w$	Profiles, potential, scale E_*	Conditional and calibrated targets	Reproducible solutions, calibration record, covariance	Out-of-sample mass predictions
Higgs scale	Dimensional combination $\sqrt{\beta_0 \hbar c / R_c}$	Collective breathing-mode interpretation	Conditional analogy; scale misses by $\sim 10^{39}$	Microscopic scale or derived hierarchy	Higgs mass and couplings
Three generations	Assumed w^2 strain ordering	Stability-threshold interpretation	Conjecture	Binding energies, decays, threshold eigenproblem	Presence/absence of further families

Table 44.4: Stage 2 physical derivation ledger: Volume II (continued).

Claim	Mathematical Input	Physical Input	Derivation Status	Missing Input	Testability
$\alpha_* = 48$	Two-term autonomous beta function and equilibrium relation	Beta-function coefficients and $\gamma_{\text{str}} = 0$	Analytically derived from stated ansatz	Four-dimensional loop derivation for physical interpretation	RG running if coupling is identified
RG/disformal matching	Distinct variables α_{RG} and α_{dis}	Equality of RG and disformal couplings	Hypothesis, not established	Scheme, operator mixing, matching conditions	Scale dependence of disformal observables
Muon $g - 2$	None beyond EFT placeholder	Proposed numerical correction	Proposed target	Loop diagrams, masses, couplings, renormalization	Direct precision comparison
Higgs coupling shift	Conditional Z_0	$O(Z_0)$ scaling	Phenomenological scaling ansatz	Coefficient, sign, operator matching	Collider coupling fit
Electron EDM	No CP calculation supplied	Proposed upper target	Proposed benchmark	CP phases, operators, loops	EDM search
Axion coupling	Local normalization of $\Phi F \tilde{F}$	Wilson coefficient $c_{\Phi F} / M_*$	Algebraic relation derived; global compact quantization open	Gauge normalization, allowed topological sectors, $c_{\Phi F}$, and M_*	Axion/photon searches

Table 44.5: Stage 2 physical derivation ledger: Volume III.

Claim	Mathematical Input	Physical Input	Derivation Status	Missing Input	Testability
R_c, f_a, a_0	Stage 1 definitions and sensitivities	$L_*, \kappa_f, \kappa_a, \alpha_{\text{eff}}^{\text{gal}}$	Conditional calibrated scales	Independent microscopic determination	Cross-system scale consistency
Rotation law	Positive monotone $\mathcal{F}(y)$	Algebraic circular-speed postulate	Asymptotics and $v^4 = GMa_0\beta_0$ postulate-derived	Covariant force/stress solution	Public out-of-sample rotation curves
SPARC numbers	Rotation formula	Catalogue fit claim	Reported fit targets, not reproduced	Data, covariance, priors, code, parameter count	Reproduction and predictive comparison
Hubble ansatz	Monotone rational phenomenological function	Collapse correction with $\alpha_H = 0.086053412\dots$	Phenomenological ansatz with conditional normalization	EFT derivation and joint cosmological likelihood	Joint redshift-dependent likelihood
Fractional dimension	Bounded monotone function $n(z)$	Effective-dimension interpretation	Mathematical properties derived; physical map open	Covariant definition and growth equations	Growth and survey forward model
Scalar cosmology	Minimal scalar stress and FLRW symmetry	Source-free minimal truncation	ρ, p, w and conservation derived conditionally	Disformal exchange, full EFT, initial data	Expansion and perturbations
Lensing	Relativistic metric potentials	Identification of rotation effect with gravity	Open — requires new physical input	Two metric potentials and slip equation	Joint dynamics and lensing
Magnetic profile	Axisymmetric vector calculus	Toroidal field ansatz	Divergence-free property proved; dynamics open	Currents, amplitude, units, MHD evolution	Polarization and Faraday maps
GW phase	Historical dimensionless targets	Modified-wave interpretation	Removed as active prediction; unreproduced historical targets	Waveform equation, source model, detector likelihood	Stacked waveform analysis
Aharonov–Bohm term	Dimensionless local-lift combinations	Extra topological phase ansatz	Local/trivial-sector only; global compact completion open	Lift-independent observable, Hamiltonian/path-integral derivation, and fixed parameters	Controlled interferometry

Table 44.6: Stage 2 physical derivation ledger: Volume III (continued).

Claim	Mathematical Input	Physical Input	Derivation Status	Missing Input	Testability
Dipole target	Reported angle and significance	Toroidal-equator interpretation	Reported target	Mask, estimator, covariance, null distribution	Common-mask multi-survey test
Bullet ratio	$Z_0\beta_0 = (5/3)\beta_0^2$	Optical-depth identification	Ratio algebraically derived; offset proposed	Transport/lensing model converting ratio to distance	Cluster dynamics and lensing
DESI/dark energy	Minimal scalar can have negative pressure	Layer interpretation	Model-compatible only	Specific $w(z)$ and correlated growth	Joint expansion/growth likelihood
Early structure	Monotone $n(z)$ ansatz	Growth-enhancement interpretation	Hypothesis; no enhancement derived	Linear/nonlinear perturbations and selection model	High-redshift mass/luminosity functions
Weak-lensing threshold	None from rotation law alone	Historical benchmark	Proposed, not a closed forecast	Sign, amplitude, relativistic potentials, likelihood	Euclid-type lensing analysis
Falsification program	General pre-registration standard	Historical thresholds and dates	Partly testable; most numbers remain benchmarks	Fixed equations, parameters, uncertainties, alternatives	Model-specific exclusion regions

Chapter 45

Stage 2 Closure Report

45.1 Claims fully derived at the stated conditional level

No physical postulate was promoted to a theorem. The following consequences are complete once their displayed inputs are accepted:

- the minimal scalar Euler–Lagrange equation, canonical stress tensor, on-shell conservation law, and exact chain-rule matter source;
- the four-dimensional disformal inverse, determinant, eigenvalue, invertibility, and algebraic signature conditions;
- the local no-ghost, gradient, and sound-speed conditions for the explicitly isolated G_2 truncation;
- the regularity and induced line element of the displayed trefoil curve;
- the standard Dirac equation and global current from the independently postulated four-dimensional Dirac action;
- the radial mass Euler–Lagrange equation and the Derrick-scaling obstruction for the displayed nonnegative canonical scalar functional;
- the two-term RG fixed point and flow results inherited from Stage 1, including $\alpha_* = 48$ under the stated parameter postulates;
- monotonicity, limiting behavior, dimensional closure, and asymptotic mass–velocity scaling of the galactic rotation postulate;
- the Hubble-ansatz normalization $\alpha_H = 0.086053412\dots$ and the demonstration that the distinct galactic value $\alpha_{\text{eff}}^{\text{gal}} = 0.36$ cannot be substituted into that normalization;
- the bounds and monotonicity of $n(z)$, the source-free minimal scalar FLRW relations, the divergence-free property of the axisymmetric toroidal magnetic profile, and the algebraic product $Z_0\beta_0 \simeq 0.03305$.

45.2 Claims conditionally derived

Conditional consequences include the scalar mass gap after the periodic potential is assumed, disformal volume correction after the spatial ansatz is assumed, rotation-curve asymptotics after the velocity law is postulated, R_c , f_a , and a_0 after their calibrations are supplied, and the normalization relation $g_{\Phi\gamma\gamma} = c_{\Phi F}/M_*$. These are valid dependency calculations, not independent physical derivations.

45.3 Phenomenological assumptions

The layered-vacuum energy interpretation, topological screening, Topological Involution as a cosmological mechanism, the galactic rotation law, the Hubble correction, fractional-dimension evolution, toroidal magnetic profile, particle mass scale E_* , collective Higgs interpretation, and the listed numerical phase, dipole, cluster, EDM, and coupling targets remain phenomenological assumptions or proposed benchmarks.

45.4 Unresolved physical constructions

The following are *Open* — *require new physical input*:

- microscopic derivation of the composite EM condensate and its EFT;
- a dynamically separable three-layer stress tensor reproducing dark-matter clustering and dark-energy pressure;
- a four-dimensional realization of winding charge, TTA representation map, winding–spin map, and spin–statistics quantization;
- complete local $U(1)$, $SU(2)$, and $SU(3)$ gauge theories with matter, anomalies, symmetry breaking, and confinement;
- a stabilizing particle-soliton action and reproducible mass solutions;
- a generation threshold and a microscopic Higgs scale;
- loop-derived full-EFT RG coefficients and matching of RG, disformal, screened, galactic, and cosmological couplings;
- induced gravity, a complete stable scalar–tensor theory, Lorentzian cone analysis, and operational emergent time;
- cosmological background/perturbation solutions, relativistic lensing, magnetic sources and evolution, modified GW waveforms, and archived numerical likelihoods.

45.5 Numerical results requiring recalibration or reproduction

- The Hubble ansatz uses the conditional normalization $\alpha_H = 0.086053412 \dots$ to map 67.4 to 73.2 at $z = 0$. The distinct galactic fit target $\alpha_{\text{eff}}^{\text{gal}} = 0.36$ would instead give approximately $91.7 \text{ km s}^{-1} \text{ Mpc}^{-1}$ if incorrectly inserted.
- The Higgs dimensional scale is approximately $1.01 \times 10^{-28} \text{ eV}$; matching 125 GeV requires an unexplained factor of about 1.2×10^{39} .
- Particle masses require a new stabilizing functional and an archived calibration of E_* , the potential, profiles, and boundary conditions.
- a_0 requires the calibration κ_a as well as R_c and $\alpha_{\text{eff}}^{\text{gal}}$; it is not independently derived.
- The SPARC χ^2 , galactic $\alpha_{\text{eff}}^{\text{gal}}$, GW phase, Bullet Cluster distance, $g - 2$, EDM, axion, neutrino, weak-lensing, and magnetic thresholds require the missing dependency chains and reproducibility archive.
- The historical 89.4 remains α_{cal} and must not be reported as the fixed point of the displayed beta function.

45.6 Candidate observables that can become genuinely falsifiable

The most direct candidates are the fixed radial shape of the rotation law in an out-of-sample public likelihood, the redshift shape of the Hubble ansatz after its coupling is separated and fixed, the angular magnetic profile after its sources and nuisance parameters are specified, and the additional Aharonov–Bohm phase after a Hamiltonian derivation fixes its coefficient. The current exact GW, dipole, cluster, particle, and lensing numbers are not closed forecasts and should not yet be treated as independent falsification thresholds.

45.7 Claims requiring removal, downgrade, or scientific rewriting

For scientific accuracy, any summary should:

- describe the three-layer energy sum as conditional rather than implied by $\Phi = \sum_i \Phi_i$;
- restrict the continuous scalar-shift current to the kinetic truncation, because the displayed potential and sources break it;
- describe the baseline model as a postulated Einstein–scalar EFT, not a derivation of emergent gravity;
- present $U(1)$, $SU(2)$, $SU(3)$, chirality, TTA, winding–spin, and spin–statistics statements as candidate correspondences or open problems;
- replace claims of static canonical-scalar particle solitons by an open stabilization problem in view of the scaling obstruction;
- present the Higgs and generation mechanisms as hypotheses without derived mass or family count;
- separate α_{RG} , α_{dis} , $\alpha_{\text{eff}}^{\text{gal}}$, and α_H until a matching calculation is supplied;
- retain $\alpha_H = 0.086053412\dots$ for the Hubble normalization and do not substitute the galactic value $\alpha_{\text{eff}}^{\text{gal}} = 0.36$ into that ansatz; and
- label unreproduced mass, GW, dipole, Bullet Cluster, weak-lensing, EDM, $g - 2$, axion, neutrino, and magnetic numbers as proposed benchmarks rather than derived predictions.

No new physical assumption was introduced in the historical Stage 2 audit. Every result in that pass was a variation, algebraic consequence, dimensional/symmetry check, stability condition, dependency audit, or obstruction derived from equations already present in the frozen V15.2 baseline. The later V15.3 metric replacement is recorded separately in Chapter 47.

Chapter 46

Stage 2 Critical Findings Report

46.1 Derrick obstruction — verified result

The tested functional is the explicitly three-dimensional radial expression

$$I_w^{\text{rad}} = 4\pi \int_0^\infty \left[\frac{1}{2}(\hat{\Phi}')^2 + \hat{V}(\hat{\Phi}) \right] \tilde{r}^2 d\tilde{r} = T + U.$$

For $\hat{\Phi}_\lambda(\tilde{r}) = \hat{\Phi}(\lambda\tilde{r})$, direct substitution gives

$$T(\lambda) = \lambda^{-1}T, \quad U(\lambda) = \lambda^{-3}U, \quad I_w^{\text{rad}}(\lambda) = \lambda^{-1}T + \lambda^{-3}U.$$

Therefore

$$\left. \frac{dI_w^{\text{rad}}}{d\lambda} \right|_{\lambda=1} = -T - 3U.$$

Stationarity requires $T + 3U = 0$. Under the precise assumptions of a static, regular, finite-energy configuration in three spatial dimensions; a canonical positive quadratic gradient term; a nonnegative local potential normalized to zero at the vacuum; admissibility of the scaling variation; and absence of an additional fixed constraint, one has $T \geq 0$ and $U \geq 0$. Hence $T = U = 0$, leaving only a constant vacuum. The nontrivial finite static soliton claim is therefore **Invalid under the current functional**. This is a proved obstruction, not a failure of numerical solution.

The separate compact internal functional $I_w^{(T^2)}$ is not subject to this specific three-dimensional scaling argument, but it also has no demonstrated map to a four-dimensional particle mass. It remains a conditional internal research ansatz.

46.2 Affected particle-mass claims

The mass equations have been preserved as proposed research directions. The main text, Stage 2 ledger, numerical tables, Volume IV claim-status ledger, and conclusions now state that no finite static particle mass is established by the current radial functional.

46.3 Hubble normalization — verified result

The original cosmological ansatz is

$$H_{\text{obs}}(z) = H_\Lambda(z) [1 + \alpha_H \Omega_{\text{coll}}(z)], \quad \Omega_{\text{coll}}(z) = \frac{1}{1 + (z/z_c)^2}, \quad z_c = 0.35.$$

At $z = 0$, $\Omega_{\text{coll}}(0) = 1$. Substitution of the stated values gives

$$73.2 = 67.4(1 + \alpha_H),$$

Claim	Final status	Reason
Derrick obstruction for I_w^{rad}	Proved	Exact scaling gives the incompatible virial condition $T + 3U = 0$ for nonnegative T, U
Nontrivial finite static radial soliton	Invalid under current functional	Only the constant vacuum satisfies stationarity
$m_w c^2 = E_* I_w^{\text{rad}}$ as a mass derivation	Conditional dimensional ansatz only	No nontrivial stationary I_w^{rad} configuration exists
$I_w^{(T^2)}$ as an internal contribution	Conditional	Dimensionally closed after E_* , but no 4D reduction or particle-state map exists
Proton, electron, muon, and tau values	Historical targets; Open Problem	Not outputs of the obstructed functional and not independently reproduced
w^2 mass scaling	Conjectural/Open	No admissible solution family or derived coefficients A, B are supplied

so

$$\alpha_H = \frac{73.2}{67.4} - 1 = 0.086053412 \dots \simeq 0.0861.$$

This number is a **conditional algebraic derivation and two-value calibration**: it follows from the phenomenological ansatz after the displayed early and local H_0 values are inherited as inputs. It is not a likelihood fit and not a parameter-free physical prediction.

The previous 0.36 ± 0.03 value entered through the reported SPARC/galactic screened-fit parameter $\alpha_{\text{eff}}^{\text{gal}}$ and the associated a_0 calibration. It did not originate in the Hubble normalization. Substituting it into the Hubble ansatz gives

$$67.4(1 + 0.36) = 91.664 \text{ km s}^{-1} \text{ Mpc}^{-1},$$

which contradicts the stated local value $73.2 \text{ km s}^{-1} \text{ Mpc}^{-1}$. The resolution is not to replace the galactic value, but to keep the two parameters distinct.

46.4 Affected Hubble claims and numerical values

The Hubble ansatz in Volumes 0 and III, the DESI benchmark row, Stage 2 ledger, dimensionless-parameter table, claim-status ledger, and closure conclusions now use α_H . The complete numerical classification is:

- $\alpha_H = 0.086053412 \dots \simeq 0.0861$: conditional Hubble normalization calibrated from 67.4 and 73.2;
- $\alpha_{\text{eff}}^{\text{gal}} = 0.36 \pm 0.03$: distinct unreproduced galactic screened-fit target, retained for the SPARC and a_0 dependency chains;
- $91.664 \text{ km s}^{-1} \text{ Mpc}^{-1}$: diagnostic inconsistent value obtained only by incorrectly inserting the galactic parameter into the Hubble ansatz;
- the early and local values 67.4 and 73.2 are unchanged inherited inputs; and
- all historical particle-mass numbers are unchanged numerically but are now classified only as targets, not derived static-soliton masses.

46.5 Remaining open problems

The particle sector still requires an admissible physical mass construction, a spacetime/winding map, boundary data, a calibration protocol, and an out-of-sample prediction. The Hubble sector

still requires an EFT derivation of $H_{\text{obs}}(z)$, an independently defined cosmological coupling, background and perturbation solutions, and a joint likelihood. No relation between α_H and $\alpha_{\text{eff}}^{\text{gal}}$ has been derived.

46.6 Closure statement

This section records the historical Stage 2 closure of the frozen V15.2 baseline. Its statement that no new physical assumption was introduced applies only to that earlier closure pass. V15.3 subsequently replaces the physical metric through the explicitly declared pure-disformal postulate.

Chapter 47

Stage 4B Pure-Disformal Reformulation Ledger

47.1 Frozen ontology and obstruction status

V15.3 contains one real compact scalar and no radial or dynamical organizational field. The unit phase $U = e^{i\Phi/f_a}$ is an equivalent representation only. The metric and organizational statuses are

$$\tilde{g}_{\mu\nu} = g_{\mu\nu} - \frac{\alpha}{M_*^4} \partial_\mu \Phi \partial_\nu \Phi, \quad \mathcal{O}_\Omega = \sqrt{\Omega_{\text{cand}}} \quad (\text{nondynamical diagnostic}).$$

The compact-shift metric obstruction F1 is removed because $d\Phi$ is representative independent. The zero-field obstruction F2 is removed because $\tilde{g} = g$ when $d\Phi = 0$. The organizational obstruction F3 is removed by deleting the global equality between Φ and a single-valued function of Ω_{cand} , not by deleting winding sectors.

47.2 Major re-derived equations

Construction	V15.3 result	Status
Inverse metric	$g^{\mu\nu} + \alpha \nabla^\mu \Phi \nabla^\nu \Phi / (M_*^4 + 2\alpha X)$	Proved algebraically
Determinant	$\det \tilde{g} = \det g \Delta_D$	Proved algebraically
Matter measure	$\sqrt{-\tilde{g}} = \sqrt{-g} \sqrt{\Delta_D}$ on $\Delta_D > 0$	Proved algebraically
Scalar matter source	$\mathcal{J}_m^{(D)} = \alpha (M_*^4 \sqrt{-g})^{-1} \partial_\nu (\sqrt{-\tilde{g}} \tilde{T}^{\mu\nu} \partial_\mu \Phi)$	Derived from S_m
g -frame matter tensor	$T_{\rho\sigma}^{(m,g)} = \sqrt{\Delta_D} g_{\rho\mu} g_{\sigma\nu} \tilde{T}^{\mu\nu}$	Derived from S_m
Metric equation	$M_{\text{Pl}}^2 G_{\mu\nu} = T_{\mu\nu}^{(\Phi)} + T_{\mu\nu}^{(m,g)} + \text{declared sources}$	Derived from active action
Homogeneous lapse	$N_D^2 = 1 + \alpha \dot{\Phi}^2 / M_*^4$	Derived
Physical Hubble rate	$\tilde{H} = H / N_D$	Derived kinematically
Spatial volume	$\det \tilde{\gamma} = a^6 (1 - Q_D)$	Proved algebraically
Spatial volume rate	$H_{\text{vol}} = H - \dot{Q}_D / [6(1 - Q_D)]$	Derived kinematically

Sector	V15.3 classifica- tion	Reason
Gravity	Conditional; postulated	Einstein–Hilbert term is assumed; emergence is not derived
Matter coupling	Derived at baseline	Exact scalar and metric variations are displayed
General G_{eff}	Open	Requires a background-dependent weak-field solution
Constant- background GR limit	Conditional derived limit	$d\Phi_0 = 0$ makes $\tilde{g} = g$ at linear order
Homogeneous cosmology	Kinematics derived; dy- namics open	Lapse and $\tilde{H}$ are known, but no background solution is supplied
Perturbations and lensing	Open	Full quadratic action and solved background are absent
Tensor waves	Minimal g - frame equation derived	Numerical phase and matter-frame propagation history remain open
PPN/Newtonian	Conditional GR limit; oth- erwise open	Nonzero-gradient source problem is unsolved
Topology, TTA, TEMP	Mathematical content pre- served	Physical representation maps remain open
Particle masses	Obstructed; open	Derrick obstruction is unchanged

Observable	Status	V15.3 conclusion
$H_{\text{obs}}(z)$	Survives phe- nomenologically	Not derived from N_D ; α_H remains a conditional calibration
SPARC numbers	Survive as un- reproduced targets	Independent of the metric rewrite until a covariant force law is derived
Rotation-law asymptotics	Conditional	Algebra survives the postulate; relativistic metric origin remains open
Weak lens- ing/clusters	Open	No solved potentials or slip relation
GW phase num- bers	Removed as ac- tive predictions	Historical targets lack waveform and likelihood derivations
Tensor speed	Re-derived con- ditionally	$c_{T,g} = 1$; matter-frame comparison depends on N_D
Dipole and mag- netic numbers	Unreproduced targets	No new derivation follows from the pure-disformal metric
PPN/fifth force	Open beyond constant back- ground	Source solution and boundary data required

47.3 Physical-sector classification

47.4 Observable-status audit

47.5 Stale-formula classification

The former active formulas are classified as follows:

- $(\Phi/M_*)^2 g_{\mu\nu}$: **historical/removed**; it appears only in explanatory statements identifying the V15.2 obstruction.
- G_N/φ^2 : **historical/removed**; no active V15.3 equation uses it.
- $\Phi = M_*\sqrt{\Omega_{\text{cand}}}$: **historical/removed**; a future matching is conditional and not presently assumed.
- conformal determinant and volume formulas: **removed**; replaced by Δ_D .
- the plus-sign spatial expression $1 + Q$: **removed**; replaced by $1 - Q_D$ for the declared sign convention.
- the previous GW phase values: **historical unreproduced targets**; not active predictions.

47.6 New assumptions and remaining open inputs

The single new physical assumption introduced by Stage 4B is the choice of the pure-disformal matter metric. Downgrading Ω_{cand} removes a former identity and does not introduce dynamics. No complex scalar, radial mode, dynamical organization field, stabilizing interaction, or new gauge field is added.

Additional physical input is still required for a global compact normalization of $\Phi F\tilde{F}$, a complete higher-operator basis, stable background solutions, perturbation and lensing equations, binary waveforms, general PPN parameters, the internal-torus/spacetime map, and any particle-mass construction.

Part V

Stage 5: Mathematical and Observational Closure Audit

Chapter 48

Stage 5.1 Active-Action and Hamiltonian Audit

48.1 Exact active-action inventory

V15.3.2 contains a fully specified lower-order vacuum gravity–scalar functional and several incompletely specified extensions. To prevent an unspecified symbol from being treated as a hidden completion, write

$$S = S_{\text{low}} + S_m[\tilde{g}, \psi_m] + S_{\Phi F} + S_{\text{top}} + S_{>4},$$

where

$$S_{\text{low}} = \int d^4x \sqrt{-g} \left[\frac{M_{\text{Pl}}^2}{2} R - \frac{1}{2} g^{\mu\nu} \partial_\mu \Phi \partial_\nu \Phi - V(\Phi) \right], \quad V(\Phi) = \Lambda^4 \left[1 - \cos \left(\frac{\Phi}{f_a} \right) \right],$$

$$\tilde{g}_{\mu\nu} = g_{\mu\nu} - \frac{\alpha_{\text{dis}}}{M_*^4} \partial_\mu \Phi \partial_\nu \Phi,$$

and

$$S_{\Phi F} = \frac{c_{\Phi F}}{4M_*} \int d^4x \sqrt{-g} \Phi F_{\mu\nu} \tilde{F}^{\mu\nu}.$$

This decomposition changes no term in the frozen theory; it only separates the functional that is explicitly known from the placeholders whose Lagrangians are absent.

Table 48.1: Action-sector closure audit.

Sector	Fields explicitly declared	Mathematical status	Consequence for the audit
$M_{\text{Pl}}^2 R/2$	$g_{\mu\nu}$	Fully specified postulate	Standard Einstein–Hilbert constraint structure can be counted.
Canonical compact scalar	One real $\Phi \sim \Phi + 2\pi f_a$	Fully specified lower-order sector	One local scalar mode; compactness changes global sectors, not the local count.
Periodic potential	Φ	Fully specified function; Λ and f_a remain inputs	Fixes the local curvature $V'' = (\Lambda^4/f_a^2) \cos(\Phi/f_a)$.
Pure-disformal metric	Algebraic function of $g_{\mu\nu}$ and $d\Phi$	Fully specified coupling map	Not an independent field and has no separate canonical momentum.

Sector	Fields explicitly declared	Mathematical status	Consequence for the audit
$S_m[\tilde{g}, \psi_m]$	Generic ψ_m only	Coupling prescription specified; matter Lagrangian unspecified	Matter DOF, kinetic Hessian, constraints, equation of state, and principal symbol cannot be counted.
$S_{\Phi F}$	Φ and a symbol $F_{\mu\nu}$	Displayed local operator; gauge kinetic term, connection, and group absent	No physical gauge DOF count follows. The compact global normalization is audited separately.
S_{top}	No field list	Unspecified placeholder	Its name does not prove that it has no local modes; no count is possible.
$S_{>4}$ or $\mathcal{L}_{\text{EFT}}^{(>4)}$	No fixed operator basis	Unspecified EFT placeholder	Derivative order, degeneracy, cutoff, and additional modes are undetermined.
Optional G_2	Φ, X	Explicit isolated $P(X)$ ansatz	Local kinetic and gradient conditions can be derived for that truncation.
Optional G_3	Function $G_3(\Phi, X)$ only	The operator multiplying G_3 is not declared	No equation, Hessian, or DOF statement follows without adding a Lagrangian definition.
Ω_{cand}	Nondynamical diagnostic	Explicitly not a spacetime field	Zero propagating DOF.

Proposition 48.1 (Maximum closed functional). The only covariant dynamical functional whose complete field content and derivative order are fixed without further input is S_{low} . The source-free minimal truncation used below is therefore a mathematically closed subsector of V15.3.2, not a declaration that $S_m = S_{\text{top}} = S_{>4} = S_{\Phi F} = 0$ in the complete physical theory.

48.2 ADM decomposition of the closed minimal subsector

Use

$$ds^2 = -N^2 dt^2 + h_{ij}(dx^i + N^i dt)(dx^j + N^j dt)$$

and

$$K_{ij} = \frac{1}{2N} \left(\dot{h}_{ij} - D_i N_j - D_j N_i \right).$$

After the usual gravitational boundary term is separated, the minimal action becomes

$$S_{\text{low}} = \int dt d^3x N \sqrt{h} \left[\frac{M_{\text{Pl}}^2}{2} \left({}^{(3)}R + K_{ij} K^{ij} - K^2 \right) + \frac{1}{2N^2} \left(\dot{\Phi} - N^i D_i \Phi \right)^2 - \frac{1}{2} D_i \Phi D^i \Phi - V(\Phi) \right].$$

The canonical momenta are

$$\pi^{ij} = \frac{M_{\text{Pl}}^2 \sqrt{h}}{2} (K^{ij} - h^{ij} K), \quad p_\Phi = \frac{\sqrt{h}}{N} (\dot{\Phi} - N^i D_i \Phi),$$

$$\pi_N \approx 0, \quad \pi_i \approx 0.$$

The first-order Hamiltonian is

$$H_{\text{low}} = \int d^3x \left(N\mathcal{H}_0 + N^i\mathcal{H}_i + \lambda_N\pi_N + \lambda^i\pi_i \right),$$

with

$$\begin{aligned} \mathcal{H}_0 &= \frac{2}{M_{\text{Pl}}^2\sqrt{h}} \left(\pi^{ij}\pi_{ij} - \frac{1}{2}\pi^2 \right) - \frac{M_{\text{Pl}}^2\sqrt{h}}{2} {}^{(3)}R \\ &\quad + \frac{p_\Phi^2}{2\sqrt{h}} + \sqrt{h} \left(\frac{1}{2}D_i\Phi D^i\Phi + V(\Phi) \right), \\ \mathcal{H}_i &= -2D_j\pi^j_i + p_\Phi D_i\Phi. \end{aligned}$$

Preservation of π_N and π_i generates

$$\mathcal{H}_0 \approx 0, \quad \mathcal{H}_i \approx 0.$$

For the diffeomorphism-invariant minimal theory the four primary and four secondary constraints are first class. There are no scalar primary constraints because

$$\frac{\partial^2 \mathcal{L}_{\text{low}}}{\partial \dot{\Phi}^2} = \frac{\sqrt{h}}{N} \neq 0$$

on the regular branch $N \neq 0$.

Theorem 48.1 (Minimal local degree count). Including (h_{ij}, N, N^i, Φ) gives eleven configuration variables and a 22-dimensional phase space per spatial point. Eight first-class constraints remove sixteen phase-space dimensions. Hence

$$N_{\text{DOF}}^{\text{low}} = \frac{22 - 2(8)}{2} = 3,$$

namely two tensor polarizations and one real compact scalar mode.

Remark 48.1 (Constraint-algebra cross-reference). The first-class statement is completed explicitly in Proposition K.10, where the smeared Hamiltonian and momentum constraints are shown to obey the standard hypersurface-deformation algebra under boundary conditions that remove surface terms.

Remark 48.2 (Pure-disformal map). $\tilde{g}_{\mu\nu}$ is not an additional canonical coordinate. Its invertibility condition

$$\Delta_D = 1 + \frac{2\alpha_{\text{dis}}X}{M_*^4} > 0$$

is necessary for a regular matter metric, but it does not determine the kinetic rank of the unspecified matter fields. Invertibility of this rank-one relation alone therefore cannot close the total coupled DOF count.

48.3 Sector-by-sector DOF boundary

Table 48.2: Maximum justified propagating-degree statement.

Sector	Count	Justification
Einstein–Hilbert metric	2	Four-dimensional diffeomorphism constraints remove gauge and nondynamical lapse/shift components.
Canonical compact Φ	1	Nondegenerate first-derivative kinetic term; target compactness does not add a radial mode.
Pure-disformal metric	0 independent	It is a composite coupling tensor, not a second metric variable.
Ω_{cand}	0	No kinetic term or spacetime variation.
S_m	Undetermined	No matter field list or kinetic action.
S_{top}	Undetermined	A label does not establish metric independence or a topological constraint algebra.
$S_{>4}$	Undetermined	Arbitrary higher time derivatives may add modes unless the basis is degenerate or treated by controlled EFT order reduction.
$F_{\mu\nu}$ sector	Undetermined	With an independently declared Maxwell action a massless Abelian field would have two modes, but that action is absent and is not supplied here.

T6 conclusion. The closed lower-order gravity–scalar subsector has exactly three local propagating modes. The total V15.3.2 count remains **C: underdetermined**, because the kinetic matrices and constraints of S_m , S_{top} , and $S_{>4}$ are not defined.

48.4 Ghost-candidate ledger

No ghost is present in the isolated canonical scalar or Einstein tensor kinetic terms when $M_{\text{Pl}}^2 > 0$. The following are risk boundaries rather than proved instabilities:

- $\Delta_D = 0$ is a singular metric boundary, not a propagating ghost calculation.
- An arbitrary nondegenerate higher-time-derivative operator in $S_{>4}$ would be an Ostrogradsky candidate; no such active operator is explicitly given.
- The undefined G_3 notation cannot be tested until its actual Lagrangian operator is declared.
- Matter–scalar kinetic mixing cannot be diagonalized without the matter Hessian.
- A negative V'' is a tachyonic background instability, not a wrong-sign kinetic ghost.

Chapter 49

Stage 5.1 Quadratic Stability and T10 Audit

49.1 Minimal scalar and tensor quadratic actions

For a background solution $\Phi_{\text{bg}}(t)$ of the source-free minimal subsector, the local scalar fluctuation has the quadratic principal action

$$S_{\delta\Phi}^{(2)} = \frac{1}{2} \int dt d^3x a^3 \left[\delta\dot{\Phi}^2 - \frac{(\partial_i \delta\Phi)^2}{a^2} - V''(\Phi_{\text{bg}}) \delta\Phi^2 \right],$$

before scalar–metric constraints are eliminated. Therefore

$$Q_{\delta\Phi} = 1 > 0, \quad c_{s,g}^2 = 1,$$

and

$$m_{\text{loc}}^2 = V''(\Phi_{\text{bg}}) = \frac{\Lambda^4}{f_a^2} \cos\left(\frac{\Phi_{\text{bg}}}{f_a}\right).$$

At $\Phi_{\text{bg}} = 2\pi n f_a$ this mass is positive. At $\Phi_{\text{bg}} = (2n+1)\pi f_a$ it is negative, which is the expected tachyonic instability of a potential maximum rather than a kinetic ghost.

When $\dot{\Phi}_{\text{bg}} \neq 0$, solving the lapse and shift constraints gives the standard single-field curvature mode ζ in the minimal source-free subsector:

$$S_{\zeta}^{(2)} = \int dt d^3x a^3 \frac{\dot{\Phi}_{\text{bg}}^2}{2H^2} \left[\dot{\zeta}^2 - \frac{(\partial_i \zeta)^2}{a^2} \right].$$

Thus the scalar curvature mode has a positive kinetic coefficient for $H \neq 0$ and $\dot{\Phi}_{\text{bg}} \neq 0$, and $c_s^2 = 1$. If $\dot{\Phi}_{\text{bg}} = 0$, comoving gauge is degenerate and the regular variable is $\delta\Phi$, whose action is the one displayed above.

The tensor action is

$$S_T^{(2)} = \frac{M_{\text{Pl}}^2}{8} \int dt d^3x a^3 \left[\dot{h}_{ij} \dot{h}^{ij} - \frac{\partial_k h_{ij} \partial^k h^{ij}}{a^2} \right].$$

Consequently

$$Q_T = \frac{M_{\text{Pl}}^2}{8} > 0, \quad c_{T,g}^2 = 1$$

when $M_{\text{Pl}}^2 > 0$.

49.2 Optional G_2 stability region

For the explicitly displayed isolated ansatz

$$P(\Phi, X) = X - V(\Phi) + \frac{\beta(\Phi)}{M_*^4} X^2, \quad \beta(\Phi) = \beta_0 \cos\left(\frac{\Phi}{f_a}\right),$$

define

$$y = \frac{\beta(\Phi)X}{M_*^4}.$$

The kinetic, gradient, and sound-speed expressions reduce to

$$P_X = 1 + 2y, \quad P_X + 2XP_{XX} = 1 + 6y, \quad c_s^2 = \frac{1 + 2y}{1 + 6y}.$$

For a homogeneous timelike background $X \geq 0$:

- if $\beta(\Phi) \geq 0$, both stability inequalities hold for all finite $X \geq 0$, and $0 < c_s^2 \leq 1$;
- if $\beta(\Phi) < 0$, the stronger condition is

$$-\frac{1}{6} < y < 0 \quad \Longleftrightarrow \quad X < \frac{M_*^4}{6|\beta(\Phi)|},$$

and the isolated truncation has $c_s^2 > 1$ relative to the g cone;

- at $y = -1/6$ the kinetic denominator vanishes, while at $y = -1/2$ the gradient coefficient vanishes. The first boundary is encountered earlier as X increases in the negative- β region.

At the periodic vacua $\Phi = 2\pi n f_a$, one has $\beta = \beta_0 > 0$, so the isolated G_2 inequalities are satisfied.

This result does not activate G_2 in the baseline and does not test G_3 , matter mixing, or $S_{>4}$.

49.3 Hyperbolicity boundary of the coupled theory

The explicit necessary conditions obtainable from V15.3.2 are

$$M_{\text{Pl}}^2 > 0, \quad \Delta_D = 1 + \frac{2\alpha_{\text{dis}}X}{M_*^4} > 0,$$

together with the appropriate scalar inequalities above if the optional G_2 truncation is selected. These conditions establish positive isolated kinetic terms and Lorentzian g and $\tilde{g}$ metrics.

They are not sufficient for the complete system. On a nonzero-gradient background, variation of $S_m[\tilde{g}, \psi_m]$ produces principal kinetic mixing between $\delta\Phi$ and matter perturbations. The full characteristic determinant has the schematic block form

$$\det \begin{pmatrix} \mathcal{K}_{\Phi\Phi} & \mathcal{K}_{\Phi m} \\ \mathcal{K}_{m\Phi} & \mathcal{K}_{mm} \end{pmatrix},$$

and none of the three matter blocks is fixed without the explicit matter action. Therefore no complete no-ghost matrix, characteristic cone, or strong-coupling scale can be calculated from the placeholder S_m .

Full stability conclusion. Minimal scalar and tensor stability is derived; the isolated G_2 parameter region is derived conditionally; full coupled stability remains **C: underdetermined**.

49.4 T10 dimensional and global derivation

In four dimensions,

$$[\Phi] = 1, \quad [F_{\mu\nu}] = 2, \quad [F_{\mu\nu}\tilde{F}^{\mu\nu}] = 4.$$

Hence

$$\frac{c_{\Phi F}}{4M_*} \Phi F_{\mu\nu} \tilde{F}^{\mu\nu}$$

has mass dimension four exactly when $c_{\Phi F}$ is dimensionless.

Under the compact shift $\Phi \mapsto \Phi + 2\pi f_a$,

$$\begin{aligned} \Delta S_{\Phi F} &= \frac{c_{\Phi F}}{4M_*} (2\pi f_a) \int d^4x \sqrt{-g} F_{\mu\nu} \tilde{F}^{\mu\nu} \\ &= 2\pi \frac{c_{\Phi F} f_a}{M_*} \mathcal{Q}_F, \end{aligned}$$

where the source defines

$$\mathcal{Q}_F = \frac{1}{4} \int d^4x \sqrt{-g} F_{\mu\nu} \tilde{F}^{\mu\nu}.$$

The path-integral phase transforms as

$$e^{iS} \mapsto e^{iS} \exp \left[i2\pi \frac{c_{\Phi F} f_a}{M_*} \mathcal{Q}_F \right].$$

It is single-valued if and only if

$$\frac{c_{\Phi F} f_a}{M_*} \mathcal{Q}_F \in \mathbb{Z}$$

for every allowed gauge configuration.

To isolate the missing normalization, suppose a completed gauge sector proves

$$\mathcal{Q}_F = q_F n, \quad n \in \mathbb{Z},$$

where q_F is fixed by the gauge coupling, trace convention, bundle, and normalization of F . Then the necessary and sufficient coefficient condition for all integer sectors is

$$\boxed{\frac{c_{\Phi F} f_a}{M_*} q_F \in \mathbb{Z}.}$$

For example, if a convention defines

$$n \equiv \frac{1}{32\pi^2} \int d^4x \sqrt{-g} F_{\mu\nu}^a \tilde{F}^{a\mu\nu},$$

then the present definition gives $\mathcal{Q}_F = 8\pi^2 n$ and the condition becomes

$$8\pi^2 \frac{c_{\Phi F} f_a}{M_*} \in \mathbb{Z}.$$

This example is a normalization identity, not a choice of ATHENA gauge group. Changing the trace normalization, absorbing a gauge coupling into F , or using an Abelian flux convention changes q_F .

On noncompact spacetime, $\mathcal{Q}_F$ is a boundary quantity unless falloff conditions or a compact spacetime domain are supplied. In a trivial Abelian bundle it may vanish, while nontrivial bundles or magnetic flux sectors require an explicit integral cohomology normalization. None of these global data is specified in V15.3.2.

T10 conclusion. The dimensional analysis, action shift, quantum phase, and general quantization condition are fully derived. Physical closure remains **B: requires an additional gauge-sector assumption**, specifically the gauge group, connection normalization, bundle/topological sectors, boundary conditions, and a value or quantization rule for $c_{\Phi F}$.

49.5 Parameter dependency and circularity chain

The active provenance graph is

$$\begin{aligned} \frac{\pi}{12} &\xrightarrow{\beta_0(2-\beta_0)=\pi/12} \beta_0 \xrightarrow{\gamma_0=\beta_0^2/2} \gamma_0, \\ (L_*, \beta_0) &\longrightarrow R_c, & (R_c, \kappa_f) &\longrightarrow f_a, \\ (R_c, \beta_0, \kappa_a, \alpha_{\text{eff}}^{\text{gal}}) &\longrightarrow a_0, & (H_0^{\text{local}}, H_0^\Lambda) &\longrightarrow \alpha_H. \end{aligned}$$

It follows that:

- β_0 and γ_0 are conditional consequences of model normalizations, not observationally derived constants;
- R_c inherits the external reference scale L_* ;
- f_a inherits both R_c and the chosen coefficient κ_f ;
- a_0 cannot be an independent prediction of the acceleration scale because κ_a and $\alpha_{\text{eff}}^{\text{gal}}$ enter its calibration;
- α_H cannot independently predict the same present-day Hubble difference used to set it;
- α_{dis} and M_* enter the matter metric through the combination $\alpha_{\text{dis}}/M_*^4$, while T10 uses the independent combination $c_{\Phi F} f_a/M_*$;
- no equation identifies α_H , $\alpha_{\text{eff}}^{\text{gal}}$, α_{dis} , α_{RG} , or α_{cal} .

Chapter 50

Stage 5.1 Background, Weak-Field, and Perturbation Closure

50.1 Homogeneous equations from the declared lower-order action

For

$$ds^2 = -dt^2 + a^2(t)d\mathbf{x}^2, \quad \Phi = \Phi(t),$$

define

$$\rho_\Phi = \frac{1}{2}\dot{\Phi}^2 + V(\Phi), \quad p_\Phi = \frac{1}{2}\dot{\Phi}^2 - V(\Phi), \quad N_D^2 = 1 + \frac{\alpha_{\text{dis}}\dot{\Phi}^2}{M_*^4}.$$

If all unspecified source sectors admit a homogeneous and isotropic stress tensor, collect them as (ρ_U, p_U) , where U denotes $S_{\text{top}} + S_{>4} + S_{\Phi F}$ after their actual variations are supplied. The exact background form of the displayed metric equation is then

$$3M_{\text{Pl}}^2 H^2 = \rho_\Phi + \rho_{m,g} + \rho_U,$$

$$-2M_{\text{Pl}}^2 \dot{H} = \rho_\Phi + p_\Phi + \rho_{m,g} + p_{m,g} + \rho_U + p_U.$$

These equations are identities in terms of the still-undefined source densities; they are not closed until their actions and states are specified.

If, conditionally, the matter-frame stress tensor has a comoving perfect-fluid form with density $\tilde{\rho}$ and pressure $\tilde{p}$, the already derived frame relation gives

$$\rho_{m,g} = \frac{\tilde{\rho}}{N_D}, \quad p_{m,g} = N_D \tilde{p}.$$

For the same homogeneous configuration, the disformal scalar source reduces to

$$\mathcal{J}_m^{(D)} = \frac{\alpha_{\text{dis}}}{M_*^4 a^3} \frac{d}{dt} \left(\frac{a^3 \tilde{\rho} \dot{\Phi}}{N_D} \right).$$

Thus the scalar equation is

$$\ddot{\Phi} + 3H\dot{\Phi} + V'(\Phi) = \mathcal{J}_m^{(D)} + \mathcal{J}_{\text{top}} + \frac{c_{\Phi F}}{4M_*} F_{\mu\nu} \tilde{F}^{\mu\nu}.$$

This is the maximum homogeneous system derivable without inventing an equation of state, gauge background, or higher-operator stress tensor.

50.2 Closed source-free minimal FLRW system

In the explicitly isolated source-free minimal subsector, the equations close:

$$\begin{aligned} 3M_{\text{Pl}}^2 H^2 &= \frac{1}{2} \dot{\Phi}^2 + V(\Phi), \\ -2M_{\text{Pl}}^2 \dot{H} &= \dot{\Phi}^2, \\ \ddot{\Phi} + 3H\dot{\Phi} + V'(\Phi) &= 0. \end{aligned}$$

Differentiating the Friedmann constraint and using the last two equations gives identically

$$\frac{d}{dt} \left[3M_{\text{Pl}}^2 H^2 - \frac{1}{2} \dot{\Phi}^2 - V(\Phi) \right] = 0.$$

The physical matter-frame clock and Hubble rate remain

$$d\tilde{t} = N_D dt, \quad \tilde{H} = \frac{H}{N_D}.$$

No algebraic manipulation of these equations produces the independent phenomenological expression $H_{\text{obs}} = H_\Lambda(1 + \alpha_H \Omega_{\text{coll}})$.

Remark 50.1 (Numerical infrastructure). The accompanying minimal-FLRW program integrates precisely this source-free system. The distributed filename is recorded in the release manifest. It requires M_{Pl} , f_a , Λ , the initial scalar data, the integration interval, and the expanding or contracting branch as explicit inputs. It records the Friedmann residual and makes no observational parameter choice. Its output is therefore a conditional trajectory, not a prediction of the incomplete total action.

50.3 Weak-field and PPN result that actually follows

Take a constant vacuum representative

$$\Phi = \Phi_n = 2\pi n f_a, \quad \partial_\mu \Phi_n = 0, \quad V(\Phi_n) = V'(\Phi_n) = 0.$$

Then

$$\tilde{g}_{\mu\nu} = g_{\mu\nu}, \quad \mathcal{J}_m^{(D)} = 0$$

at linear order. With matter retained through its universal coupling but $S_{>4}$ and the unspecified topological/gauge source stresses omitted in the isolated lower-order limit, the metric equation is the Einstein equation and the scalar perturbation is not linearly sourced by static nonrelativistic matter through the pure-disformal term. Subject to the usual asymptotically flat boundary condition selecting $\delta\Phi = 0$, the solution is the GR weak-field solution and

$$\gamma_{\text{PPN}} = 1, \quad \beta_{\text{PPN}} = 1, \quad G_{\text{eff}} = G_N = \frac{1}{8\pi M_{\text{Pl}}^2}.$$

This is a derived conditional limit, not a general prediction for a nontrivial scalar background.

For a background with $\dot{\Phi}_{\text{bg}} \neq 0$, the physical metric already contains linear scalar perturbations:

$$\begin{aligned} \delta\tilde{g}_{00} &= -2\Psi - \frac{2\alpha_{\text{dis}}\dot{\Phi}_{\text{bg}}}{M_*^4} \delta\dot{\Phi}, \\ \delta\tilde{g}_{0i} &= -\frac{\alpha_{\text{dis}}\dot{\Phi}_{\text{bg}}}{M_*^4} \partial_i \delta\Phi, \quad \delta\tilde{g}_{ij} = -2a^2 \Phi_N \delta_{ij}. \end{aligned}$$

The last equality uses a homogeneous background; the product $\partial_i \delta\Phi \partial_j \delta\Phi$ starts at second perturbative order. These lapse and shift mixings show why a nonzero-gradient PPN analysis requires the explicit matter action, a local background solution, source boundary conditions, and a choice of physical coordinates. No unique fifth force or preferred-frame PPN parameters follow from the rank-one metric alone.

50.4 Minimal cosmological perturbation equations

Use Newtonian gauge

$$ds^2 = -(1 + 2\Psi)dt^2 + a^2(t)(1 - 2\Phi_N)d\mathbf{x}^2$$

and Fourier wavenumber k . For the source-free minimal scalar, define

$$\delta\rho_\Phi = \dot{\Phi} \delta\dot{\Phi} - \dot{\Phi}^2 \Psi + V'(\Phi)\delta\Phi,$$

$$\delta p_\Phi = \dot{\Phi} \delta\dot{\Phi} - \dot{\Phi}^2 \Psi - V'(\Phi)\delta\Phi.$$

The scalar perturbation equation is

$$\delta\ddot{\Phi} + 3H\delta\dot{\Phi} + \left(\frac{k^2}{a^2} + V''\right)\delta\Phi = \dot{\Phi}(\dot{\Psi} + 3\dot{\Phi}_N) - 2V'\Psi.$$

The independent linear Einstein constraints may be written as

$$\frac{k^2}{a^2}\Phi_N + 3H(\dot{\Phi}_N + H\Psi) = -\frac{\delta\rho_\Phi}{2M_{\text{Pl}}^2},$$

$$\dot{\Phi}_N + H\Psi = \frac{\dot{\Phi}\delta\Phi}{2M_{\text{Pl}}^2},$$

and the canonical scalar has no linear anisotropic stress:

$$\Phi_N - \Psi = 0.$$

At a constant periodic vacuum, $\dot{\Phi} = V' = 0$, the scalar and metric sectors decouple at first order and

$$\delta\ddot{\Phi} + 3H\delta\dot{\Phi} + \left(\frac{k^2}{a^2} + \frac{\Lambda^4}{f_a^2}\right)\delta\Phi = 0.$$

These equations close only the vacuum minimal perturbation problem. Once $S_m[\tilde{g}, \psi_m]$ is restored, its density, velocity, pressure, anisotropic stress, and kinetic principal symbol are required. Consequently the following quantities remain undefined for the full ATHENA model:

$$G_{\text{eff}}(k, a), \quad \eta(k, a) = \frac{\Phi_N}{\Psi}, \quad f\sigma_8(z),$$

as do the transfer functions used by CMB, lensing, BAO, and galaxy surveys.

50.5 Completion matrix

Table 50.1: Result of the Stage 5.1 mathematical work.

Task	Result	Remaining boundary
Action inventory	COMPLETED	Every specified term and placeholder is separated.
Minimal ADM/constraint analysis	COMPLETED	Three local gravity–scalar DOF.
Total coupled DOF	OPEN (C)	S_m , S_{top} , and $S_{>4}$ kinetic structures absent.
Minimal scalar/tensor stability	COMPLETED	$M_{\text{Pl}}^2 > 0$, canonical scalar coefficient positive, $c_s^2 = c_{T,g}^2 = 1$.

Task	Result	Remaining boundary
Optional G_2 region	COMPLETED	Applies only to the isolated displayed
	CONDITIONALLY	$P(X)$ ansatz.
Full coupled stability/hyperbolicity	OPEN (C)	Matter and higher-operator Hessians absent.
T10 mathematical condition	COMPLETED	Physical gauge normalization and coefficient quantization remain B.
Parameter provenance/circularity	COMPLETED	No calibrated quantity is promoted to an independent prediction.
Minimal FLRW equations	COMPLETED	Source-free numerical integrator supplied.
Full ATHENA background	OPEN (C)	Matter, topological, gauge, and higher EFT sectors unspecified.
Constant-vacuum PPN limit	COMPLETED	General nonzero-gradient PPN/fifth force remains open.
Minimal vacuum perturbations	COMPLETED	Full matter growth, slip, lensing, and $f\sigma_8$ remain open.

Chapter 51

Independent Observational Update — 25 August 2026

51.1 Scope and status vocabulary

This chapter reports observational facts available by 25 August 2026 and keeps them separate from ATHENA interpretations. It does not alter the frozen V15.3.1 action. The labels used below distinguish derived and conditional predictions, calibrations, fits, historical targets, open mappings, constrained claims, and quantities that are not yet testable.

51.2 V15.3.1 to V15.3.2 observational update

Table 51.1: Observational facts and their controlled relevance to ATHENA.
Reference keys are listed in Section 53.5.

Date	Source and observable	Observational result	ATHENA relevance and status	Impact on ATHENA
2020	Planck 2018 base- Λ CDM parameters [S5-1]	$H_0 = 67.4 \pm 0.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and $\Omega_m = 0.315 \pm 0.007$ in the stated model.	External cosmological reference; not an ATHENA derivation.	Constrains any completed background model but does not by itself test the phenomenological H_{obs} mapping.
2022	SH0ES distance ladder [S5-2]	$H_0 = 73.04 \pm 1.04 \text{ km s}^{-1} \text{ Mpc}^{-1}$.	The discrepancy with early-universe inference motivates, but does not derive, the ATHENA Hubble ansatz.	α_H remains a calibration and cannot be validated by the datum used to set it.
2024	JWST/HST Cepheid cross-check [S5-3]	Combined JWST estimate $H_0 = 72.6 \pm 2.0$ and full HST estimate $73.2 \pm 0.9 \text{ km s}^{-1} \text{ Mpc}^{-1}$ in that analysis.	Independent observational context for the distance ladder.	Does not supply a covariant derivation of $H_{\text{obs}}(z)$.
2025	DESI DR2 BAO and cosmology [S5-4,S5-5]	Three years and about fourteen million galaxy/quasar measurements; combined analyses show a dataset-dependent preference for evolving dark energy, approximately $2.8\text{--}4.2\sigma$ when supernova samples are included.	Potentially constrains a completed ATHENA background; currently not yet testable .	No ATHENA-specific $w(z)$ or joint likelihood is derived.

Date	Source and observable	Observational result	ATHENA relevance and status	Impact on ATHENA
2026	DESI DR2 Lyman- α full-shape/AP release [S5-6]	Tighter high-redshift Lyman- α constraints, with central behavior closer to Λ CDM than some BAO-only combinations.	Neutral observational update.	Prevents treating one DESI combination as unique confirmation of ATHENA.
2025	ACT DR6 CMB parameters [S5-7,S5-8]	Representative combinations give $H_0 \simeq 68.2 \pm 0.4$; the base Λ CDM fit is good and tested extensions do not resolve the high local value.	Constrains future ATHENA background and perturbation calculations.	The frozen formulation has no CMB transfer functions, so no direct likelihood test is presently possible.
2026	DES Year 6 3×2 -point analysis [S5-9]	About 140 million source shapes and nine million lens positions over roughly 5000 square degrees, with percent-level late-time clustering constraints.	Directly relevant to lensing, slip, and growth, but the ATHENA forward map is open.	Not yet computable from V15.3.1.
2026	Euclid Quick releases [S5-10]	Quick-release imaging and catalogues are public, but the core cosmology weak-lensing/clustering release and associated parameter results are not yet available at the audit date.	No Euclid cosmological result can yet support or disfavor ATHENA.	Status is neutral and prospective.
2022	MICROSCOPE universality of free fall [S5-11]	$\eta(\text{Ti, Pt}) = (-1.5 \pm 2.3_{\text{stat}} \pm 1.5_{\text{syst}}) \times 10^{-15}$, consistent with zero.	Conditionally applicable to non-universal matter couplings.	The frozen universal disformal metric needs a sourced scalar background before this bound can be mapped to α/M_*^4 .
2003	Cassini Shapiro delay [S5-12]	$\gamma_{\text{PPN}} - 1 = (2.1 \pm 2.3) \times 10^{-5}$.	Conditionally applicable.	ATHENA has no general weak-field solution or PPN reduction, so the datum is not yet a numerical bound on α_{dis} .
2021–2025	LVK propagation and strong-field tests [S5-13,S5-14]	No statistically significant deviation from general relativity was found in the reported dispersion, polarization, or waveform tests.	The frozen Einstein-frame result $c_{T,g} = 1$ is consistent; binary phase and matter-frame propagation remain open.	Historical ATHENA phase numbers are not restored as predictions.
2026	Fermilab Muon $g-2$ final result [S5-15]	$a_\mu^{\text{Exp}} = 0.001165920715(145)$, a 127-ppb measurement.	ATHENA supplies no renormalized loop calculation for a_μ .	Not yet testable ; numerical proximity is not a prediction.
2026	CMS Higgs-pair search [S5-16]	The reported 95% intervals include $-2.5 < \kappa_\lambda < 9.4$ and $0.02 < \kappa_{2V} < 2.1$.	Relevant to any explicit ATHENA Higgs interaction, but no such coefficient is derived in V15.3.1.	Higgs statements remain open or historical targets.
2016	SPARC rotation-curve database [S5-17]	Public photometry and rotation curves for 175 late-type galaxies.	The reported ATHENA $\chi^2 = 731.1$ is a fit claim without archived likelihood, covariance, priors, and reproduction package.	Fit/pending independent reproduction.
2021–2025	Event Horizon Telescope M87 polarization [S5-18,S5-19]	Polarization morphology and inferred magnetic structure evolve between epochs, including a reversal of the spiral pattern.	Qualitative toroidal appearance is non-unique; a fixed ATHENA magnetic profile has no time-dependent forward model.	Not yet testable ; a naive static reading is qualitatively in tension, not falsified.

Date	Source and observable	Observational result	ATHENA relevance and status	Impact on ATHENA
2025	Cosmic dipole review [S5-20]	Several matter tracers exhibit a dipole amplitude larger than the simple kinematic expectation, with unresolved systematics and interpretations.	This is not documentation of the ATHENA-specific 134° claim.	The 134° number remains a historical/open target without a reproducible dataset, estimator, or trial-factor analysis.

51.3 Domain conclusions

Planck and ACT infer a lower H_0 within base Λ CDM, while calibrated late-time distance ladders infer a higher value. DESI DR2 combinations provide important evidence about expansion history, but the preference for evolving dark energy depends on supernova compilation and model space. DES Year 6 supplies stronger late-time lensing and clustering information. Euclid had not released its core cosmological parameter analysis by the audit date.

The frozen expression

$$H_{\text{obs}}(z) = H_{\Lambda}(z)[1 + \alpha_H \Omega_{\text{coll}}(z)]$$

is a phenomenological ansatz. V15.3.1 does not derive $\Omega_{\text{coll}}(z)$ from a covariant solution or provide its perturbations, and it calibrates α_H using the present-day normalization. The same $z = 0$ discrepancy is therefore not an independent prediction.

Solar-System, equivalence-principle, lensing, cluster, and growth measurements are stringent, but application to the pure-disformal metric requires a scalar profile, source solution, matter content, boundary conditions, perturbation equations, and survey forward model. A constant scalar returns the general-relativistic weak-field limit at leading order; this special-background check is not a general PPN theory.

The Einstein–Hilbert tensor sector gives $c_{T,g} = 1$ on the minimal background already considered. This is compatible with multimessenger and LVK propagation tests, but it does not determine arbitrary matter-frame propagation or binary waveforms. Precision Higgs, muon, SPARC, M87, magnetic, and dipole observations cannot be turned into ATHENA tests without the corresponding amplitudes, source evolution, selection functions, and likelihoods. The Derrick obstruction and historical status of target particle masses remain unchanged.

Chapter 52

ATHENA Claim and Parameter Status Audit

52.1 Observational claim-status table

Table 52.1: Stage 5 status of major ATHENA observational quantities.

Quantity or claim	Status	Audit conclusion
$\alpha_H = 0.086053412 \dots$	CALIBRATION	Chosen from the present-day Hubble normalization; distinct from every other α .
$\alpha_{\text{eff}}^{\text{gal}} = 0.36 \pm 0.03$	FIT	Galaxy-scale phenomenological value; not derived from α_{dis} or α_H .
α_{RG}	CONDITIONAL PREDICTION	The nonzero fixed point follows algebraically from the displayed beta function after accepting that beta function and its coefficients.
α_{dis}	OPEN	Coefficient of the physical disformal metric; no independent numerical value or observational determination is supplied.
α_{cal}	HISTORICAL TARGET	Calibration label retained for provenance; it is not a universal coupling.
β_0	CONDITIONAL PREDICTION	Algebraic root of the equilibrium postulate, not a Casimir minimization result.
γ_0	CONDITIONAL PREDICTION	Defined by the normalization $\gamma_0 = \beta_0^2/2$ once the postulate is accepted.
R_c	CALIBRATION	Requires the external length $L_* = 1$ kpc in the displayed normalization.
f_a	CALIBRATION	Requires R_c and $\kappa_f = 0.10$; topology alone does not fix the numerical energy scale.
a_0	CALIBRATION	Requires κ_a and $\alpha_{\text{eff}}^{\text{gal}}$ and is matched to the historical acceleration scale.
$H_{\text{obs}}(z)$	CONDITIONAL PREDICTION	Conditional only after the phenomenological ansatz and a specified $\Omega_{\text{coll}}(z)$ are accepted; not derived from the action.
SPARC $\chi^2 = 731.1$	FIT	Unreproduced model-fit claim; not an independent prediction.
M87 profile	NOT YET TESTABLE	No frozen time-dependent plasma/magnetic forward model or likelihood.
Lensing and gravitational slip	OPEN	No solved potentials or slip relation.
Structure growth $f\sigma_8$	OPEN	No linear perturbation system or survey forward model.

Quantity or claim	Status	Audit conclusion
GW phase targets	HISTORICAL TARGET	No binary solution or waveform derivation in V15.3.1.
134° dipole	HISTORICAL TARGET	The underlying statistic and reproducible independent dataset are not identified.
Magnetic profiles	NOT YET TESTABLE	Shape-level ansatz without amplitude, source evolution, or likelihood.
Muon $g - 2$	NOT YET TESTABLE	No ATHENA loop calculation.
Higgs couplings	OPEN	No complete gauge-invariant interaction and coefficient derivation.
Particle masses and mass ladder	HISTORICAL TARGET	Numerical targets are not derived masses; Derrick obstruction remains active.

52.2 Parameter provenance and circularity

Table 52.2: Provenance of parameters entering the principal active equations. Derived here means algebra after stated model inputs, not independent empirical validation.

Symbol	Definition / dimension	Value in source	Provenance	Independence and circularity
M_*	EFT scale; mass	Unfixed	Normalizes scalar energy and disformal operator.	Unconstrained; only $\alpha_{\text{dis}}/M_*^4$ is relevant to the metric.
α_{dis}	Disformal coefficient; dimensionless	Unfixed	$\tilde{g}_{\mu\nu} = g_{\mu\nu} - \alpha_{\text{dis}} M_*^{-4} \partial_\mu \Phi \partial_\nu \Phi$.	Not derived, fitted, or independently measured.
f_a	Compact period scale; mass	$2.7 \times 10^{-29} \text{ eV}$ target	$\kappa_f \hbar c / R_c$, with $\kappa_f = 0.10$.	Calibrated chain; not topologically fixed.
Λ	Periodic-potential scale; mass	Unfixed	$V = \Lambda^4 [1 - \cos(\Phi/f_a)]$.	Unconstrained; fixes $m_\Phi^2 = \Lambda^4 / f_a^2$.
$c_{\Phi F}$	Topological-operator coefficient; convention dependent	Unfixed	Optional $\Phi F \tilde{F}$ operator.	Global quantization, gauge normalization, and independent value remain open.
β_0	Aspect parameter; dimensionless	0.140814...	Selected root of the postulate $\beta_0(2 - \beta_0) = \pi/12$.	Conditionally derived from a postulate; not fitted to Stage 5 observations.
γ_0	Normalization; dimensionless	0.009914...	$\gamma_0 = \beta_0^2/2$.	Conditionally defined, not independently measured.
L_*	Reference length; length	1 kpc	External normalization of R_c .	Calibration input.
R_c	Galactic scale; length	23.67 kpc	$R_c = L_* 10 / (3\beta_0)$.	Calibrated through L_* ; later galactic use is not independent.
κ_f	Scale coefficient; dimensionless	0.10	Phenomenological normalization.	Calibration; unconstrained.
κ_a	Acceleration coefficient; dimensionless	1.44×10^{-5}	Chosen in the a_0 normalization.	Calibrated to the acceleration target.
a_0	Acceleration scale; L/T^2	$1.103 \times 10^{-10} \text{ m s}^{-2}$	Uses R_c , β_0 , κ_a , and $\alpha_{\text{eff}}^{\text{gal}}$.	Calibration; cannot independently predict the same acceleration scale.
$\alpha_{\text{eff}}^{\text{gal}}$	Galaxy effective coefficient; dimensionless	0.36 ± 0.03	Phenomenological galaxy fit.	Fitted; distinct from α_H and α_{dis} .
α_H	Hubble-ansatz coefficient; dimensionless	0.0860534...	Present-day normalization of H_{obs} .	Calibrated using the Hubble discrepancy; circular for the same $z = 0$ datum.
α_{RG}	Running coupling; dimensionless	48 for $\gamma_{\text{str}} = 0$ at the displayed fixed point	Algebraic root of the postulated beta function.	Conditional, not identified with the disformal or galactic coefficient.

Symbol	Definition / dimension	Value in source	Provenance	Independence and circularity
α_{cal}	Calibration label; dimensionless	Context dependent	Historical phenomenological normalization.	Not a derived universal constant.
γ_{str}	String/beta-function parameter; dimensionless	Often set to 0 in the example	Input to the displayed beta function.	Unconstrained model input.
$\Omega_{\text{coll}}(z)$	Collapsed fraction; dimensionless	Unspecified	Input to the Hubble ansatz.	No covariant definition or independent frozen calculation.

The parameters obey no derived equality $\alpha_H = \alpha_{\text{eff}}^{\text{gal}} = \alpha_{\text{dis}} = \alpha_{\text{RG}}$. They label different equations and provenances. The arithmetic

$$67.4(1 + 0.36) \simeq 91.7$$

is only a diagnostic multiplication. It is not an ATHENA prediction because 0.36 is a galactic fit and no equation maps it to the Hubble observable.

52.3 Pure-disformal constraint audit

Table 52.3: Applicability of existing constraints to the frozen pure-disformal sector.

Observation/check	Classification	Reason
$\Delta_D > 0$ and invertibility	DIRECTLY CONSTRAINABLE	This is an algebraic admissibility condition on any proposed background, not an observational parameter bound.
Einstein-frame tensor speed	DIRECTLY CONSTRAINABLE	The minimal Einstein–Hilbert tensor principal part gives $c_{T,g} = 1$; current GW observations are consistent with it.
Matter-frame GW/photon comparison	CONDITIONALLY CONSTRAINABLE	Requires N_D , a background $\dot{\Phi}$, and a specified matter clock.
MICROSCOPE equivalence principle	CONDITIONALLY CONSTRAINABLE	Requires a sourced scalar profile and the composition dependence of the full matter action.
Cassini/PPN and redshift tests	CONDITIONALLY CONSTRAINABLE	Require weak-field solutions, boundary conditions, and PPN matching.
Laboratory/collider disformal searches	CONDITIONALLY CONSTRAINABLE	Published limits assume particular scalar spectra, operator normalizations, and matter completions not fixed by V15.3.1.
BAO, supernovae, CMB $H(z)$	NOT YET COMPUTABLE	No solved covariant cosmological background or likelihood map.
Weak lensing and gravitational slip	NOT YET COMPUTABLE	No two-potential perturbation equations or photon/matter transfer functions.
Growth, clusters, and $f\sigma_8$	NOT YET COMPUTABLE	No perturbation, bias, nonlinear, or survey forward model.
SPARC/M87 dynamics	NOT YET COMPUTABLE	The covariant force law and source solutions are absent; existing expressions are phenomenological.
Binary inspiral and waveform phase	NOT YET COMPUTABLE	No compact-object solution, scalar charge, radiation sector, or waveform.

Chapter 53

Open Problems, Falsification, and Release Controls

53.1 Open-problem status

The classifications are: A, derivable from the frozen theory; B, requires an additional physical assumption; C, underdetermined by the frozen source; and D, mathematically obstructed by the stated construction.

Table 53.1: Stage 5 open-problem inventory.

Problem	Class	Maximum justified conclusion
Total propagating degrees of freedom	C	Einstein gravity contributes two tensor modes and the displayed canonical compact scalar contributes one local scalar mode. The total cannot be counted because S_m , S_{top} , and $\mathcal{L}^{(>4)}$ are not fully specified.
Global $\Phi F\tilde{F}$ normalization	B	Local variation is meaningful; quantum single-valuedness needs the gauge group, bundle, trace convention, topological-charge normalization, boundary conditions, and quantized coefficient.
Full coupled quadratic stability	C	The scalar and minimal tensor checks do not determine the Hessian and constraints of the unspecified sectors.
General weak-field/fifth-force/PPN theory	C	A source solution and full matter action are missing.
Lensing and gravitational slip	C	The two metric potentials and perturbation equations are missing.
Structure growth	C	Background and linear/nonlinear perturbation dynamics are missing.
Solved cosmological backgrounds	C	Initial data, matter content, parameters, and complete operators are insufficiently specified.
Covariant H_{obs} matching	B	Would require an additional physical identification between the covariant solution and the phenomenological collapsed-fraction ansatz.
Binary/GW waveform calculation	C	Compact-object solutions, scalar charges, radiation boundary conditions, and the complete dynamical sectors are absent.
Spacetime realization of internal winding	B	Requires a stated bundle or dynamical map from internal configuration-space T^2 data to four-dimensional spacetime observables.

Problem	Class	Maximum justified conclusion
Particle masses beyond Derrick obstruction	D	The displayed static two-derivative scalar functional does not support the claimed stable localized particle construction. No permitted replacement is present in V15.3.1.

53.2 No-go and falsification screen

Table 53.2: Controlled comparison of active or historical ATHENA statements with current tests.

Claim	Observation/test	Status	Conclusion
$c_{T,g} = 1$	GW propagation tests	CONSISTENT	The minimal Einstein-frame tensor result is compatible; this does not close matter-frame propagation or waveform questions.
$H_{\text{obs}}(z)$ ansatz	BAO, supernovae, CMB, local H_0	NOT YET TESTABLE	The calibrated $z = 0$ normalization and undefined $\Omega_{\text{coll}}(z)$ prevent an independent full-redshift test.
SPARC statistic	Public SPARC database	NOT YET TESTABLE	The reported fit lacks the complete reproducibility package.
Lensing/dark shadow claims	DES Y6 and other lensing data	NOT YET TESTABLE	No lensing/slip forward model exists.
Static M87 magnetic form	Multi-epoch EHT polarization	NOT YET TESTABLE	Observed evolution is a qualitative warning for a static interpretation, but there is no valid ATHENA likelihood to falsify.
GW phase numbers	LVK waveform tests	NOT YET TESTABLE	The numbers are historical targets, not V15.3.1 waveform predictions.
134° dipole	Dipole surveys and reviews	NOT YET TESTABLE	The exact ATHENA estimator, sample, uncertainty, and look-elsewhere treatment are not documented.
Muon $g - 2$ target	Fermilab final result	NOT YET TESTABLE	No model amplitude connects ATHENA to a_μ .
Higgs quantities	LHC coupling/search results	NOT YET TESTABLE	No complete interaction coefficient or cross-section prediction.
Stable scalar particle/mass construction	Derrick scaling	OBSTRUCTED (D)	The obstruction applies to the displayed static two-derivative construction; no workaround is introduced in this release.

The frozen source does not prove a ghost, Ostrogradsky instability, loss of hyperbolicity, or

excessive total degree count: unspecified sectors make those questions underdetermined. Likewise, missing global data for $\Phi F\tilde{F}$ establish conditionality, not a proof of inconsistency.

53.3 V15.3.1 to V15.3.2 changelog

- Added the Stage 5.1 active-action inventory and ADM constraint analysis, including the exact three-mode count for the closed lower-order gravity–scalar subsector and the explicit boundary preventing a total count.
- Added minimal scalar/tensor quadratic actions, the conditional G_2 stability region, the coupled-hyperbolicity boundary, and the complete T10 action-shift/normalization derivation.
- Added the closed source-free FLRW equations, a conditional numerical integrator, the constant-vacuum PPN result, and the minimal cosmological perturbation equations.
- Added independently sourced observational context through 25 August 2026.
- Added explicit separation of observational facts from ATHENA interpretations.
- Added claim-status, parameter-provenance, circularity, pure-disformal-applicability, open-problem, and falsification tables.
- Preserved α_H , $\alpha_{\text{eff}}^{\text{gal}}$, α_{dis} , α_{RG} , and α_{cal} as distinct quantities.
- Preserved all Stage 4B equations, parameters, OPEN classifications, the Derrick obstruction, and the historical status of numerical GW targets.
- Added release limitations and reproducibility requirements. No new physics, parameter tuning, or Stage 5 repair was introduced.

53.4 Reproducibility and limitations

Observational comparisons require immutable data releases, catalogue and mask versions, covariance matrices, likelihood code, priors, nuisance models, environment locks, exact commands, and output checksums. Until those artifacts exist for an ATHENA forward model, proximity between an ATHENA target and a measured number is not independent evidence.

This release is not experimentally validated; it is not a Standard Model derivation, an emergent-gravity derivation, a complete cosmological model, a complete particle-mass theory, or a complete perturbation theory. Total degrees of freedom, global topological normalization, general stability, weak-field gravity, lensing, growth, cosmological backgrounds, binary waveforms, internal-winding realization, and particle masses remain unresolved at the statuses listed above.

53.5 Stage 5 observational references

- [S5-1] Planck Collaboration, Planck 2018 results. VI. Cosmological parameters, *Astron. Astrophys.* **641**, A6 (2020), arXiv:1807.06209.
- [S5-2] A. G. Riess et al., A Comprehensive Measurement of the Local Value of the Hubble Constant with $1 \text{ km s}^{-1} \text{ Mpc}^{-1}$ Uncertainty from the Hubble Space Telescope and the SH0ES Team, *Astrophys. J. Lett.* **934**, L7 (2022), arXiv:2112.04510.
- [S5-3] A. G. Riess et al., JWST Validates HST Distance Measurements: Selection of JWST Data Explains Differences in the Hubble Constant, *Astrophys. J.* **977**, 120 (2024), arXiv:2408.11770.

- [S5-4] DESI Collaboration, DESI DR2 Results II: Measurements of Baryon Acoustic Oscillations and Cosmological Constraints, arXiv:2503.14738 (2025).
- [S5-5] DESI Collaboration, DESI DR2 Results V: Extended Dark Energy Analysis, arXiv:2503.14743 (2025).
- [S5-6] DESI Collaboration, DESI DR2 Results IV: Alcock–Paczynski Measurements from the Lyman Alpha Forest and Cosmological Constraints, official DESI release, 30 July 2026.
- [S5-7] ACT Collaboration, The Atacama Cosmology Telescope: DR6 Constraints on Extended Cosmological Models, arXiv:2503.14454 (2025).
- [S5-8] ACT Collaboration, The Atacama Cosmology Telescope: DR6 Power Spectra, Likelihoods and Λ CDM Parameters, ACT DR6 collaboration release (2025).
- [S5-9] DES Collaboration, Dark Energy Survey Year 6 Results: Cosmological Constraints from Galaxy Clustering and Weak Lensing, arXiv:2601.14559 (2026).
- [S5-10] Euclid Collaboration and ESA, Euclid Quick Data Releases Q1 and Q2, official archive and mission releases (2025–2026).
- [S5-11] MICROSCOPE Collaboration, MICROSCOPE Mission: Final Results of the Test of the Equivalence Principle, *Phys. Rev. Lett.* **129**, 121102 (2022), arXiv:2209.15487.
- [S5-12] B. Bertotti, L. Iess, and P. Tortora, A test of general relativity using radio links with the Cassini spacecraft, *Nature* **425**, 374–376 (2003).
- [S5-13] LIGO Scientific, Virgo, and KAGRA Collaborations, Tests of General Relativity with GWTC-3, *Phys. Rev. D* **106**, 042003 (2022), arXiv:2112.06861.
- [S5-14] LIGO Scientific, Virgo, and KAGRA Collaborations, O4a tests of general relativity and compact-binary catalog documentation (2025).
- [S5-15] Muon $g - 2$ Collaboration, final Fermilab measurement of the positive muon magnetic anomaly, official publication and arXiv:2606.17323 (2026).
- [S5-16] CMS Collaboration, combined Higgs-boson pair-production search and coupling constraints, official CMS release, 10 August 2026.
- [S5-17] F. Lelli, S. S. McGaugh, and J. M. Schombert, SPARC: Mass Models for 175 Disk Galaxies with Spitzer Photometry and Accurate Rotation Curves, *Astron. J.* **152**, 157 (2016), arXiv:1606.09251.
- [S5-18] Event Horizon Telescope Collaboration, First M87 Event Horizon Telescope Results. VIII. Magnetic Field Structure near the Event Horizon, *Astrophys. J. Lett.* **910**, L13 (2021), arXiv:2105.01173.
- [S5-19] Event Horizon Telescope Collaboration, multi-epoch M87 polarization update, official collaboration release, 16 September 2025.
- [S5-20] N. Secrest et al., Colloquium: The cosmic dipole anomaly, *Rev. Mod. Phys.* (2025).

Part VI

Volume IV: References, Appendices, and Status of Claims

Chapter 54

References

54.1 Experimental and Observational References

54.1.1 Cosmology and Large-Scale Structure

1. Planck Collaboration, Planck 2018 Results. VI. Cosmological Parameters, *Astron. Astrophys.* 641, A6 (2020).
2. DESI Collaboration, Extended Dark Energy Analysis using DESI DR2 BAO Measurements, arXiv:2503.14743 (2025).
3. Riess, A.G. et al., JWST Observations Reject Unrecognized Crowding of Cepheid Photometry as an Explanation for the Hubble Tension at 8σ , *Astrophys. J. Lett.* (2026).
4. Lelli, F., McGaugh, S.S., Schombert, J.M., SPARC: Mass Models for 175 Disk Galaxies, *Astron. J.* 152, 157 (2016).
5. McGaugh, S.S., Lelli, F., Schombert, J.M., Radial Acceleration Relation in Rotationally Supported Galaxies, *Phys. Rev. Lett.* 117, 201101 (2016).
6. Capozziello, S. et al., Evidence of dynamical dark energy from the DESI DR2 Lyman α forest, *Astron. Astrophys.* (2026).
7. Krishnan, J.R. et al., Statistics meet systematics: Resolution of the massive early JWST galaxy tension, *Phys. Rev. D* (2026).
8. Secrest, N. et al., Colloquium: The cosmic dipole anomaly, *Rev. Mod. Phys.* (2025).
9. Bashir, M. et al., A Reassessment of the Cosmic Dipole Anomaly, *Astrophys. J.* (2026).
10. Murai, K. et al., Isotropic cosmic birefringence from an oscillating axionlike field, *Phys. Rev. D* (2025).
11. Anonymous, Dark Matter as Geometric Tension: Galactic Rotation Curves from Potential Gradient Scaling, Zenodo 10.5281/zenodo.18637244 (2026).

54.1.2 Galactic and Astrophysical Observations

1. Event Horizon Telescope Collaboration, First M87* Event Horizon Telescope Results. VIII. Magnetic Field Structure, *Astrophys. J. Lett.* 910, L13 (2021).
2. Loi, F. et al., The MeerKAT Fornax Survey VII: Characterisation of the Fornax Cluster's Magnetic Field, arXiv:2604.18338 (2026).

3. Dinsmore, J.T. et al., IXPE Polarizations of the Lighthouse Pulsar, Trail, and Filament, *Astrophys. J.* **1005**, 237 (2026), arXiv:2604.22914.
4. Stewart, R.E. et al., Vacuum birefringence and the polarized X-ray emission from a radio magnetar, *Nature* **656**, 590–594 (2026); arXiv:2509.19446.
5. Gómez, J.L. et al., Spatially resolved polarization swings in the supermassive binary black hole candidate OJ 287, *Astron. Astrophys.* (2026).
6. Hees, A. et al., Combined Solar system and rotation curve constraints on MOND, *Mon. Not. R. Astron. Soc.* 455, 449 (2016).

54.1.3 Gravitational Waves and High-Energy Physics

1. LIGO/Virgo Collaboration, Observation of Gravitational Waves from a Binary Black Hole Merger, *Phys. Rev. Lett.* 116, 061102 (2016).
2. Fermilab Muon $g-2$ Collaboration, Measurement of the Positive Muon Anomalous Magnetic Moment, *Phys. Rev. Lett.* 131, 161802 (2023).
3. Liberal, I. and Manjavacas, A., Synthetic Crystal Rotation with Spacetime Metamaterials, *Phys. Rev. Lett.* **136**, 146903 (2026); arXiv:2506.02495.

54.1.4 Quantum and Optical Physics

1. Optica, Quantum Entanglement from Sunlight, *Optica* 13, 1508 (2026).

54.2 Theoretical and Foundational References

1. Dirac, P.A.M., *The Principles of Quantum Mechanics*, Oxford (1930).
2. Misner, C.W., Thorne, K.S., Wheeler, J.A., *Gravitation*, Freeman (1973).
3. Weinberg, S., *The Quantum Theory of Fields*, Cambridge (1995–2000).
4. Peskin, M.E., Schroeder, D.V., *An Introduction to Quantum Field Theory*, Westview (1995).
5. Nakahara, M., *Geometry, Topology and Physics*, IOP (1990).
6. Horndeski, G.W., Second-Order Scalar-Tensor Field Equations, *Int. J. Theor. Phys.* 10, 363 (1974).
7. Arnowitt, R., Deser, S., Misner, C.W., The Dynamics of General Relativity, in *Gravitation: An Introduction to Current Research*, Wiley (1962); arXiv:gr-qc/0405109.
8. Garriga, J., Mukhanov, V.F., Perturbations in k-inflation, *Phys. Lett. B* 458, 219–225 (1999); arXiv:hep-th/9904176.
9. Domènech, G. et al., Derivative-dependent metric transformation and physical degrees of freedom, *Phys. Rev. D* 92, 084027 (2015); arXiv:1507.05390.
10. Minamitsuji, M., Disformal transformation of cosmological perturbations, *Phys. Lett. B* 737, 139–150 (2014); arXiv:1409.1566.
11. Sakharov, A.D., Vacuum Quantum Fluctuations in Curved Space, *Sov. Phys. Dokl.* 12, 1040 (1968).

Appendix A

Mathematical Symbols

Symbol	Chapter	Description
T^2	All	Toroidal manifold
φ	0	Periodic part of a chosen real lift of the compact scalar
Φ	All	Real compact scalar of canonical mass dimension one
$U = e^{i\Phi/f_a}$	0–I	Equivalent unit-modulus representation of Φ
$\mathcal{O}_\Omega$	0–I	Nondynamical organizational diagnostic $\sqrt{\Omega_{\text{cand}}}$
Φ_n	0–I	Local vacuum representatives $2\pi n f_a$
w	0–II	Winding number
β_0	0–I	Dimensionless equilibrium-postulate parameter
Z_0	0–I	Conditional ratio $5\beta_0/3$
γ_0	0–I	Conditional ratio $\beta_0^2/2$
R_c	0–III	Calibrated screening length
L_*	0–III	Phenomenological length used to define R_c
f_a	0–II	Dimension-one compact-field periodicity scale
κ_f	0–II	Dimensionless calibration in $f_a = \kappa_f \hbar c / R_c$
κ_a	0–III	Dimensionless acceleration-scale calibration
a_0	I	Calibrated MOND-scale target
M_*	0–II	EFT normalization/suppression mass
E_*	0–II	Explicit energy scale in the soliton-mass ansatz
α	I–III	Dimensionless disformal coupling
α_*	0–II	Conditional root of the truncated beta ansatz
α_{cal}	0–II	Historical calibrated value 89.4
α_{eff}	I–III	Environment-dependent phenomenological coupling
$\alpha_{\text{eff}}^{\text{gal}}$	0–III	Unreproduced galactic/SPARC screened-fit target
α_H	0, III	Conditional Hubble-ansatz calibration
γ_{str}	0–II	Dimensionless string-susceptibility input
$n(z)$	I–III	Dimensionless phenomenological effective-dimension ansatz
$g_{\mu\nu}$	I	Spacetime metric
$\tilde{g}_{\mu\nu}$	I	Pure-disformal physical matter metric
Δ_D	I	$1 + 2\alpha X/M_*^4$; determinant ratio and admissibility factor
G_N	I	Constant g -frame Newton coupling $1/(8\pi M_{\text{Pl}}^2)$
$\tilde{H}$	I–III	Physical homogeneous Hubble rate H/N_D
H_{vol}	0–I	Spatial-gradient volume rate
m_w	II	Conditional mass symbol for a proposed winding sector
H_{topo}	0	Topological entropy
$\hat{C}$	0	Consciousness operator

Symbol	Chapter	Description
θ_{res}	I, III	Residual dipole misalignment

Appendix B

Physical Constants

Constant	Symbol	Value
Speed of light	c	$2.99792458 \times 10^8 \text{ m/s}$
Planck constant	h	$6.62607015 \times 10^{-34} \text{ J}\cdot\text{s}$
Reduced Planck constant	$\hbar$	$1.054571817 \times 10^{-34} \text{ J}\cdot\text{s}$
Gravitational constant	G	$6.67430 \times 10^{-11} \text{ m}^3\text{kg}^{-1}\text{s}^{-2}$
Reduced Planck mass	M_{Pl}	$4.34136 \times 10^{-9} \text{ kg}$
Elementary charge	e	$1.602176634 \times 10^{-19} \text{ C}$
Electron mass	m_e	$9.1093837 \times 10^{-31} \text{ kg}$
Proton mass	m_p	$1.67262192 \times 10^{-27} \text{ kg}$

Here M_{Pl} denotes the reduced Planck mass,

$$M_{\text{Pl}} = \sqrt{\frac{\hbar c}{8\pi G}} = 4.34136 \times 10^{-9} \text{ kg},$$

consistent with $M_{\text{Pl}}^{-2} = 8\pi G$ in natural units $\hbar = c = 1$. The unreduced Planck mass $\sqrt{\hbar c/G}$ is not used in the action normalization.

Appendix C

Dimensionless Parameters

Parameter	Symbol	Value
Galactic fit target	$\alpha_{\text{eff}}^{\text{gal}}$	0.36 ± 0.03 (not reproduced)
Hubble normalization	α_H	$0.086053412 \dots$ (conditional calibration)
Truncated-beta root	α_*	48 for $\gamma_{\text{str}} = 0$
Historical UV calibration	α_{cal}	89.4 (provenance not independently documented)
Equilibrium-postulate parameter	β_0	0.1408
Conditional normalization parameter	Z_0	0.2347
Model-defined normalization parameter	γ_0	0.009914
String susceptibility	γ_{str}	0 (model input; not EFT-derived)
Acceleration calibration	κ_a	1.44×10^{-5} (target matching)
Hubble transition parameter	z_c	0.35 (phenomenological; provenance not documented)
Effective-dimension transition	z_n	0.80 (phenomenological; provenance not documented)
Historical organizational/EM diagnostic	$\mathcal{O}_0$	0.782 ± 0.015 (not a value of Φ/M_*)
Inactive V15.2 nonminimal parameter	ξ	≈ 0.14 (not part of the V15.3 action)

C.1 Parameter Provenance Summary

Parameter	Current status	Provenance in V15.3
$\alpha_{\text{cal}} = 89.4$	Historical calibration	Original dataset, normalization, and uncertainty not independently documented
$z_c = 0.35$	Phenomenological parameter	Hubble-ansatz transition redshift; source or fit not independently documented
$z_n = 0.80$	Phenomenological parameter	Fractional-dimension shape scale; source or fit not independently documented
$\mathcal{O}_0 = 0.782 \pm 0.015$	Historical diagnostic target	Calculation and uncertainty provenance not independently documented; not identified with Φ/M_*
$\xi \approx 0.14$	Inactive historical parameter	Not part of the V15.3 pure-disformal action
B_0	Phenomenological parameter	Magnetic amplitude not numerically specified or independently documented
L_B	Phenomenological parameter	Radial length whose historical numerical representation is 20 in an undeclared radial unit
a	Phenomenological parameter	Magnetic-profile shape parameter to be fitted; no value or fit is documented
$\theta_{\text{res}} \simeq 5.40^\circ$, $p = 0.0022$	Reported analysis target	Estimator, covariance, null distribution, and calculation not archived
134° dipole threshold	Proposed benchmark	Observable definition and falsification rule incomplete in the present version

Appendix D

Numerical Values

Quantity	Symbol	Value
Calibrated screening length	R_c	23.67 kpc ($L_* = 1$ kpc)
Compact-field scale	f_a	2.7×10^{-29} eV ($\kappa_f = 0.10$)
Calibrated MOND-scale target	a_0	1.1×10^{-10} m/s ²
Muon mass target	m_μ	105.66 ± 0.02 MeV
Electron mass target	m_e	0.511 MeV
Proton mass target	m_p	938.272 MeV
Tau mass target	m_τ	1.776 GeV
External local H_0 input	H_0^{local}	73.2 km/s/Mpc
External early H_0 input	H_0^{early}	67.4 km/s/Mpc
Reported dipole-misalignment target	θ_{res}	5.40°

Appendix E

Units and Conventions

Metric signature: $(-, +, +, +)$. Natural units: $c = \hbar = 1$ (restored for numerical values).

Greek indices μ, ν, ρ, σ run from 0 to 3. Latin indices i, j, k run from 1 to 3. Latin indices a, b, c are vierbein indices.

Appendix F

Status of Claims

Table F.1: ATHENA V15.3 claim-status ledger.

Claim	Mathematical status	Physical status	Evidence status	Required test
β_0	Algebraic root of an equilibrium postulate	Dimensionless model parameter	No direct measurement	Full spectral Casimir and stability calculation
TTA	Bookkeeping postulate plus open construction	Candidate representation map	None specific	Explicit functor to Lorentz/internal representations
Spin-1/2	Four torus spin structures are standard; winding map unproved	Winding-spin hypothesis	None specific	4D Dirac spectrum and embedding
Spin-statistics	Not derived from linking	Open QFT problem	Fermion statistics known independently	Local Lorentzian quantization and micro-causality
SU(3)	Candidate trefoil correspondence	Incomplete color sector	None specific	Eight modes, Lie bracket, matter reps, confinement
U(1)	Global compact shift only	Electromagnetism not derived	Standard U(1) known independently	Connection, covariant derivative, charges
SU(2)	Candidate metastable triplet	Electroweak sector incomplete	Standard SU(2) known independently	Chirality, breaking, masses, anomaly cancellation

Table F.2: ATHENA V15.3 claim-status ledger (continued).

Claim	Mathematical status	Physical status	Evidence status	Required test
Particle masses	Dimensionally closed only after E_* is added	Historical mass-target hypothesis	Targets not reproduced	Out-of-sample mass with archived solver
Higgs	Breathing-mode hypothesis	No valid 125-GeV normalization	Higgs observed independently	Microscopic scale and couplings
Three generations	Conjecture; no $w = 6$ threshold proof	Candidate truncation mechanism	Three generations observed independently	Stability spectrum and decay thresholds
α_*	Conditional proposition: 48 for $\gamma_{\text{str}} = 0$	Root of un-derived beta ansatz	No direct measurement	EFT loop derivation and scheme definition
$\alpha_{\text{cal}} = 89.4$	Not obtained from displayed beta function	Historical calibrated input	Not independently reproduced	Document correction/normalization or retire value
Topological screening	Dimensionless ansatz with Ξ_w	Candidate environment dependence	SPARC target pending	Public likelihood and cross-validation
Pure-disformal metric	Rank-one inverse and determinant proved	Matter metric postulate	No direct validation	Coupled stability, PPN, waves, and backgrounds
Topological Involution	Re-derived spatial determinant with $1 - Q_D$	Cosmological interpretation remains an ansatz	No discriminating evidence	Background and perturbation solutions
TEMP	Postulate; entropy argument did not prove powers of two	Candidate arrow/stability rule	None specific	Microscopic dynamics and Lyapunov proof

Table F.3: ATHENA V15.3 claim-status ledger (continued).

Claim	Mathematical status	Physical status	Evidence status	Required test
Emergent gravity	Einstein–Hilbert dynamics are postulated, not emergent	Emergence remains open	Compatible phenomena non-unique	Induced-gravity derivation and precision tests
Organizational–EM map	Global identity removed	Conditional diagnostic interpretation only	No operational evidence	Local or expectation-value matching calculation
Dark matter as EM shadow	No derivation from EFT solutions	Hypothesis	Model-compatible observations only	Joint lensing, dynamics, and cluster likelihood
Dark energy as vacuum pressure	No derived $w(z)$	Hypothesis	DESI trend is a convergent anomaly	Specific $w(z)$ plus growth prediction
Hubble tension	Phenomenological $H_{\text{obs}}(z)$ ansatz with $\alpha_H = 0.086053412\dots$	Conditional normalization; candidate explanation	Non-unique tension	Joint pre-registered cosmology fit
$n(z)$	Dimensionless phenomenological ansatz	Effective dimension undefined covariantly	No growth calculation	Perturbation theory and survey forward model
SPARC $\chi^2 = 731.1$	Reported target	Unreproduced model-fit claim	Pending independent reproduction	Catalogue, covariance, code, priors, checksums

Table F.4: ATHENA V15.3 claim-status ledger (continued).

Claim	Mathematical status	Physical status	Evidence status	Required test
Spectral Determinism	Axiom	Organizational assumption	Not empirical	Representation and uniqueness theorem
Rigidity and uniformization	Conjectures	Candidate topology selection	None specific	Regularized global extremization
Emergent time	Monotone clock exists but is non-unique	Proper time not derived	None specific	Lorentzian causal/clock construction
Consciousness operator	Formal hypothesis with undefined domain	Speculative analogy	No operational evidence	Observable, blinded protocol, null model

Appendix G

ATHENA Development History

Version	Date	Key Developments
V4.0	14 May 2026	OMM-TOE: Gamma-Condensate Process Cosmology
V4.9	23 May 2026	Ultimate Civilization Manifesto
V7.2	19 June 2026	Disformal Toroidal Vacuum Framework
V8.0	28 June 2026	Complete Topo-Geometric Unification
V13.1	July 2026	Final Monograph, Volumes 0–IV
V14.0	July 2026	Extended with 8 phases and observational comparisons
V15.0	July 2026	Consistency Theorem, Spectral Determinism, Emergent Time
V15.1	7 Aug 2026	Observational status, compatibility assessment, and complete monograph
V15.2	25 Aug 2026	Final scientific audit, corrected fixed point, dimensional closure, and claim-status ledger
V15.3	25 Aug 2026	Pure-disformal reformulation, compact-field ontology repair, and Stage 4B re-derivation
V15.3.1	25 Aug 2026	Minimal compact-shift corrections for the Kuramoto and local-lift phase ansatzes; explicit open quantization condition for $\Phi F \tilde{F}$
V15.3.2	25 Aug 2026	Stage 5 action/DOF, stability, T10, FLRW, weak-field, perturbation, observational, provenance, circularity, and falsification audits; no new physics
V15.3.2.1	26 Aug 2026	Zenodo release and reproducibility patch; exact disformal, FLRW, provenance, and release-control identities
V15.3.3	27 Aug 2026	Mathematical foundation update: ten derivation packages N1–N10, with no new fields, interactions, parameters, or physical assumptions

Appendix H

Speculative Mycelial-Network Analogy

Recent studies have mapped vast underground fungal networks. In the topsoil ($\sim 10\text{--}15$ cm), the total length of fungal hyphae is estimated at 1.10×10^{17} km. These networks:

- Act as distributed communication and resource transport systems.
- Facilitate carbon sequestration (~ 300 megatons of carbon stored annually).
- Exhibit emergent intelligence through chemical signaling.

These biological networks provide a qualitative analogy for distributed organization. They do not constitute evidence for toroidal vacuum dynamics, the ATHENA scalar field, or a theory of consciousness.

Appendix I

Synthetic Crystal Rotation and Rotational Wave Exchange

In April 2026, Liberal and Manjavacas published a spacetime-metamaterial construction that emulates rapid crystal rotation through spatiotemporal modulation. The analysis exhibits frequency and spin-angular-momentum exchange without mechanically rotating a bulk crystal.

This result belongs to time-modulated electrodynamics. It is not a direct laboratory realization of black-hole energy extraction and is not evidence for ATHENA. Connecting it to ATHENA would require a quantitative prediction of the pure-disformal coupling dependence that differs from the standard metamaterial description.

Appendix J

Reproducibility Protocol

The detailed status is given in Chapter 43. The present source package does not include the external scripts and data needed for an end-to-end numerical reproduction. The required archival checklist is:

1. versioned source code and an environment lock file;
2. immutable input data with licences and checksums;
3. covariance matrices, priors, likelihoods, and nuisance parameters;
4. commands or workflows that regenerate every table and figure; and
5. reference outputs with numerical tolerances and checksums.

Until that archive is supplied, numerical claims are classified as pending independent reproduction.

Final Remarks

This volume records the ATHENA Ultimate V15.3.2 Stage 5 release candidate. It contains mathematical-closure and observational-context audits. The frozen V15.3.1 theory presented in Volumes 0–III is documented with references, appendices, explicit assumptions, a claim-status ledger, and a provenance-labeled observational audit. The author invites the scientific community to scrutinize, reproduce, test, and refine these ideas.

The framework is offered not as a finished dogma, but as a scientific proposal open to scrutiny, refinement, and experimental test. Independent confirmation of a pre-specified, model-discriminating conditional prediction would support the assumptions responsible for it; falsification would constrain those assumptions and guide further revision.

İzmir, August 2026
Mustafa Gökhan Yılmaz

Status of the Framework — August 2026

ATHENA is a unified topo-geometric research framework containing mathematically defined structures, conditional derivations, phenomenological hypotheses, proposed empirical tests, and conditional predictions.

- The flat-torus spectrum, compact-scalar winding pair, and elementary algebraic consequences of stated equations are mathematically controlled.
- The Casimir normalization, equilibrium rule for β_0 , TEMP, TTA, gauge correspondences, particle assignments, and cosmological mappings remain postulates, hypotheses, conjectures, or open problems at the levels stated in Table F.1.
- The displayed beta function gives $\alpha_* = 48$ for $\gamma_{\text{str}} = 0$; 89.4 is not a consequence of that equation.
- R_c , f_a , mass scales, and observational fits contain explicit phenomenological calibrations. The present package does not yet permit independent numerical reproduction.
- August 2026 observations discussed here are classified as model-compatible observations, convergent anomalies, or pending precision tests. None is presented as a unique confirmation of ATHENA.

The V15.3.2 observational release candidate is therefore a controlled research record, not a claim that a complete theory of fundamental physics has been proved or experimentally validated. It adds no new physical assumption to the frozen V15.3.1 theory and leaves unresolved observable mappings explicitly OPEN.

V15.3.2.1 Zenodo Release and Reproducibility Patch

Release identity and scientific freeze

This patch promotes the current V15.3.2 Final Candidate, whose pre-patch SHA-256 is:

```
ce2748e3404bc541988c15990cd72bfd
c505dc372d0b02ecfdcee8e5326837e9
```

The frozen lower-order core is a mathematically controlled physical theory in the sense of a specified action and derived local degrees of freedom; it is not, however, a completed and experimentally validated unified theory.

The unique release identity is *ATHENA Ultimate V15.3.2.1: Zenodo Release & Reproducibility Patch*. It is not asserted to be byte-identical to an earlier V15.3.2 artifact. V15.2 and V15.3.1 remain frozen.

The ontology is unchanged: there is one real compact scalar $\Phi \sim \Phi + 2\pi f_a$; $U = e^{i\Phi/f_a}$ is an equivalent unit-modulus representation and introduces no radial mode; Ω_{cand} is nondynamical diagnostic data; the physical matter metric is pure disformal; Einstein–Hilbert gravity is postulated; and the internal T^2 is configuration space rather than additional spacetime. Winding sectors remain candidate sectors. No winding-to-spin, charge, or particle identity is added.

The active computational core is

$$S = S_{\text{EH}} + S_{\Phi} + S_m[\tilde{g}, \psi], \quad V(\Phi) = \Lambda_{\Phi}^4 \left[1 - \cos\left(\frac{\Phi}{f_a}\right) \right],$$

$$\tilde{g}_{\mu\nu} = g_{\mu\nu} - \frac{\alpha_{\text{dis}}}{M_*^4} \partial_{\mu} \Phi \partial_{\nu} \Phi.$$

S_{top} and $\mathcal{L}^{(>4)}$ remain optional/open and are not silently inserted into this core.

Pure-disformal algebra

Set

$$c_D \equiv \frac{\alpha_{\text{dis}}}{M_*^4}, \quad u_{\mu} \equiv \partial_{\mu} \Phi, \quad u^2 = g^{\mu\nu} u_{\mu} u_{\nu} = -2X,$$

so that

$$\tilde{g}_{\mu\nu} = g_{\mu\nu} - c_D u_{\mu} u_{\nu}, \quad \Delta_D = 1 - c_D u^2 = 1 + 2c_D X.$$

The rank-one inverse and determinant identities are

$$\tilde{g}^{\mu\nu} = g^{\mu\nu} + \frac{c_D}{\Delta_D} u^{\mu} u^{\nu} = g^{\mu\nu} + \frac{\alpha_{\text{dis}}}{M_*^4 + 2\alpha_{\text{dis}} X} u^{\mu} u^{\nu},$$

$$\det \tilde{g} = \det g \Delta_D, \quad \sqrt{-\tilde{g}} = \sqrt{-g} \sqrt{\Delta_D}$$

on the positive branch. Hence

$$\tilde{g}_{\mu\nu} \text{ is invertible} \iff \Delta_D \neq 0.$$

For a non-null gradient the mixed map has three transverse eigenvalues equal to one and longitudinal eigenvalue Δ_D . Thus $\Delta_D > 0$ preserves Lorentzian inertia on the connected branch.

For a null gradient, $u^2 = 0$ and $\Delta_D = 1$. Define

$$N^\mu{}_\nu = -c_D u^\mu u_\nu.$$

Then

$$(N^2)^\mu{}_\nu = c_D^2 u^\mu (u_\rho u^\rho) u_\nu = 0.$$

The update is nilpotent, $A = I + N$, $A^{-1} = I - N$, all eigenvalues are one, and $\det A = 1$. The metric remains nondegenerate and Lorentzian for finite c_D , although its null cone can shift.

Status: DERIVED.

Matter variation and Bianchi exchange

With

$$\tilde{T}^{\mu\nu} = \frac{2}{\sqrt{-\tilde{g}}} \frac{\delta S_m}{\delta \tilde{g}_{\mu\nu}},$$

variation at fixed $g_{\mu\nu}$ gives

$$\delta_\Phi \tilde{g}_{\mu\nu} = -c_D (u_\mu \partial_\nu \delta\Phi + u_\nu \partial_\mu \delta\Phi).$$

After integration by parts,

$$\mathcal{J}_m^{(D)} = \frac{\alpha_{\text{dis}}}{M_*^4 \sqrt{-g}} \partial_\nu \left[\sqrt{-\tilde{g}} \tilde{T}^{\mu\nu} \partial_\mu \Phi \right].$$

At fixed Φ ,

$$\delta \tilde{g}_{\mu\nu} = -g_{\mu\rho} g_{\nu\sigma} \delta g^{\rho\sigma},$$

and therefore

$$T_{\rho\sigma}^{(m,g)} = \sqrt{\Delta_D} g_{\rho\mu} g_{\sigma\nu} \tilde{T}^{\mu\nu}.$$

Under the frozen scalar convention

$$\square_g \Phi - V'(\Phi) + \mathcal{J}_m^{(D)} = 0,$$

the scalar stress tensor obeys

$$\nabla^\mu T_{\mu\nu}^{(\Phi)} = -\mathcal{J}_m^{(D)} \partial_\nu \Phi.$$

The diffeomorphism identity requires

$$\nabla_\mu T^{(m,g)\mu}{}_\nu = +\mathcal{J}_m^{(D)} \partial_\nu \Phi.$$

Consequently the scalar-plus-matter stress tensor is conserved on shell. The plus sign agrees with the frozen source; no sign correction is required.

Status: DERIVED, conditional on the declared stress-tensor convention and matter equations.

Homogeneous and spatial-gradient kinematics

For

$$ds^2 = -dt^2 + a^2(t)d\mathbf{x}^2, \quad \Phi = \Phi(t),$$

one has $X = \dot{\Phi}^2/2$ and

$$d\tilde{s}^2 = -N_D^2 dt^2 + a^2 d\mathbf{x}^2, \quad N_D^2 = \Delta_D = 1 + c_D \dot{\Phi}^2.$$

Choosing the future-directed positive lapse,

$$\tilde{dt} = N_D dt, \quad \tilde{H} = \frac{H}{N_D}.$$

One further derivative gives

$$\frac{1}{a} \frac{d^2 a}{d\tilde{t}^2} = \frac{1}{N_D^2} \left(\dot{H} + H^2 - H \frac{\dot{N}_D}{N_D} \right).$$

For a spatial scalar gradient,

$$\tilde{\gamma}_{ij} = a^2 \delta_{ij} - c_D \partial_i \Phi \partial_j \Phi, \quad Q_D = \frac{c_D}{a^2} \delta^{ij} \partial_i \Phi \partial_j \Phi.$$

The determinant and volume rate are

$$\det \tilde{\gamma} = a^6 (1 - Q_D), \quad H_{\text{vol}} = H - \frac{\dot{Q}_D}{6(1 - Q_D)}.$$

Necessary algebraic conditions include

$$\Delta_D > 0, \quad 1 - Q_D > 0,$$

together with finite fields and derivatives. These establish metric invertibility/signature only, not full coupled hyperbolicity.

Status: DERIVED.

Minimal scalar dynamics and kinetic regime

For the periodic potential,

$$V'(\Phi) = \frac{\Lambda_\Phi^4}{f_a} \sin\left(\frac{\Phi}{f_a}\right), \quad V''(\Phi) = \frac{\Lambda_\Phi^4}{f_a^2} \cos\left(\frac{\Phi}{f_a}\right).$$

Near a minimum,

$$V = \frac{1}{2} m_\Phi^2 (\delta\Phi)^2 + O(\delta\Phi^4), \quad m_\Phi^2 = \frac{\Lambda_\Phi^4}{f_a^2}.$$

If the potential gradient and every source term are negligible on a homogeneous expanding interval,

$$\ddot{\Phi} + 3H\dot{\Phi} \simeq 0 \implies \dot{\Phi} \propto a^{-3}, \quad \dot{\Phi}^2 \propto a^{-6}.$$

For constant c_D ,

$$N_D^2 - 1 = c_D \dot{\Phi}^2 \propto a^{-6} \propto (1+z)^6$$

when $a_0 = 1$. This scaling is not the full cosmological solution and fails when the potential, matter exchange, optional operators, or inhomogeneous dynamics matter.

Status: KINETIC-REGIME CONDITIONAL DERIVATION.

Modular T1/T2 FLRW system

Let $\tilde{\rho}_A, \tilde{p}_A$ denote matter-frame perfect-fluid quantities and define totals by summing over A . The frame conversion is

$$\rho_{m,g} = \frac{\tilde{\rho}}{N_D}, \quad p_{m,g} = N_D \tilde{p}.$$

The homogeneous equations are

$$\begin{aligned} 3M_{\text{Pl}}^2 H^2 &= \frac{1}{2} \dot{\Phi}^2 + V(\Phi) + \sum_A \frac{\tilde{\rho}_A}{N_D}, \\ -2M_{\text{Pl}}^2 \dot{H} &= \dot{\Phi}^2 + \sum_A \left(\frac{\tilde{\rho}_A}{N_D} + N_D \tilde{p}_A \right), \\ \ddot{\tilde{\rho}}_A + 3H(\tilde{\rho}_A + \tilde{p}_A) &= 0. \end{aligned}$$

For a homogeneous comoving perfect fluid,

$$\mathcal{J}_m^{(D)} = \frac{c_D}{a^3} \frac{d}{dt} \left(\frac{a^3 \tilde{\rho} \dot{\Phi}}{N_D} \right) = \frac{c_D \tilde{\rho}}{N_D^3} \ddot{\Phi} - \frac{3c_D H \tilde{p}}{N_D} \dot{\Phi}.$$

Consequently

$$K_D \ddot{\Phi} + 3H \dot{\Phi} \left(1 + \frac{c_D \tilde{p}}{N_D} \right) + V'(\Phi) = 0, \quad K_D = 1 - \frac{c_D \tilde{\rho}}{N_D^3}.$$

For T1, $A \in \{b, \gamma, \nu\}$; for T2, $A \in \{b, c, \gamma, \nu\}$. There is no independent cosmological constant in either configuration; Λ_Φ is the scalar-potential scale. T2 is a control, not an ATHENA dark-sector prediction.

The source does not uniquely specify neutrino masses, hierarchy, temperature law, or transition equation of state. Therefore numerical closure is

OPEN — matter-sector closure required.

Admissibility map

For any proposed homogeneous point define

$$\mathcal{A}(\Phi, \dot{\Phi}, \tilde{\rho}, \tilde{p}) = \left\{ \Delta_D > 0, \quad K_D \neq 0, \quad \tilde{\rho}_A \geq 0, \quad \frac{1}{2} \dot{\Phi}^2 + V + \frac{\tilde{\rho}}{N_D} \geq 0, \quad \text{all quantities finite} \right\}.$$

The release code implements this Boolean map and distinguishes algebraically inadmissible points from numerically unresolved trajectories. No physical parameter range follows from the frozen theory. The finite grid in `reference_outputs` is an implementation-verification domain, not a prior or observational constraint.

Status: DERIVED map; numerical domain is an IMPLEMENTATION INPUT.

Conditional scalar regime classification

The homogeneous equation of state is

$$w_\Phi = \frac{\dot{\Phi}^2/2 - V}{\dot{\Phi}^2/2 + V}.$$

Near a quadratic minimum, if $m_\Phi/H \gg 1$, disformal exchange is negligible, and coherent oscillations persist over many Hubble times,

$$\langle w_\Phi \rangle \simeq 0.$$

If $m_\Phi/H \ll 1$ and independently specified initial conditions place the field in a potential-dominated slow-evolution regime,

$$w_\Phi \simeq -1.$$

These are distinct conditional regimes, not a derivation that one parameter point supplies both clustered dark matter and dark energy.

Status: CONDITIONAL DYNAMICAL REGIME CLASSIFICATION.

Minimal stability and conditional weak-field limit

The isolated canonical scalar has positive quadratic principal part and

$$c_{s,g}^2 = 1.$$

The Einstein–Hilbert transverse-traceless tensor action has positive coefficient for $M_{\text{Pl}}^2 > 0$ and gives

$$c_{T,g}^2 = 1.$$

These statements do not establish full scalar–metric–matter stability, constraint closure, or hyperbolicity with unspecified sectors.

For a constant scalar background,

$$\partial_\mu \Phi_{\text{bg}} = 0,$$

the disformal correction begins quadratically in scalar perturbation gradients. Thus $\tilde{g}_{\mu\nu} = g_{\mu\nu}$ at linear order. If scalar stress, higher operators, and unspecified source sectors are also neglected, the linear weak-field equations are those of general relativity and

$$\gamma_{\text{PPN}} = \beta_{\text{PPN}} = 1.$$

This conclusion is restricted to that declared background. General PPN, fifth-force, and compact-source calculations remain open.

Status: DERIVED for isolated principal parts; CONDITIONAL for the constant-background PPN limit; full stability and general PPN are OPEN.

T10 compact-shift condition

For

$$S_{\Phi F} = \frac{c_{\Phi F}}{4M_*} \int d^4x \sqrt{-g} \Phi F_{\mu\nu} \tilde{F}^{\mu\nu},$$

the shift $\Phi \mapsto \Phi + 2\pi f_a$ gives

$$\Delta S = 2\pi \frac{c_{\Phi F} f_a}{M_*} Q_F, \quad Q_F = \frac{1}{4} \int d^4x \sqrt{-g} F_{\mu\nu} \tilde{F}^{\mu\nu}.$$

Single-valuedness of the quantum phase requires

$$\frac{c_{\Phi F} f_a}{M_*} Q_F \in \mathbb{Z}$$

for every allowed topological sector after gauge and trace normalization has been fixed. No gauge group, bundle, trace convention, charge normalization, or allowed sector is supplied by the frozen theory.

Status: local shift algebra DERIVED; global completion OPEN.

Derrick obstruction

For a static three-dimensional canonical functional with nonnegative gradient and potential pieces T, U , the rescaling $\Phi_\lambda(\mathbf{x}) = \Phi(\lambda\mathbf{x})$ gives

$$I(\lambda) = \lambda^{-1}T + \lambda^{-3}U, \quad \left. \frac{dI}{d\lambda} \right|_{\lambda=1} = -T - 3U.$$

Stationarity demands

$$T + 3U = 0.$$

If $T, U \geq 0$, then $T = U = 0$. Thus a nontrivial static finite-energy canonical scalar soliton is obstructed under the declared assumptions. No workaround is added.

Status: NO-GO / OBSTRUCTED.

Exact completion identities from the frozen core

This section records further consequences that require no additional field, operator, parameter identification, or observational input.

Inverse disformal kinetic map and stress trace

Contracting the exact inverse metric with $u_\mu = \partial_\mu \Phi$ gives

$$\tilde{u}^\mu \equiv \tilde{g}^{\mu\nu} u_\nu = \frac{u^\mu}{\Delta_D}, \quad \tilde{X} \equiv -\frac{1}{2} \tilde{g}^{\mu\nu} u_\mu u_\nu = \frac{X}{\Delta_D}.$$

Consequently

$$X = \frac{\tilde{X}}{1 - 2c_D \tilde{X}}, \quad \Delta_D = \frac{1}{1 - 2c_D \tilde{X}},$$

where either expression is valid only away from its singular denominator. The map satisfies

$$\frac{d\tilde{X}}{dX} = \frac{1}{\Delta_D^2},$$

so it is locally one-to-one on every regular branch.

The matter stress tensors obey the exact trace identity

$$T^{(m,g)} = \sqrt{\Delta_D} \left(\tilde{T} + c_D \tilde{T}^{\mu\nu} u_\mu u_\nu \right),$$

with $\tilde{T} = \tilde{g}_{\mu\nu} \tilde{T}^{\mu\nu}$. This is an algebraic frame relation, not an equation of state.

Status: DERIVED.

General ADM form of the physical metric

Write the g -frame metric as

$$ds^2 = -\mathcal{N}^2 dt^2 + h_{ij}(dx^i + N^i dt)(dx^j + N^j dt)$$

and define

$$q_i = D_i \Phi, \quad A = \frac{\dot{\Phi} - N^i q_i}{\mathcal{N}}, \quad Q_D = c_D h^{ij} q_i q_j.$$

Then

$$\Delta_D = 1 + c_D A^2 - Q_D.$$

The exact spatial metric, inverse, and determinant are

$$\tilde{h}_{ij} = h_{ij} - c_D q_i q_j, \quad \tilde{h}^{ij} = h^{ij} + \frac{c_D}{1 - Q_D} q^i q^j,$$

$$\det \tilde{h} = (\det h)(1 - Q_D).$$

The physical shift and lapse are

$$\begin{aligned} \tilde{N}_i &= N_i - c_D \dot{\Phi} q_i, \\ \tilde{N}^i &= N^i - \frac{c_D q^i}{1 - Q_D} (\dot{\Phi} - N^j q_j), \\ \tilde{\mathcal{N}}^2 &= \mathcal{N}^2 \frac{\Delta_D}{1 - Q_D} = \mathcal{N}^2 \left(1 + \frac{c_D A^2}{1 - Q_D} \right). \end{aligned}$$

For positive-definite h_{ij} , a regular Lorentzian ADM decomposition on the branch connected to $c_D = 0$ therefore requires

$$1 - Q_D > 0, \quad \Delta_D > 0.$$

The homogeneous formulas follow by setting $q_i = N^i = 0$ and $\mathcal{N} = 1$. These conditions establish the metric signature; they do not prove the principal-symbol hyperbolicity of unspecified matter sectors.

Status: DERIVED.

Energy exchange and propagation of the Friedmann constraint

For the homogeneous perfect-fluid closure used above, define

$$\rho_{m,g} = \frac{\tilde{\rho}}{N_D}, \quad p_{m,g} = N_D \tilde{p}.$$

The scalar equation and the Bianchi identity imply

$$\begin{aligned} \dot{\rho}_\Phi + 3H(\rho_\Phi + p_\Phi) &= \mathcal{J}_m^{(D)} \dot{\Phi}, \\ \dot{\rho}_{m,g} + 3H(\rho_{m,g} + p_{m,g}) &= -\mathcal{J}_m^{(D)} \dot{\Phi}. \end{aligned}$$

Thus

$$\frac{d}{dt}(\rho_\Phi + \rho_{m,g}) + 3H(\rho_\Phi + p_\Phi + \rho_{m,g} + p_{m,g}) = 0.$$

Differentiating

$$\mathcal{C}_F = 3M_{\text{Pl}}^2 H^2 - \rho_\Phi - \rho_{m,g}$$

and using the acceleration equation gives

$$\dot{\mathcal{C}}_F = 0.$$

Hence an initially satisfied Friedmann constraint is propagated by the exact homogeneous equations; its numerical drift is an implementation error diagnostic.

The corresponding acceleration law is

$$\frac{\ddot{a}}{a} = \frac{V - \dot{\Phi}^2}{3M_{\text{Pl}}^2} - \frac{1}{6M_{\text{Pl}}^2} \sum_A \left(\frac{\tilde{\rho}_A}{N_D} + 3N_D \tilde{p}_A \right).$$

Status: DERIVED for the declared perfect-fluid closure.

Closed dimensionless system

Use reduced-Planck normalizations

$$\begin{aligned}\varphi &= \frac{\Phi}{M_{\text{Pl}}}, \quad \tau = M_{\text{Pl}}t, \quad v = \frac{d\varphi}{d\tau} = \frac{\dot{\Phi}}{M_{\text{Pl}}^2}, \quad h = \frac{H}{M_{\text{Pl}}}, \\ r_A &= \frac{\tilde{\rho}_A}{M_{\text{Pl}}^4}, \quad q_A = \frac{\tilde{p}_A}{M_{\text{Pl}}^4}, \quad f = \frac{f_a}{M_{\text{Pl}}}, \quad \ell = \frac{\Lambda_\Phi}{M_{\text{Pl}}}, \\ d &= c_D M_{\text{Pl}}^4 = \alpha_{\text{dis}} \left(\frac{M_{\text{Pl}}}{M_*} \right)^4.\end{aligned}$$

All displayed variables are dimensionless. Define

$$U(\varphi) = \ell^4 \left[1 - \cos \left(\frac{\varphi}{f} \right) \right], \quad n_D = \sqrt{1 + dv^2}, \quad k_D = 1 - \frac{dr}{n_D^3},$$

where $r = \sum_A r_A$ and $q = \sum_A q_A$. The Friedmann constraint fixes

$$h = s_H \sqrt{\frac{v^2/2 + U + r/n_D}{3}}, \quad s_H \in \{+1, -1\}.$$

Thus h_i is derived after an expanding or contracting branch is declared; it is not an independent fitted initial condition.

With the e-fold variable $\mathcal{N}_e = \ln(a/a_i)$ and $h \neq 0$,

$$\begin{aligned}\frac{d\varphi}{d\mathcal{N}_e} &= \frac{v}{h}, \\ \frac{dv}{d\mathcal{N}_e} &= -\frac{3hv(1 + dq/n_D) + U_{,\varphi}}{k_D h}, \\ \frac{dr_A}{d\mathcal{N}_e} &= -3(r_A + q_A),\end{aligned}$$

and

$$\frac{dh}{d\mathcal{N}_e} = -\frac{1}{2h} \left[v^2 + \frac{r}{n_D} + n_D q \right].$$

The last equation is redundant when the constraint is imposed and is useful as an independent numerical check. Closure still requires every function $q_A(\mathcal{N}_e, r_A)$, particularly the neutrino sector.

The exact theoretical domain includes

$$f > 0, \quad \ell \geq 0, \quad r_A \geq 0, \quad n_D^2 > 0, \quad k_D \neq 0.$$

If $d < 0$, lapse admissibility imposes

$$|v| < \frac{1}{\sqrt{-d}}.$$

If $d \geq 0$, the lapse condition alone gives no finite velocity bound. If $d \leq 0$ and $r \geq 0$, then $k_D > 0$ automatically. If $d > 0$, the positive reduced kinetic branch is

$$r < \frac{n_D^3}{d},$$

with $k_D = 0$ as a singular boundary. Positivity of k_D is a property of the reduced homogeneous equation, not a proof of full coupled ghost freedom.

The external initial data are $(\varphi_i, v_i, r_{A,i})$ and the matter closures. Compactness permits the representative choice $0 \leq \varphi_i < 2\pi f$, but supplies no probability measure on that interval. No finite numerical scan range or sampling distribution is derived.

Status: DERIVED dimensionless equations; matter closure and numerical sampling domain remain OPEN / IMPLEMENTATION INPUT.

Exact scalar bounds, stationary backgrounds, and kinetic solution

For $\Lambda_\Phi^4 \geq 0$,

$$0 \leq V(\Phi) \leq 2\Lambda_\Phi^4.$$

At the minima $\Phi = 2\pi n f_a$, $V = 0$ and $V'' = +m_\Phi^2$. In the source-free scalar-only system the corresponding constant solution is Minkowski, $H = 0$. At the maxima $\Phi = (2n+1)\pi f_a$,

$$V = 2\Lambda_\Phi^4, \quad V'' = -m_\Phi^2, \quad H^2 = \frac{2\Lambda_\Phi^4}{3M_{\text{Pl}}^2}.$$

This is an exact de Sitter background but is tachyonically unstable in the isolated scalar sector. The frozen potential therefore contains no stable positive-energy vacuum minimum. A permanent dark-energy asymptote is not derived; a potential-dominated phase requires displaced initial data or an additional sector.

For the source-free kinetic limit $V = V' = 0$ on an expanding branch,

$$a(t) \propto (t - t_0)^{1/3}, \quad \Phi(t) = \Phi_0 \pm \sqrt{\frac{2}{3}} M_{\text{Pl}} \ln |t - t_0|,$$

up to normalization constants. Hence

$$\dot{\Phi} = \pm \sqrt{\frac{2}{3}} \frac{M_{\text{Pl}}}{t - t_0} \propto a^{-3}.$$

This exact solution realizes the previously stated kinetic scaling.

For any homogeneous source-free solution,

$$\dot{\rho}_\Phi = -3H\dot{\Phi}^2.$$

Thus scalar energy is non-increasing on an expanding branch. Since

$$\rho_\Phi + p_\Phi = \dot{\Phi}^2 \geq 0, \quad -1 \leq w_\Phi \leq 1$$

whenever $\rho_\Phi > 0$, the canonical scalar does not cross the phantom value $w = -1$. In a scalar-only universe acceleration occurs exactly when

$$V > \dot{\Phi}^2.$$

The global consequence of this monotonicity on the compact scalar phase cylinder is stated and proved in Proposition K.8.

Status: DERIVED; the cosmological interpretation remains CONDITIONAL on initial data and omitted sectors.

Discrete shift, approximate current, and kinetic scaling

The periodic potential and pure-disformal metric are invariant under the discrete transformation $\Phi \mapsto \Phi + 2\pi f_a$. A discrete symmetry does not by itself produce a continuous Noether current. In the source-free regime where V' is negligible, the action acquires the approximate continuous shift symmetry with current

$$j^\mu = \nabla^\mu \Phi, \quad \nabla_\mu j^\mu \simeq 0.$$

For a homogeneous field this is equivalent to

$$\frac{d}{dt}(a^3 \dot{\Phi}) \simeq 0,$$

which supplies an independent derivation of $\dot{\Phi} \propto a^{-3}$.

Status: exact discrete symmetry; CONDITIONAL approximate continuous current.

Matter-frame redshift, distances, and relative wave speed

The homogeneous physical metric becomes standard flat FLRW after the time change $\tilde{dt} = N_D dt$. Therefore comoving matter clocks and photons give

$$1 + z = \frac{a_0}{a}, \quad \tilde{H} = \frac{H}{N_D},$$

and the physical deceleration parameter is

$$\tilde{q} = -\frac{1}{\tilde{H}^2 a} \frac{d^2 a}{d\tilde{t}^2} = q + \frac{\dot{N}_D}{H N_D}, \quad q = -1 - \frac{\dot{H}}{H^2}.$$

If photons follow null geodesics of $\tilde{g}_{\mu\nu}$ and their number is conserved, the flat-background distance relations are

$$\chi(z) = \int_0^z \frac{dz'}{\tilde{H}(z')}, \quad D_A = \frac{\chi}{1+z}, \quad D_L = (1+z)\chi.$$

These are forward kinematic relations, not an observational fit.

Einstein-frame tensor characteristics are g -null. On the homogeneous background their speed measured with the matter proper time and physical distance is

$$c_{T,\text{matter}} = \frac{1}{N_D},$$

while matter-frame light has unit speed. This relation can constrain a solved background, but it cannot determine α_{dis} without $\dot{\Phi}(t)$ and a complete propagation/source model.

Status: DERIVED homogeneous kinematics; distance and multimessenger applications are CONDITIONAL.

Kinetic-profile obstruction to the collapsed-fraction shape

Define the forward lapse excess

$$R_D(z) = N_D^2(z) - 1.$$

In a sustained kinetic regime with nonzero constant c_D ,

$$\frac{d \ln |R_D|}{d \ln(1+z)} = 6.$$

The independent phenomenological benchmark

$$\Omega_{\text{coll}}(z) = \frac{1}{1 + (z/z_c)^2}$$

with $z_c > 0$ has logarithmic slope

$$\frac{d \ln \Omega_{\text{coll}}}{d \ln(1+z)} = -\frac{2z(1+z)}{z_c^2 + z^2},$$

which is negative for $z > 0$ and approaches -2 at high redshift. Therefore no nonzero parameter-independent constant can make $R_D(z)$ proportional to $\Omega_{\text{coll}}(z)$ on an open positive-redshift interval while the kinetic approximation remains valid. For $c_D < 0$ the lapse excess also has the opposite sign to a positive collapsed fraction; for $c_D = 0$ it vanishes.

This does not exclude a non-kinetic trajectory with potential or matter exchange. It proves that the sustained kinetic asymptotic cannot globally reproduce the benchmark shape.

Status: CONDITIONAL NO-GO / SHAPE OBSTRUCTION.

Observational status lock

Quantity	Status	Release conclusion
α_H	CALIBRATED	Present-day Hubble-ansatz normalization; forbidden as an inverse input to α_{dis} .
$\alpha_{\text{eff}}^{\text{gal}}$	FIT	Galaxy-scale fitted quantity; not the bare disformal coefficient.
α_{RG}	CONDITIONAL	Algebraic result of the postulated beta-function truncation.
α_{dis}	OPEN	Active bare coefficient with no independently frozen value.
R_D versus Ω_{coll}	CONDITIONAL NO-GO	Sustained kinetic scaling has positive logarithmic slope 6, whereas the benchmark slope is negative for $z > 0$.
Matter-frame GW/light speed	CONDITIONAL	$c_{T,\text{matter}} = 1/N_D$ on a solved homogeneous background; it does not fix α_{dis} by itself.
SPARC result	FIT	Reported fit is not independently reproduced and cannot set Stage-5 theory inputs.
GW phase values	HISTORICAL TARGET	No binary solution or waveform derivation.
Lensing/slip	OPEN	No complete perturbation-to-observable map.
134° dipole	HISTORICAL TARGET	Dataset, estimator, and independent significance remain unreproduced.
Magnetic profiles	BENCHMARK	Shape ansatz without complete amplitude/evolution likelihood.
Muon $g - 2$	OPEN	No ATHENA loop calculation.
Higgs couplings	OPEN	No complete gauge-invariant interaction derivation.
Particle masses	NO-GO / OPEN	Static canonical construction is Derrick-obstructed; numerical targets are historical.

Parameter provenance and circularity firewall

Symbol	Definition/dimension	Provenance	Status	Active?	Circularity
α_{RG}	RG coefficient; dimensionless	Postulated beta function	CONDITIONAL	No	No map to disformal or observational alphas.
α_{dis}	Bare disformal coefficient; dimensionless	Frozen metric ansatz	OPEN	Yes, through c_D	Must not be inferred from α_H , SPARC, or $\alpha_{\text{eff}}^{\text{gal}}$.
d	$c_D M_{\text{Pl}}^4$; dimensionless	Derived normalization	DERIVED / OPEN VALUE	Yes in dimensionless system	Contains no new equality among alpha parameters.
$\alpha_{\text{eff}}^{\text{gal}}$	Effective galaxy coefficient; dimensionless	Phenomenological fit	FIT	No	Reuse for the same observable is circular.
α_H	Hubble coefficient; dimensionless	Present-day normalization	CALIBRATED	No	Cannot predict the datum that fixed it.
α_{cal}	Historical coefficient; dimensionless	Legacy normalization	HISTORICAL TARGET	No	Not a universal coupling.
β_0	Torus/postulate parameter; dimensionless	Equilibrium postulate	CONDITIONAL	No in minimal core	Depends on accepted postulate.
γ_0	$\beta_0^2/2$; dimensionless	Conditional normalization	CONDITIONAL	No in minimal core	Not independently fixed.

Symbol	Definition/dimension	Provenance	Status	Active?	Circularity
R_c	Length scale	Uses external L_*	CALIBRATED	No in FLRW core	Galactic reuse is not independent.
f_a	Compact period; mass	Field definition plus calibrated target	CALIBRATED / OPEN	Yes	Numerical value is not topologically fixed.
a_0	Acceleration	Phenomenological chain	CALIBRATED	No	Includes fitted galactic inputs.
M_{Pl}	Reduced Planck mass; mass	Einstein–Hilbert normalization	EXTERNAL	Yes	Not derived by ATHENA.
M_*	EFT scale; mass	Suppression scale	OPEN	Yes	Only $\alpha_{\text{dis}}/M_*^4$ enters the metric.
Λ_Φ	Potential scale; mass	Periodic potential	OPEN	Yes	No frozen numerical value.
$c_{\Phi F}$	Optional topological coefficient	Local EFT operator	OPEN	No in minimal core	Depends on unspecified gauge normalization.

No relation

$$\alpha_{\text{RG}} = \alpha_{\text{dis}} = \alpha_{\text{eff}}^{\text{gal}} = \alpha_H = \alpha_{\text{cal}}$$

is derived or assumed.

Computational and release status

The accompanying code implements the frozen algebra, potential derivatives, dimensionless FLRW right-hand sides, admissibility diagnostics, and source-free limiting checks. The original minimal solver is retained under `code/legacy`. T1 and T2 manifests keep neutrino closure explicitly unresolved; no observational data or fitted theory parameters are embedded. Reference outputs are code-verification artifacts, not observational predictions.

Final scientific verdict

Item	Status	Scope
Minimal FLRW equations	DERIVED	Conditional on declared perfect-fluid matter variables.
Disformal algebra	VERIFIED	Exact rank-one identities, including null-gradient case.
General ADM disformal split	VERIFIED	Exact lapse, shift, spatial inverse, and determinant.
Matter exchange	DERIVED	Sign verified under frozen scalar convention.
Friedmann propagation	DERIVED	Total exchange preserves an initially satisfied constraint.
Dimensionless T1/T2 system	DERIVED / OPEN	Equations fixed; neutrino closure and finite scan domain remain open.
Admissibility	DERIVED	Algebraic map; physical parameter domain remains open.
Scalar regimes	CONDITIONAL	Rapid oscillation and slow-evolution limits only.

Item	Status	Scope
Exact scalar-only backgrounds	DERIVED	Minkowski minima, unstable de Sitter maxima, and kinetic power law.
Minimal stability	DERIVED	Isolated canonical scalar and Einstein tensor sectors.
Conditional PPN	CONDITIONAL	Constant scalar background only.
Matter-frame distances	CONDITIONAL	Derived after standard photon-geodesic and number-conservation closure.
Kinetic benchmark shape	CONDITIONAL NO-GO	Cannot track Ω_{coll} on an open $z > 0$ interval.
T10 quantization	OPEN	Necessary condition derived; normalization unspecified.
Derrick obstruction	NO-GO	Nontrivial static canonical finite-energy soliton obstructed.
T1	NOT COMPUTABLE FROM FROZEN THEORY	Neutrino and complete matter closure required.
T2	NOT COMPUTABLE FROM FROZEN THEORY	Control only; same closure limitation.
Full degrees of freedom	OPEN	Unspecified matter/topological/higher sectors.
Full stability	OPEN	Coupled quadratic action and hyperbolicity absent.
General PPN	OPEN	Sourced background and complete matter action absent.
Lensing	OPEN	Perturbation/slip map absent.
Growth	OPEN	Linear/nonlinear perturbation system absent.
GW waveform	OPEN	Binary solutions and radiation calculation absent.
Winding-spin map	OPEN	No four-dimensional representation map.
Particle masses	NO-GO / OPEN	Canonical soliton route obstructed; no replacement supplied.

ATHENA V15.3.2.1 is a controlled theoretical-physics research release. It is not experimentally validated, not a completed unified theory, not a Standard Model derivation, and not an emergent-gravity derivation. Negative results and open problems are preserved.

Appendix K

V15.3.3 Mathematical Foundation Update

K.1 Release boundary and provenance

V15.3.3 is a mathematical-foundation update of the frozen V15.3.2.1 baseline. It adds only consequences of definitions, ansatzes, and equations already present in that baseline. No new field, interaction, gauge group, potential, spacetime dimension, stabilization operator, free parameter, particle interpretation, or observational calibration is introduced.

ATHENA contains a mathematically controlled physical core but is not a completed and experimentally validated unified theory. The physical TTA map, winding-spin and winding-particle identifications, gauge emergence, spin-statistics, Higgs and generation constructions, emergent gravity, complete matter and neutrino sectors, general PPN, lensing, growth, and a complete gravitational-wave waveform remain OPEN or CONDITIONAL. The Derrick obstruction is unchanged.

The ten packages N1–N10 below form the V15.3.3 **NEWLY DERIVED** ledger. Entries carrying a NO-GO qualifier are newly proved obstructions, not new physical assumptions.

K.2 N1 — Exact Poisson-resummed Epstein-zeta decomposition

Proposition K.1 (Poisson-resummed rectangular-torus Epstein continuation). For the declared spectral zeta

$$\mathcal{Z}_\beta(s) = \sum'_{(m,n) \in \mathbb{Z}^2} \left(m^2 + \frac{n^2}{\beta^2} \right)^{-s/2}, \quad \text{Re } s > 2,$$

define

$$\mathcal{S}(\beta) = \sum'_{(m,n) \in \mathbb{Z}^2} (m^2 + \beta^2 n^2)^{-3/2}, \quad \beta > 0.$$

Then

$$\mathcal{S}(\beta) = 2\zeta(3) + \frac{2\pi^2}{3\beta^2} + \frac{16\pi}{\beta} \sum_{k,n \geq 1} \frac{k}{n} K_1(2\pi k\beta n). \quad (\text{K.1})$$

Consequently,

$$\begin{aligned} \mathcal{Z}_\beta(-1) &= -\frac{\beta}{4\pi^2} \mathcal{S}(\beta) \\ &= -\frac{\beta\zeta(3)}{2\pi^2} - \frac{1}{6\beta} - \frac{4}{\pi} \sum_{k,n \geq 1} \frac{k}{n} K_1(2\pi k\beta n). \end{aligned} \quad (\text{K.2})$$

Equivalently,

$$\mathcal{Z}_\beta(-1) = -\frac{1}{6} - \frac{\zeta(3)}{2\pi^2\beta^2} - \frac{4}{\pi\beta} \sum_{n,p \geq 1} \frac{n}{p} K_1\left(\frac{2\pi pn}{\beta}\right). \quad (\text{K.3})$$

At $\beta = 1$,

$$\boxed{\mathcal{Z}_1(-1) = -0.228824310377218953\dots}.$$

Assumptions. The flat product T^2 , periodic scalar boundary conditions, deletion of the zero mode, and the existing Epstein-zeta definition and functional equation.

Proof. The $n = 0$, $m \neq 0$ contribution to $\mathcal{S}$ is $2\zeta(3)$. For $a > 0$,

$$(m^2 + a^2)^{-3/2} = 2\pi \int_0^\infty t^{1/2} e^{-\pi t(m^2 + a^2)} dt.$$

Poisson summation,

$$\sum_{m \in \mathbb{Z}} e^{-\pi t m^2} = t^{-1/2} \sum_{k \in \mathbb{Z}} e^{-\pi k^2/t},$$

separates the $k = 0$ term from the paired nonzero terms. Using

$$\int_0^\infty t^{\nu-1} e^{-At-B/t} dt = 2 \left(\frac{B}{A}\right)^{\nu/2} K_\nu(2\sqrt{AB})$$

with $\nu = 1$, $A = \pi a^2$, and $B = \pi k^2$ gives

$$\sum_{m \in \mathbb{Z}} (m^2 + a^2)^{-3/2} = \frac{2}{a^2} + \frac{8\pi}{a} \sum_{k=1}^\infty k K_1(2\pi k a).$$

Pairing n and $-n$, setting $a = \beta n$, and using $\zeta(2) = \pi^2/6$ yields

$$\begin{aligned} \mathcal{S}(\beta) &= 2\zeta(3) + 2 \sum_{n=1}^\infty \left[\frac{2}{\beta^2 n^2} + \frac{8\pi}{\beta n} \sum_{k=1}^\infty k K_1(2\pi k \beta n) \right] \\ &= 2\zeta(3) + \frac{2\pi^2}{3\beta^2} + \frac{16\pi}{\beta} \sum_{k,n \geq 1} \frac{k}{n} K_1(2\pi k \beta n). \end{aligned}$$

The existing functional equation $\mathcal{Z}_\beta(-1) = -\beta \mathcal{S}(\beta)/(4\pi^2)$ proves Equation (K.2). Repeating the resummation on the original lattice gives Equation (K.3). The two forms are related by exchanging the cycles. Their exponentially convergent series both give the displayed square-torus value. $\square$

Dimension check. $\mathcal{Z}_\beta$ and $\mathcal{S}$ are dimensionless; $E_{\text{vac}}^\zeta = \mathcal{Z}_\beta(-1)/(2R)$ has energy dimension.

Dependency. Existing T^2 metric and Epstein-zeta definition.

Status: DERIVED.

Physical interpretation. Equations (K.2) and (K.3) are exact mathematical representations of the same analytic continuation.

Limitations. No physical subtraction prescription or value of β_0 is derived.

K.3 N2 — Winding-energy aspect-ratio minimum

Proposition K.2 (Aspect-ratio minimum of a fixed winding sector). For

$$E_{\mathbf{w}}(\beta) = C \left(\beta w_{\theta}^2 + \frac{w_{\phi}^2}{\beta} \right), \quad C = \frac{2\pi^2 f_a^2}{M_*} > 0,$$

mixed winding has the unique global minimum

$$\boxed{\beta_* = \left| \frac{w_{\phi}}{w_{\theta}} \right|}, \quad \boxed{E_{\min} = 2C |w_{\theta} w_{\phi}|}.$$

Single-winding sectors have no finite interior minimum, and the winding minimum does not equal or derive the universal postulated β_0 .

Assumptions. The existing quadratic winding-gradient energy, $C > 0$, and $\beta > 0$.

Proof.

$$\frac{dE_{\mathbf{w}}}{d\beta} = C \left(w_{\theta}^2 - \frac{w_{\phi}^2}{\beta^2} \right), \quad \frac{d^2 E_{\mathbf{w}}}{d\beta^2} = \frac{2C w_{\phi}^2}{\beta^3}.$$

For $w_{\theta} w_{\phi} \neq 0$, stationarity gives the displayed β_* , the second derivative is positive, and the energy diverges at both positive domain boundaries. Substitution gives $E_{\min}$. If $w_{\phi} = 0$, the energy is strictly increasing; if $w_{\theta} = 0$, it is strictly decreasing. Thus no single-winding interior minimum exists. For mixed integer winding, β_* is rational. The postulate $\pi = 12\beta_0(2 - \beta_0)$ makes β_0 transcendental, so $\beta_* \neq \beta_0$. $\square$

Dimension check. $[C] = M$ and the bracket is dimensionless.

Dependency. Existing compact-scalar winding pair and sector-gradient energy.

Status: DERIVED; PHYSICAL IDENTIFICATION CONDITIONAL.

Physical interpretation. The optimum is winding-sector dependent.

Limitations. The universal equilibrium postulate for β_0 remains an independent conditional input and its dynamical derivation remains OPEN.

K.4 N3 — Modular and duality limit

Proposition K.3 (Reduced symmetry versus exact Epstein duality). The reduced ansatz

$$\mathcal{Z}_{\beta}^{\text{model}}(-1) = -\frac{\pi}{12} \left(\beta + \frac{1}{\beta} \right)$$

is invariant under $\beta \leftrightarrow \beta^{-1}$, whereas the exact zeta obeys

$$\boxed{\mathcal{Z}_{\beta}(s) = \beta^s \mathcal{Z}_{1/\beta}(s)}.$$

At $\beta = 1$ the exact and reduced values are respectively $-0.228824310377218953\dots$ and $-\pi/6$.

Assumptions. The existing exact zeta and separately declared reduced ansatz.

Proof. The reduced invariance is immediate. For the exact sum,

$$\begin{aligned} \beta^s \mathcal{Z}_{1/\beta}(s) &= \sum'_{m,n} \left(\frac{m^2}{\beta^2} + n^2 \right)^{-s/2} \\ &= \sum'_{m,n} \left(m^2 + \frac{n^2}{\beta^2} \right)^{-s/2} = \mathcal{Z}_{\beta}(s), \end{aligned}$$

after exchanging the dummy lattice labels. Analytic continuation preserves the identity. Proposition K.1 and direct substitution give the unequal self-dual values. $\square$

Dimension check. All displayed zeta functions and β are dimensionless.

Dependency. Existing exact Epstein sum and reduced Casimir ansatz.

Status: DERIVED.

Physical interpretation. Cycle exchange constrains the transformation law.

Limitations. Duality does not determine the absolute phenomenological normalization or derive the equilibrium equation for β_0 .

K.5 N4 — Conditional RG fixed-point stiffness

Proposition K.4 (Linear stiffness of the truncated RG fixed point). For

$$\beta_{\text{RG}}(\alpha) = A\alpha^2 \left(1 - \frac{\alpha}{\alpha_*}\right), \quad A = \frac{1}{2\pi}, \quad \alpha_* = 48,$$

$$\boxed{\beta'_{\text{RG}}(\alpha_*) = -\frac{24}{\pi}}, \quad \boxed{\omega = \frac{24}{\pi}}.$$

Assumptions. The existing truncated beta-function ansatz and $\gamma_{\text{str}} = 0$.

Proof.

$$\beta'_{\text{RG}}(\alpha) = 2A\alpha - \frac{3A}{\alpha_*}\alpha^2, \quad \beta'_{\text{RG}}(\alpha_*) = -A\alpha_* = -\frac{24}{\pi}.$$

For $t = \ln(\mu/\mu_0)$ and $\alpha = \alpha_* + \delta\alpha$,

$$\frac{d\delta\alpha}{dt} = -\omega\delta\alpha + O(\delta\alpha^2), \quad \delta\alpha(t) = \delta\alpha(0)e^{-\omega t}$$

at linear order. □

Dimension check. α , β_{RG} , A , and ω are dimensionless.

Dependency. Existing truncated beta-function ansatz only.

Status: DERIVED / ANSATZ-CONDITIONAL.

Physical interpretation. ω is the local forward- $\ln \mu$ stiffness of the adopted flow.

Limitations. This is not a nonperturbative RG proof. It supplies no loop matching, operator-mixing result, or equality $\alpha_* = \alpha_{\text{dis}}$.

K.6 N5 — Homogeneous simultaneous dark-matter/dark-energy no-go

Proposition K.5 (No simultaneous homogeneous dust and vacuum equation of state). For one homogeneous canonical scalar with

$$K = \frac{1}{2}\dot{\Phi}^2 \geq 0, \quad V \geq 0, \quad \rho_\Phi = K + V > 0, \quad p_\Phi = K - V,$$

one has $-1 \leq w_\Phi \leq 1$, and the same configuration cannot be simultaneously dust-like and vacuum-like with error below $1/2$.

Assumptions. The existing canonical homogeneous scalar and nonnegative cosine potential.

Proof. Since

$$w_\Phi = \frac{K - V}{K + V},$$

$|K - V| \leq K + V$ proves the range. Exact $w_\Phi = 0$ requires $K = V$, whereas exact $w_\Phi = -1$ requires $K = 0$; both imply the excluded zero-density point. If $0 \leq \varepsilon < 1/2$ and both $|w_\Phi| \leq \varepsilon$ and $|w_\Phi + 1| \leq \varepsilon$, then

$$1 = |(w_\Phi + 1) - w_\Phi| \leq |w_\Phi + 1| + |w_\Phi| \leq 2\varepsilon < 1,$$

a contradiction. $\square$

Corollary K.1 (Sequential regimes). If $m_\Phi/H \gg 1$ and coherent near-minimum oscillations persist, then $\langle w_\Phi \rangle \simeq 0$. If $m_\Phi/H \ll 1$ and the field is potential dominated, then $w_\Phi \simeq -1$. Entering these regimes at different epochs is conditional sequential evolution; its direction and basin are not fixed by the algebraic result.

Dimension check. K, V, ρ_Φ, p_Φ have dimension four; w_Φ is dimensionless.

Dependency. Existing homogeneous scalar stress tensor and declared regime limits.

Status: NO-GO for simultaneous homogeneous DM/DE; **CONDITIONAL** for sequential regimes.

Physical interpretation. One homogeneous scalar stress tensor has one total equation of state per epoch.

Limitations. No multi-component decomposition or omitted sector is added or assessed.

K.7 N6 — Instability of the positive potential maximum

Proposition K.6 (Tachyonic instability of the cosine-potential maximum). For

$$V(\Phi) = \Lambda_\Phi^4 \left[1 - \cos \left(\frac{\Phi}{f_a} \right) \right],$$

$$V'(\Phi) = \frac{\Lambda_\Phi^4}{f_a} \sin \left(\frac{\Phi}{f_a} \right), \quad V''(\Phi) = \frac{\Lambda_\Phi^4}{f_a^2} \cos \left(\frac{\Phi}{f_a} \right).$$

At $\Phi_{\max} = (2n + 1)\pi f_a$,

$$V = 2\Lambda_\Phi^4, \quad V'' = -m_\Phi^2, \quad m_\Phi^2 = \frac{\Lambda_\Phi^4}{f_a^2} > 0.$$

Assumptions. The existing source-free homogeneous scalar equation and $H \geq 0$.

Proof. Direct differentiation proves the derivative formulas. Writing $\Phi = \Phi_{\max} + \delta\Phi$ gives

$$\delta\ddot{\Phi} + 3H\delta\dot{\Phi} - m_\Phi^2\delta\Phi = 0.$$

For $\delta\Phi \propto e^{\lambda t}$,

$$\lambda_\pm = \frac{-3H \pm \sqrt{9H^2 + 4m_\Phi^2}}{2}.$$

Because $\sqrt{9H^2 + 4m_\Phi^2} > 3H$, one has $\lambda_+ > 0$. The maximum is unstable. At every minimum $\Phi_{\min} = 2n\pi f_a$, $V(\Phi_{\min}) = 0$ and $V''(\Phi_{\min}) = +m_\Phi^2$. $\square$

Dimension and sign check. $[V'] = M^3$ and $[V''] = M^2$; negative curvature is not a kinetic ghost.

Dependency. Existing periodic potential and homogeneous scalar perturbation equation.

Status: DERIVED / NO-GO for a stable permanent de Sitter attractor from this potential alone.

Physical interpretation. The positive maximum is an exact but unstable de Sitter background; stable minima have zero potential energy.

Limitations. The result makes no claim about omitted sectors. A fine-tuned constant maximum or transient potential-dominated trajectory remains possible but unstable.

K.8 N7 — Exact pure-disformal differential geometry

Proposition K.7 (Connection, curvature, and geodesic identities). Let

$$\tilde{g}_{\mu\nu} = g_{\mu\nu} - c_D u_\mu u_\nu, \quad u_\mu = \nabla_\mu \Phi, \quad H_{\mu\nu} = \nabla_\mu \nabla_\nu \Phi,$$

where $c_D = \alpha_{\text{dis}}/M_*^4$ is constant and $\Delta_D = 1 - c_D u^\mu u_\mu = 1 + 2c_D X \neq 0$. The two Levi-Civita connections differ by

$$C^\rho{}_{\mu\nu} \equiv \tilde{\Gamma}^\rho{}_{\mu\nu} - \Gamma^\rho{}_{\mu\nu} = -\frac{c_D}{\Delta_D} u^\rho H_{\mu\nu}. \quad (\text{K.4})$$

With the convention $[\nabla_\mu, \nabla_\nu]v^\rho = R^\rho{}_{\sigma\mu\nu}v^\sigma$, the exact curvature relations are

$$\tilde{R}^\rho{}_{\sigma\mu\nu} = R^\rho{}_{\sigma\mu\nu} + \nabla_\mu C^\rho{}_{\nu\sigma} - \nabla_\nu C^\rho{}_{\mu\sigma} + C^\rho{}_{\mu\lambda} C^\lambda{}_{\nu\sigma} - C^\rho{}_{\nu\lambda} C^\lambda{}_{\mu\sigma}, \quad (\text{K.5})$$

$$\tilde{R}_{\sigma\nu} = R_{\sigma\nu} + \nabla_\rho C^\rho{}_{\nu\sigma} - \nabla_\nu C^\rho{}_{\rho\sigma} + C^\rho{}_{\rho\lambda} C^\lambda{}_{\nu\sigma} - C^\rho{}_{\nu\lambda} C^\lambda{}_{\rho\sigma}, \quad (\text{K.6})$$

$$\tilde{R} = \left(g^{\mu\nu} + \frac{c_D}{\Delta_D} u^\mu u^\nu \right) \tilde{R}_{\mu\nu}. \quad (\text{K.7})$$

Consequently, an affinely parametrized $\tilde{g}$ -geodesic with tangent v^μ obeys the exact g -connection equation

$$v^\mu \nabla_\mu v^\rho = \frac{c_D}{\Delta_D} u^\rho H_{\mu\nu} v^\mu v^\nu. \quad (\text{K.8})$$

Assumptions. The existing pure-disformal metric, constant c_D , a twice differentiable scalar representative on the local chart, and the regular branch $\Delta_D \neq 0$.

Proof. Metric compatibility of ∇ and constancy of c_D give

$$\nabla_\mu \tilde{g}_{\nu\sigma} = -c_D (H_{\mu\nu} u_\sigma + u_\nu H_{\mu\sigma}).$$

Since $H_{\mu\nu} = H_{\nu\mu}$ for a scalar,

$$\begin{aligned} \nabla_\mu \tilde{g}_{\nu\sigma} + \nabla_\nu \tilde{g}_{\mu\sigma} - \nabla_\sigma \tilde{g}_{\mu\nu} \\ = -2c_D H_{\mu\nu} u_\sigma. \end{aligned}$$

The already derived inverse metric implies

$$\tilde{g}^{\rho\sigma} u_\sigma = u^\rho + \frac{c_D}{\Delta_D} u^\rho u^2 = \frac{u^\rho}{\Delta_D},$$

because $\Delta_D = 1 - c_D u^2$. Substitution into the defining formula for the Levi-Civita connection proves Equation (K.4). Applying the commutator of $\tilde{\nabla} = \nabla + C$ to a vector gives Equation (K.5); contraction gives Equations (K.6) and (K.7). Finally, $v^\mu \tilde{\nabla}_\mu v^\rho = 0$ is equivalent to $v^\mu \nabla_\mu v^\rho = -C^\rho{}_{\mu\nu} v^\mu v^\nu$, which proves Equation (K.8). $\square$

Dimension and sign check. $[c_D] = M^{-4}$, $[u_\mu] = M^2$, and $[H_{\mu\nu}] = M^3$, so $[C^\rho{}_{\mu\nu}] = M$ as required for a connection in natural units. The minus sign in Equation (K.4) follows from the minus sign in the declared metric update.

Dependency. Existing pure-disformal metric, inverse metric, and scalar Hessian symmetry.
Status: DERIVED.

Physical interpretation. Equation (K.8) is the exact relative acceleration formula when a matter trajectory is assumed to be a $\tilde{g}$ -geodesic.

Limitations. For a null but nonconstant gradient, $\Delta_D = 1$ but the connection difference generally remains nonzero. The connection agrees with that of g when $c_D = 0$, $u_\mu = 0$, or $H_{\mu\nu} = 0$, with further accidental degeneracies possible. These geometric identities do not establish the hyperbolicity or stability of an unspecified matter action.

K.9 N8 — Global limit set of the closed scalar-only FLRW system

Proposition K.8 (Monotone energy and future limit set). Consider the existing spatially flat, source-free, scalar-only system on its expanding branch,

$$3M_{\text{Pl}}^2 H^2 = \frac{1}{2} \dot{\Phi}^2 + V(\Phi), \quad \dot{H} = -\frac{\dot{\Phi}^2}{2M_{\text{Pl}}^2}, \quad \ddot{\Phi} + 3H\dot{\Phi} + V'(\Phi) = 0,$$

with the compact identification $\Phi \sim \Phi + 2\pi f_a$ and the existing cosine potential. Every future-complete finite-energy trajectory has a single equilibrium as its omega-limit. That equilibrium is either a zero-energy minimum or a positive-energy maximum of V . A trajectory not belonging to the stable set of a maximum approaches a minimum; convergence to a maximum is a fine-tuned saddle trajectory rather than an attractor.

Assumptions. The closed source-free minimal FLRW equations, $H \geq 0$, $M_{\text{Pl}}^2 > 0$, $f_a > 0$, $\Lambda_\Phi^4 > 0$, compact scalar target, and finite initial scalar energy.

Proof. Let

$$\mathcal{E}(\Phi, \dot{\Phi}) = \frac{1}{2} \dot{\Phi}^2 + V(\Phi) = 3M_{\text{Pl}}^2 H^2.$$

Using the scalar equation,

$$\begin{aligned} \dot{\mathcal{E}} &= \dot{\Phi} \ddot{\Phi} + V'(\Phi) \dot{\Phi} \\ &= \dot{\Phi} [-3H\dot{\Phi} - V'(\Phi)] + V'(\Phi) \dot{\Phi} = -3H\dot{\Phi}^2 \leq 0. \end{aligned}$$

Thus $\mathcal{E}$ is a Lyapunov function. Its initial value bounds $|\dot{\Phi}|$, while compactness bounds Φ modulo its period; the Friedmann equation bounds H . The future orbit is therefore precompact on the expanding constraint surface.

The set $\dot{\mathcal{E}} = 0$ consists of $H = 0$ or $\dot{\Phi} = 0$. The Friedmann constraint and $V \geq 0$ show that $H = 0$ implies $\dot{\Phi} = 0$ and $V = 0$, hence a potential minimum. If $\dot{\Phi} = 0$ with $H > 0$, invariance additionally requires $\ddot{\Phi} = -V'(\Phi) = 0$. The largest invariant subset of $\dot{\mathcal{E}} = 0$ is therefore precisely the finite set of potential minima and maxima on one compact period. LaSalle invariance places the omega-limit set inside this set. Since an omega-limit set of one continuous trajectory is connected, it consists of one equilibrium.

At a maximum, Proposition K.6 gives one positive and one negative homogeneous characteristic exponent. The maximum is a saddle, so only its stable set can approach it; it is not a future attractor. All remaining trajectories approach a minimum. $\square$

Dimension and sign check. $[\mathcal{E}] = M^4$ and $[\dot{\mathcal{E}}] = M^5$. On the expanding branch the right-hand side is nonpositive. The sign reverses as an anti-friction effect on a contracting branch, which is excluded from the statement.

Dependency. Existing source-free scalar-only Friedmann and scalar equations, compactness, and the N6 maximum-instability result.

Status: DERIVED for the declared closed scalar-only expanding subsector.

Physical interpretation. The minimal isolated scalar dissipates toward a stationary point; the positive de Sitter maximum is not the generic late-time endpoint.

Limitations. The theorem does not apply after restoring matter exchange, topological sources, higher operators, spatial inhomogeneity, or a contracting branch. It therefore does not determine the full ATHENA cosmological history.

K.10 N9 — Exact Epstein shape derivatives and asymptotics

Proposition K.9 (Self-dual slope, fixed-area stationarity, and endpoint limits). Let $Z(\beta) = \mathcal{Z}_\beta(-1)$ be the exact continuation in Proposition K.1. Then

$$Z(\beta) = \beta^{-1}Z(\beta^{-1}), \quad \boxed{Z'(1) = -\frac{1}{2}Z(1)} = 0.114412155188609\dots \quad (\text{K.9})$$

Hence the exact continuation at fixed R is not stationary at the square torus. By contrast, the fixed-area shape factor

$$F(\beta) = \sqrt{\beta} Z(\beta)$$

obeys $F(\beta) = F(\beta^{-1})$ and $F'(1) = 0$. With $x = \ln \beta$,

$$F(e^x) = -\frac{1}{4\pi^2} \sum'_{m,n \in \mathbb{Z}} (m^2 e^{-x} + n^2 e^x)^{-3/2}, \quad (\text{K.10})$$

$$\begin{aligned} \left. \frac{d^2 F(e^x)}{dx^2} \right|_{x=0} &= -\frac{1}{4\pi^2} \sum'_{m,n \in \mathbb{Z}} \left[\frac{15(n^2 - m^2)^2}{4(m^2 + n^2)^{7/2}} - \frac{3}{2(m^2 + n^2)^{3/2}} \right] \\ &= -0.221871\dots < 0. \end{aligned} \quad (\text{K.11})$$

Thus the square torus is a strict local maximum, not a minimum, of the exact zeta-regularized fixed-area shape energy. Moreover,

$$Z(\beta) = -\frac{\zeta(3)}{2\pi^2}\beta - \frac{1}{6\beta} + O\left(\beta^{-1/2}e^{-2\pi\beta}\right), \quad \beta \rightarrow \infty, \quad (\text{K.12})$$

$$Z(\beta) = -\frac{\zeta(3)}{2\pi^2\beta^2} - \frac{1}{6} + O\left(\beta^{-1/2}e^{-2\pi/\beta}\right), \quad \beta \rightarrow 0^+. \quad (\text{K.13})$$

Assumptions. The existing exact Epstein continuation, positive aspect ratio, and, for the interpretation of F , variation at fixed torus area $A_{T^2} = 4\pi^2 R^2 \beta$.

Proof. Setting $s = -1$ in the exact cycle-exchange law of Proposition K.3 gives $Z(\beta) = \beta^{-1}Z(\beta^{-1})$. Differentiation gives

$$Z'(\beta) = -\beta^{-2}Z(\beta^{-1}) - \beta^{-3}Z'(\beta^{-1}),$$

so $2Z'(1) = -Z(1)$, proving Equation (K.9). Multiplication by $\sqrt{\beta}$ makes the cycle exchange an exact invariance and therefore $F'(1) = 0$.

Using $Z(\beta) = -\beta\mathcal{S}(\beta)/(4\pi^2)$ in F yields Equation (K.10). The series and its first two x derivatives converge absolutely, so termwise differentiation gives Equation (K.11). Its sign can be certified by truncating at $\max(|m|, |n|) \leq 100$: the omitted absolute contribution is bounded using at most $8j$ lattice sites on the j th max-norm shell and $|(d^2/dx^2)A^{-3/2}| \leq (21/4)(m^2 + n^2)^{-3/2}$. The resulting bound is smaller than the magnitude of the negative partial result. The exponentially convergent Bessel form gives the displayed numerical value.

Finally, $K_1(y) = O(y^{-1/2}e^{-y})$ as $y \rightarrow \infty$ applied to Equation (K.2) proves the large- β expansion. The small- β expansion follows from exact cycle exchange. $\square$

Dimension check. Z , F , β , and x are dimensionless. At fixed area, $E_{\text{vac}}^\zeta = \pi F(\beta)/\sqrt{A_{T^2}}$ has energy dimension.

Dependency. Existing exact Epstein definition, N1 Poisson/Bessel representation, N3 cycle-exchange law, and the declared torus area.

Status: DERIVED.

Physical interpretation. The result separates fixed- R and fixed-area variations and shows that exact duality supplies stationarity only after the correct fixed-area rescaling.

Limitations. The result neither selects the postulated β_0 nor fixes a physical vacuum subtraction. It does not turn the reduced Casimir ansatz into the exact Epstein continuation.

K.11 N10 — Explicit Hamiltonian constraint closure

Proposition K.10 (Hypersurface-deformation algebra of the closed minimal sector). For the already declared Einstein–Hilbert plus canonical-scalar Hamiltonian, define the smeared secondary constraints

$$\mathcal{H}[N] = \int d^3x N \mathcal{H}_0, \quad \mathcal{D}[\xi] = \int d^3x \xi^i \mathcal{H}_i.$$

Their Poisson brackets close as

$$\{\mathcal{D}[\xi], \mathcal{D}[\eta]\} = \mathcal{D}[[\xi, \eta]], \quad (\text{K.14})$$

$$\{\mathcal{D}[\xi], \mathcal{H}[N]\} = \mathcal{H}[\mathcal{L}_\xi N], \quad (\text{K.15})$$

$$\{\mathcal{H}[N], \mathcal{H}[M]\} = \mathcal{D}[h^{ij}(N\partial_j M - M\partial_j N)]. \quad (\text{K.16})$$

Together with $\pi_N \approx 0$ and $\pi_i \approx 0$, which have vanishing brackets with the secondary constraints, this explicitly proves that the eight constraints used in the minimal degree count are first class.

Assumptions. The existing closed lower-order Einstein–scalar Hamiltonian, the canonical brackets, and either a compact spatial hypersurface without boundary or falloff/boundary conditions that remove surface terms.

Proof. Use

$$\{h_{ij}(\mathbf{x}), \pi^{kl}(\mathbf{y})\} = \frac{1}{2} \left(\delta_i^k \delta_j^l + \delta_i^l \delta_j^k \right) \delta^3(\mathbf{x} - \mathbf{y}), \quad \{\Phi(\mathbf{x}), p_\Phi(\mathbf{y})\} = \delta^3(\mathbf{x} - \mathbf{y}).$$

The momentum constraint then generates spatial Lie derivatives:

$$\{h_{ij}, \mathcal{D}[\xi]\} = \mathcal{L}_\xi h_{ij}, \quad \{\Phi, \mathcal{D}[\xi]\} = \mathcal{L}_\xi \Phi,$$

with the corresponding density transformations for the momenta. This proves Equations (K.14) and (K.15).

For the scalar contribution

$$\mathcal{H}_\Phi[N] = \int d^3x N \left[\frac{p_\Phi^2}{2\sqrt{h}} + \sqrt{h} \left(\frac{1}{2} D_i \Phi D^i \Phi + V \right) \right],$$

the functional derivatives are

$$\frac{\delta \mathcal{H}_\Phi[N]}{\delta p_\Phi} = \frac{N p_\Phi}{\sqrt{h}}, \quad \frac{\delta \mathcal{H}_\Phi[N]}{\delta \Phi} = -D_i (N \sqrt{h} D^i \Phi) + N \sqrt{h} V'(\Phi).$$

Antisymmetrizing in N and M cancels the potential and nondifferentiated lapse terms, leaving

$$\{\mathcal{H}_\Phi[N], \mathcal{H}_\Phi[M]\} = \int d^3x p_\Phi D_i \Phi h^{ij} (N \partial_j M - M \partial_j N).$$

This is precisely the scalar part of the momentum constraint on the right-hand side of Equation (K.16). The standard Einstein–Hilbert calculation supplies the gravitational part $-2D_j \pi^j_i$ with the same structure function; mixed metric–scalar terms cancel because the scalar Hamiltonian is a spatial scalar density. This proves the final bracket. The secondary constraints contain neither N nor N^i , so their brackets with the corresponding primary momenta vanish. $\square$

Dimension and sign check. The smearing functions carry their standard ADM weights, and every bracket has the dimension of the constraint on its right-hand side. The sign in Equation (K.16) follows the displayed convention $\mathcal{H}_i = -2D_j \pi^j_i + p_\Phi D_i \Phi$.

Dependency. Existing ADM action, canonical momenta, Hamiltonian and momentum constraints.

Status: DERIVED.

Physical interpretation. The closed lower-order gravity–scalar subsector has the standard diffeomorphism constraint structure and therefore the previously counted three local propagating degrees of freedom.

Limitations. This calculation does not determine the constraint algebra after unspecified S_m , S_{top} , or $S_{>4}$ sectors are supplied. The total coupled degree count and hyperbolicity remain OPEN.

K.12 Final NEWLY DERIVED and no-go audit table

ID	Result	Mathematical status	Dependency	Physical status
N1	Exact Poisson/Bessel Epstein continuation	DERIVED	Flat T^2 and existing Epstein zeta	Physical subtraction remains CONDITIONAL
N2	Winding-energy aspect minimum	DERIVED	Existing winding-gradient energy	Identification with β_0 remains CONDITIONAL
N3	Exact duality versus reduced symmetry	DERIVED	Exact zeta and reduced ansatz	Normalization remains independent
N4	RG stiffness $24/\pi$	DERIVED	Truncated beta-function ansatz	ANSATZ-CONDITIONAL
N5	Simultaneous homogeneous DM/DE	NO-GO	One canonical homogeneous scalar	Sequential regimes CONDITIONAL
N6	Positive maximum instability	DERIVED / NO-GO	Existing potential and scalar equation	No stable permanent de Sitter attractor from this potential alone
N7	Exact disformal connection and curvature identities	DERIVED	Pure-disformal metric and inverse	Geodesic reading CONDITIONAL on matter motion
N8	Scalar-only FLRW future limit set	DERIVED	Closed expanding source-free scalar system	Full cosmological endpoint remains OPEN
N9	Exact Epstein derivatives and asymptotics	DERIVED	N1, N3, and fixed-area kinematics	Vacuum subtraction remains CONDITIONAL
N10	Minimal Hamiltonian constraint algebra	DERIVED	Closed Einstein–scalar ADM Hamiltonian	Total coupled algebra remains OPEN

K.13 Preserved OPEN and NO-GO boundaries

The physical TTA functor, winding-to-Spin(1, 3) map, winding-to-particle identity or mass, gauge emergence, spin–statistics, Higgs and generation derivations, emergent gravity, complete S_m , complete S_{top} , complete $\mathcal{L}^{(>4)}$, neutrino closure, general PPN, complete cosmological likelihood, lensing, growth, and complete waveform generation remain OPEN or CONDITIONAL. The Derrick obstruction and all calibration/fit distinctions are retained.

The dependency map is

$$\begin{aligned}
\{T^2, \mathcal{Z}_\beta\} &\rightarrow \text{N1} \rightarrow \text{N3}, & \{T^2, \Phi, E_{\mathbf{w}}\} &\rightarrow \text{N2}, \\
\{\beta_{\text{RG}}, \gamma_{\text{str}} = 0\} &\rightarrow \text{N4}, & \{S_\Phi, V, \text{homogeneous FLRW}\} &\rightarrow \{\text{N5}, \text{N6}, \text{N8}\}, \\
\{\tilde{g}, g, \Phi\} &\rightarrow \text{N7}, & \{\text{N1}, \text{N3}, A_{T^2}\} &\rightarrow \text{N9}, & \{S_{\text{low}}, \text{ADM}\} &\rightarrow \text{N10}.
\end{aligned}$$

No observational datum is used as a backward theoretical input.

ATHENA Ultimate V15.3.3 therefore records ten controlled mathematical derivation packages from previously declared structures. It adds no new physics and does not claim a completed or experimentally validated unified theory.